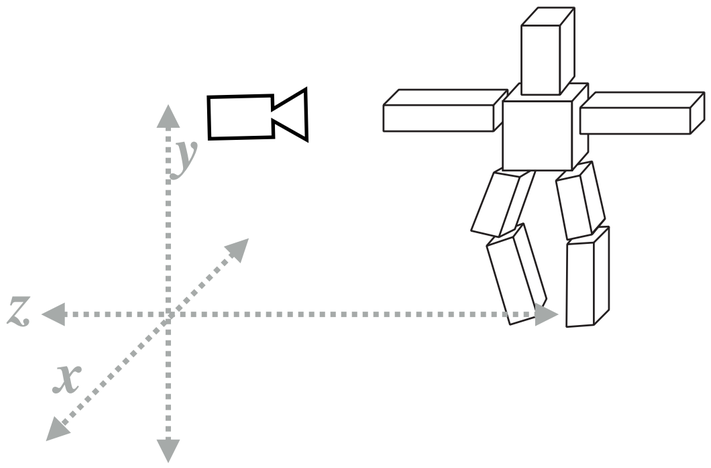
Section 6:

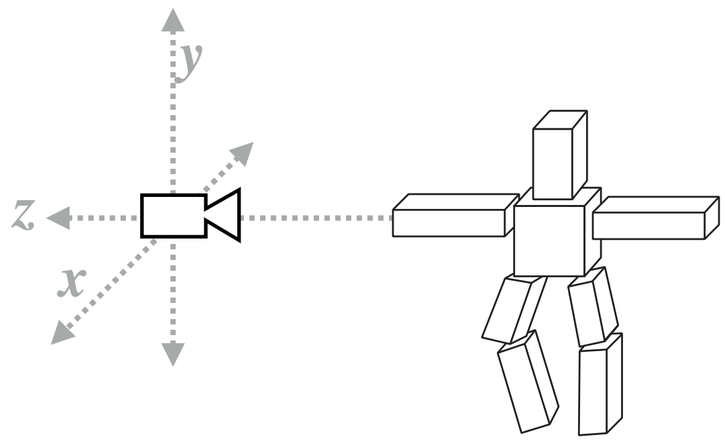
The Rasterization Pipeline

Computer Graphics and Imaging
UC Berkeley CS184/284A

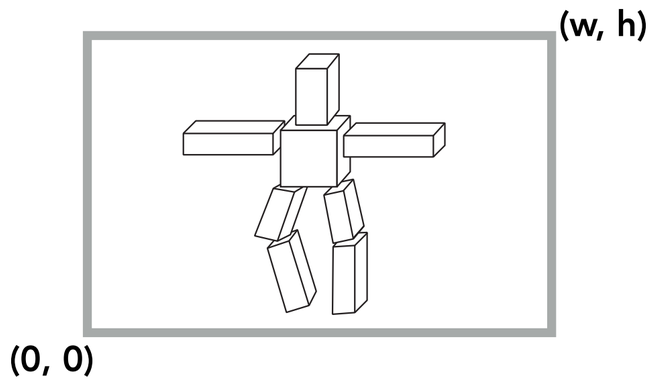
What We've Covered So Far



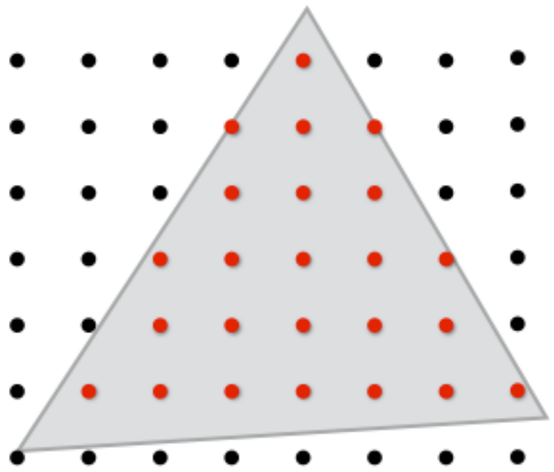
Position objects and the camera in the world



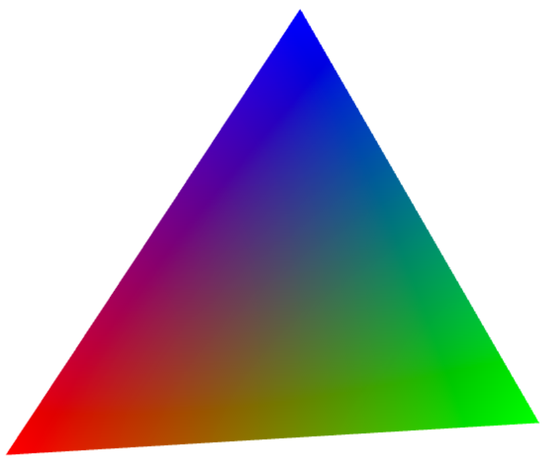
Compute position of objects relative to the camera



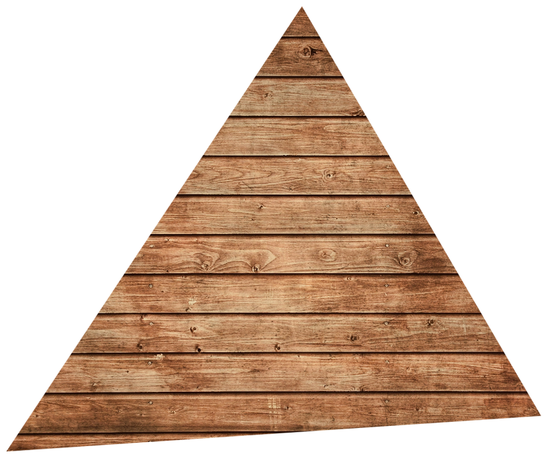
Project objects onto the screen



Sample triangle coverage

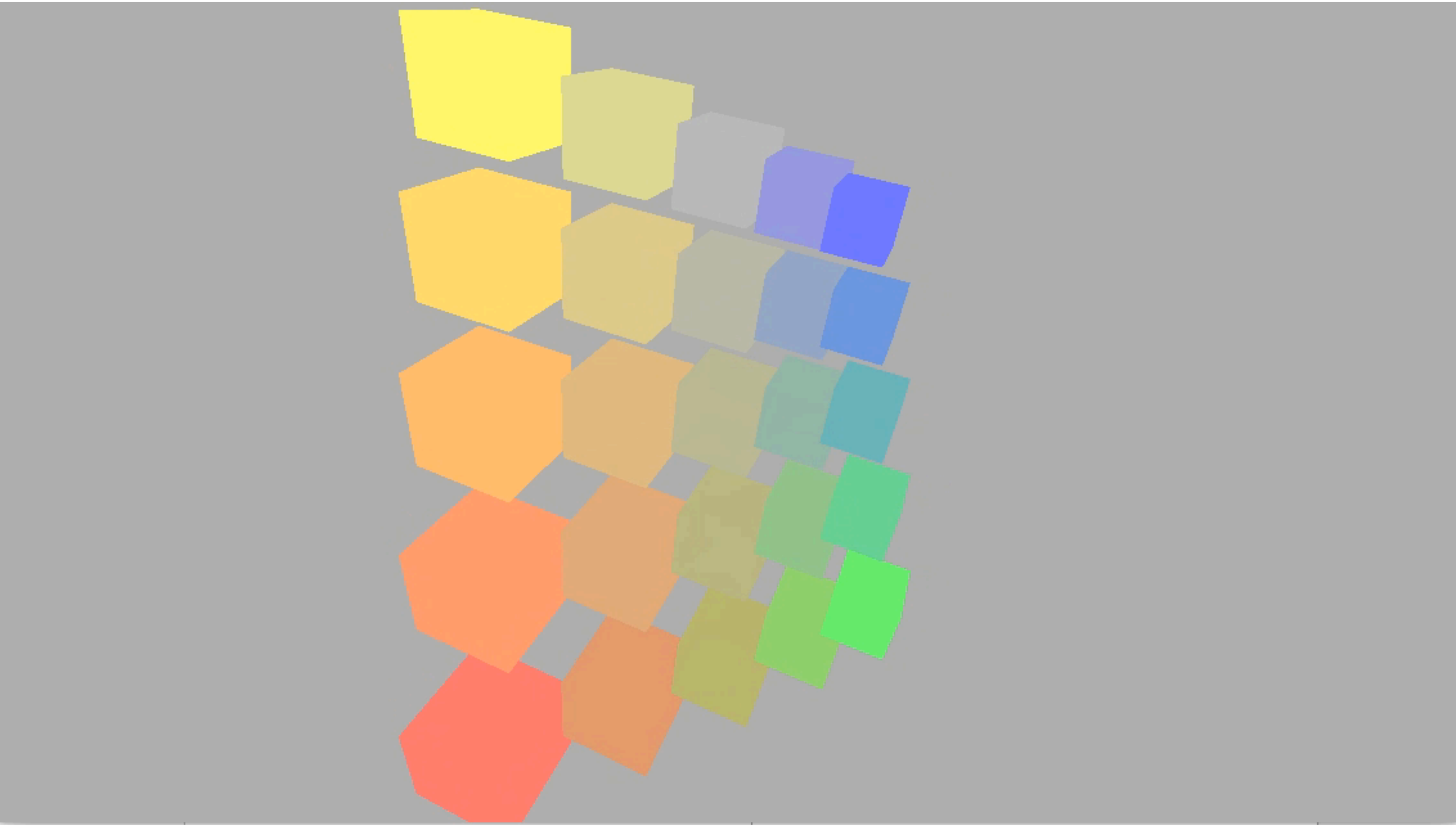


Interpolate triangle attributes



Sample texture maps

Rotating Cubes in Perspective



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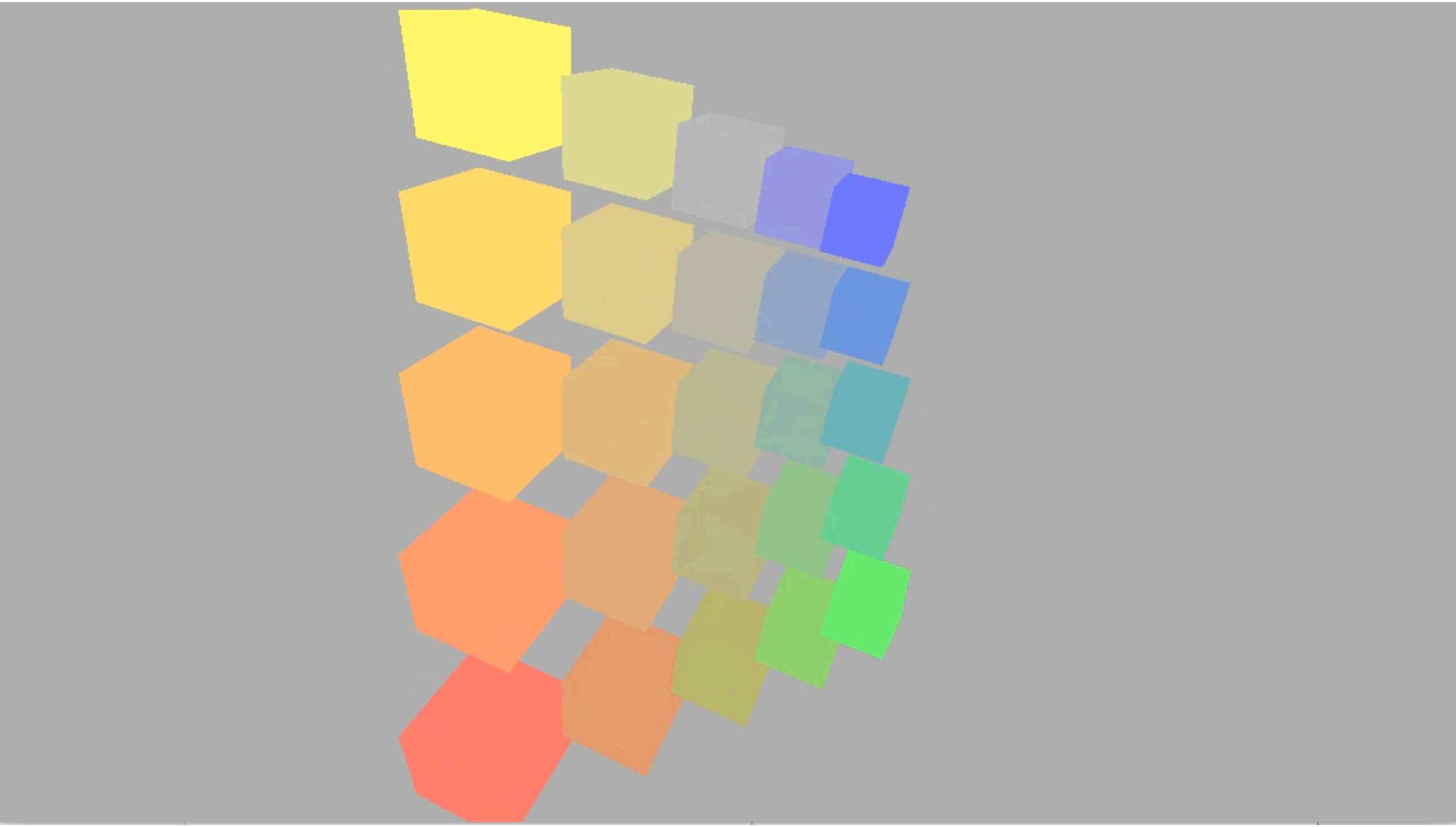

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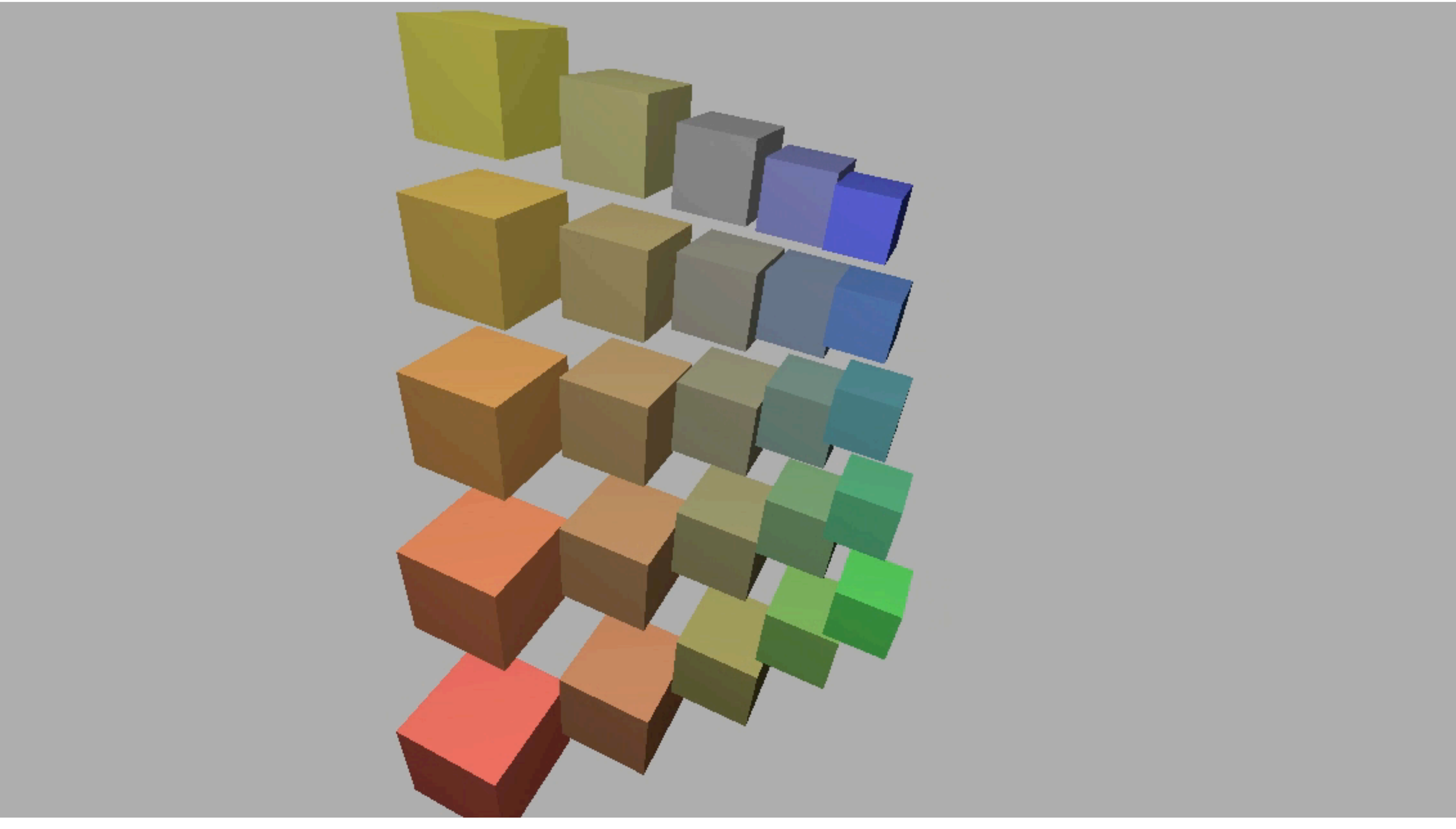

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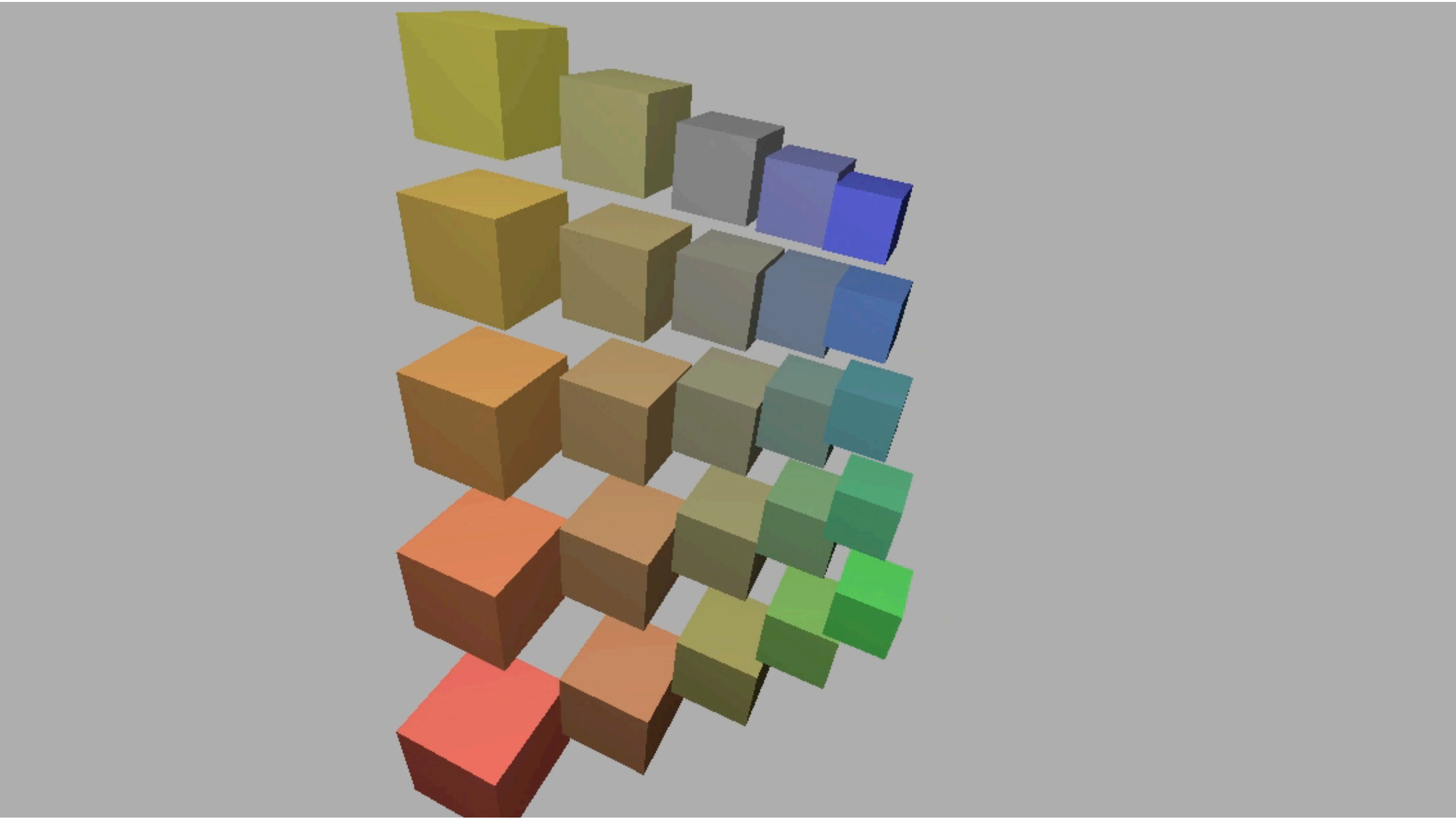

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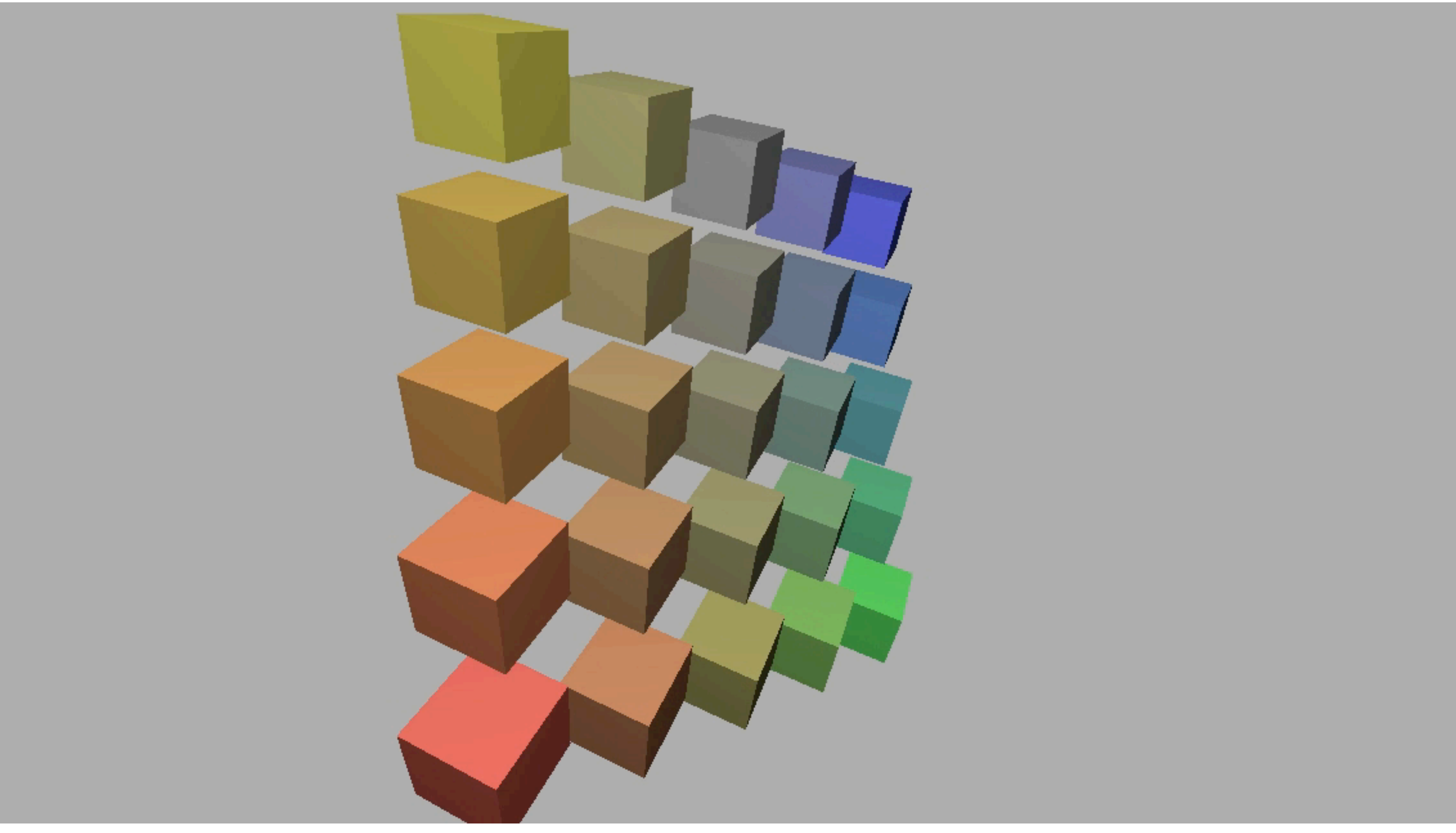
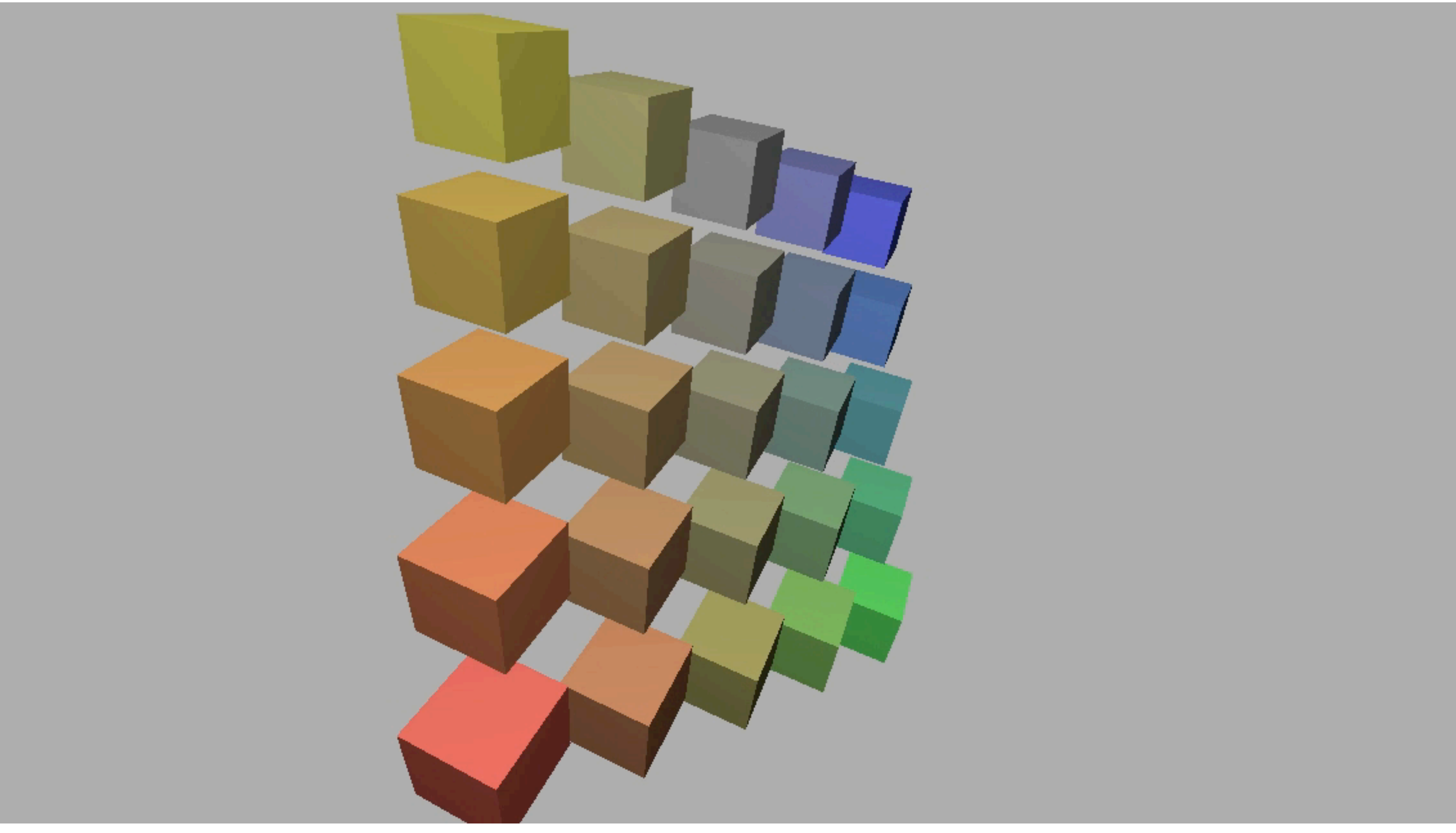


Rotating Cubes in Perspective



What Else Are We Missing?

Credit: Bertrand Benoit. “Sweet Feast,” 2009. [Blender /VRay]

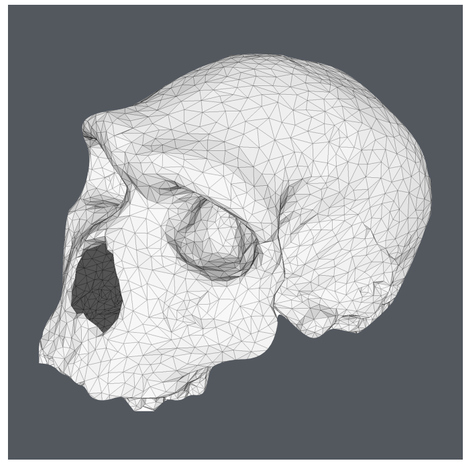
What Else Are We Missing?

Credit: Giuseppe Albergo. “Colibri” [Blender]

What Else Are We Missing?

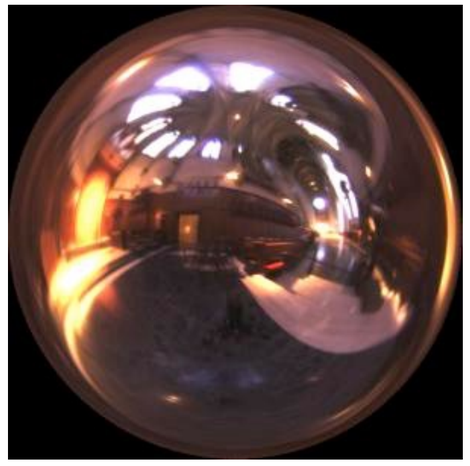
Surface representations

- Objects in the real world exhibit highly complex geometric details



Lighting and materials

- Appearance is a result of how light sources reflect off complex materials



Camera models

- Real lenses create images with focusing and other optical effects

Course Roadmap

Rasterization Pipeline

Core Concepts

- Sampling
- Antialiasing
- Transforms

Intro

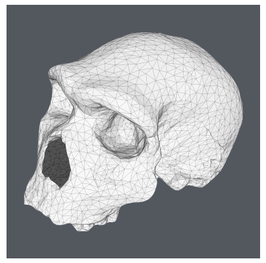
Rasterization

Transforms & Projection

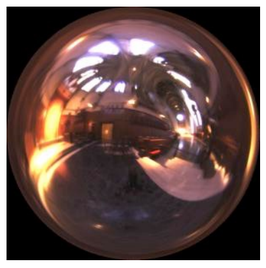
Texture Mapping

Today: Visibility, Shading, Overall Pipeline

Geometric Modeling



Lighting & Materials



Cameras & Imaging

Visibility

Painter's Algorithm

Inspired by how painters paint

Paint from back to front, overwrite in the framebuffer

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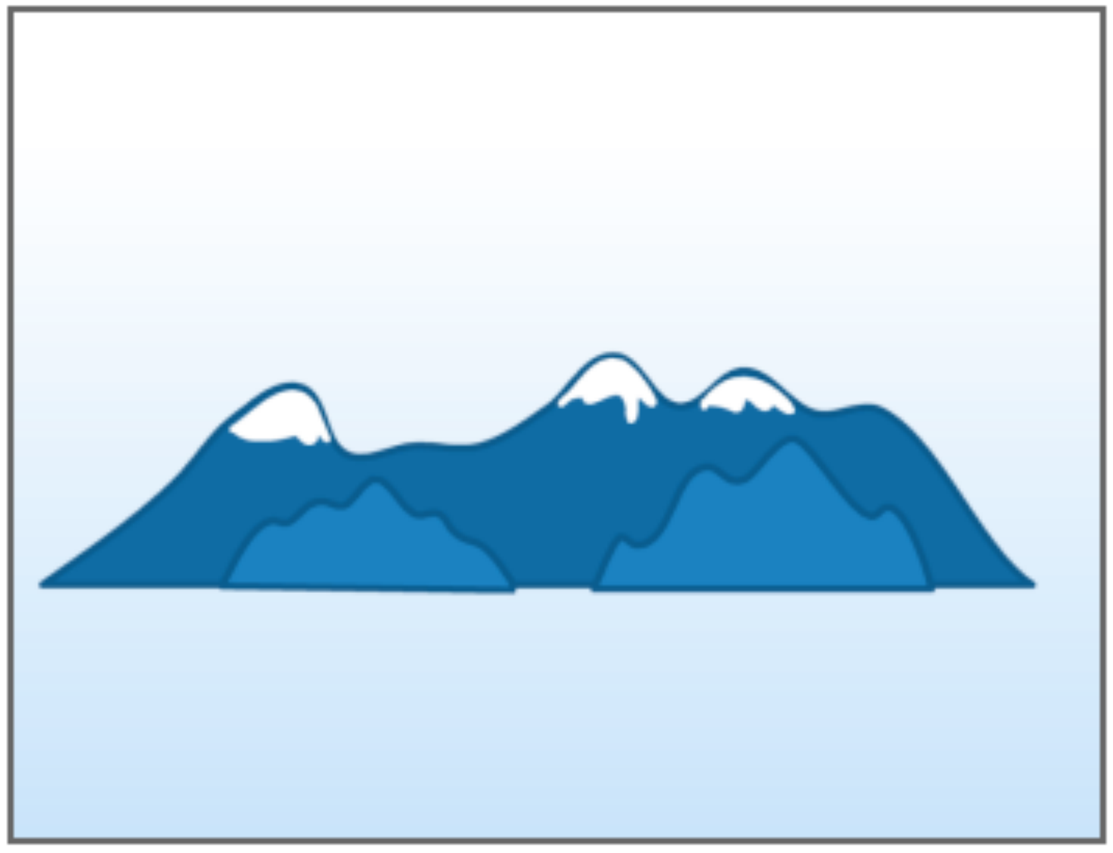

[Wikipedia]

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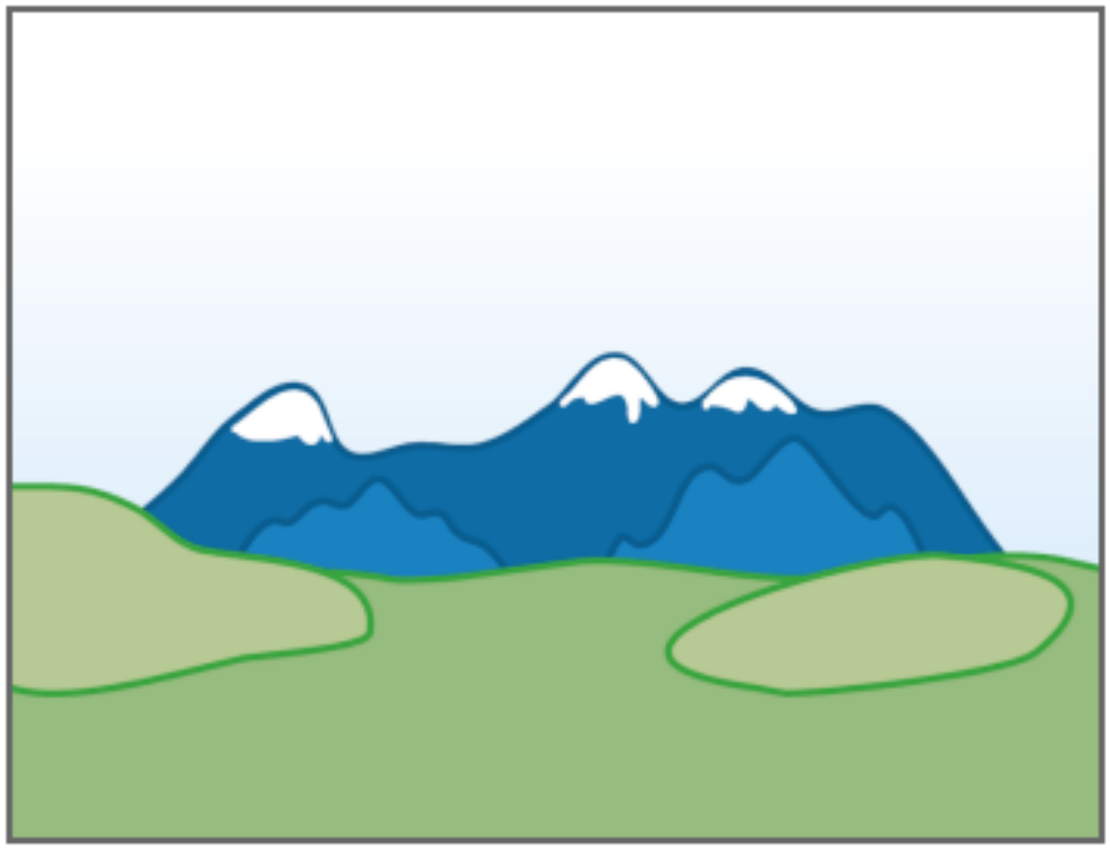

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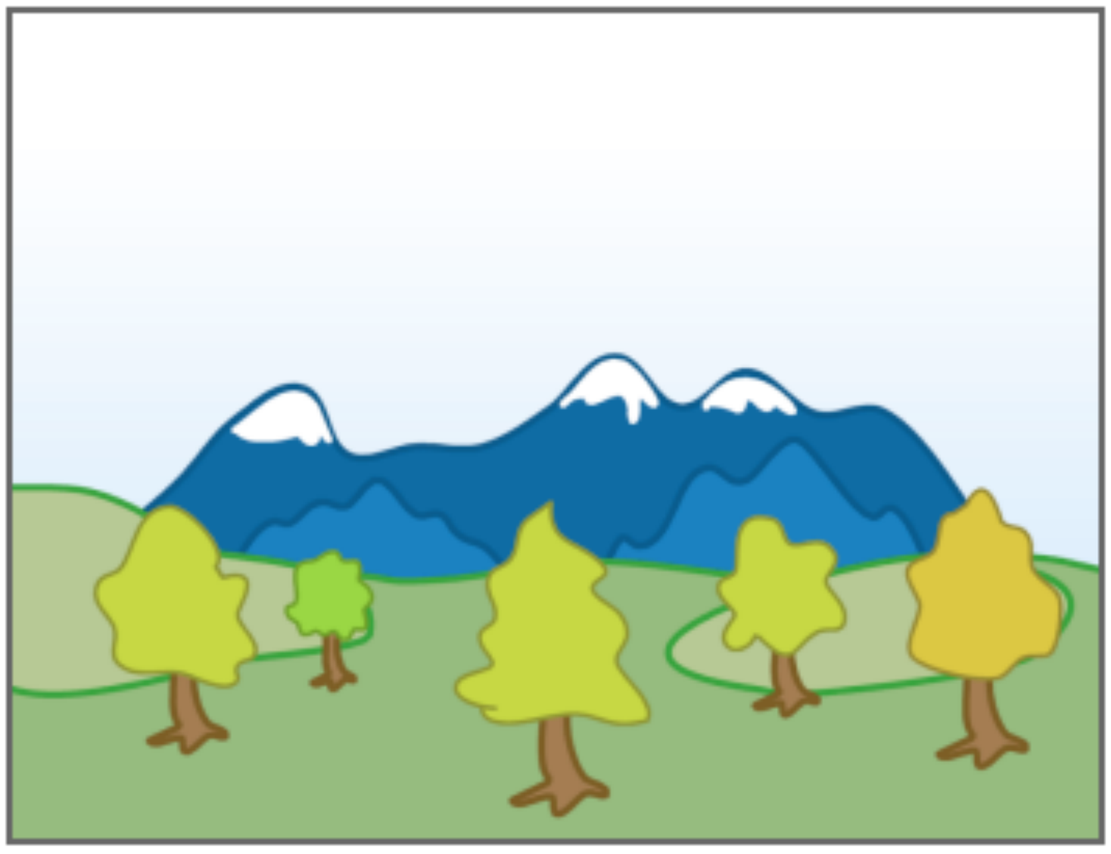

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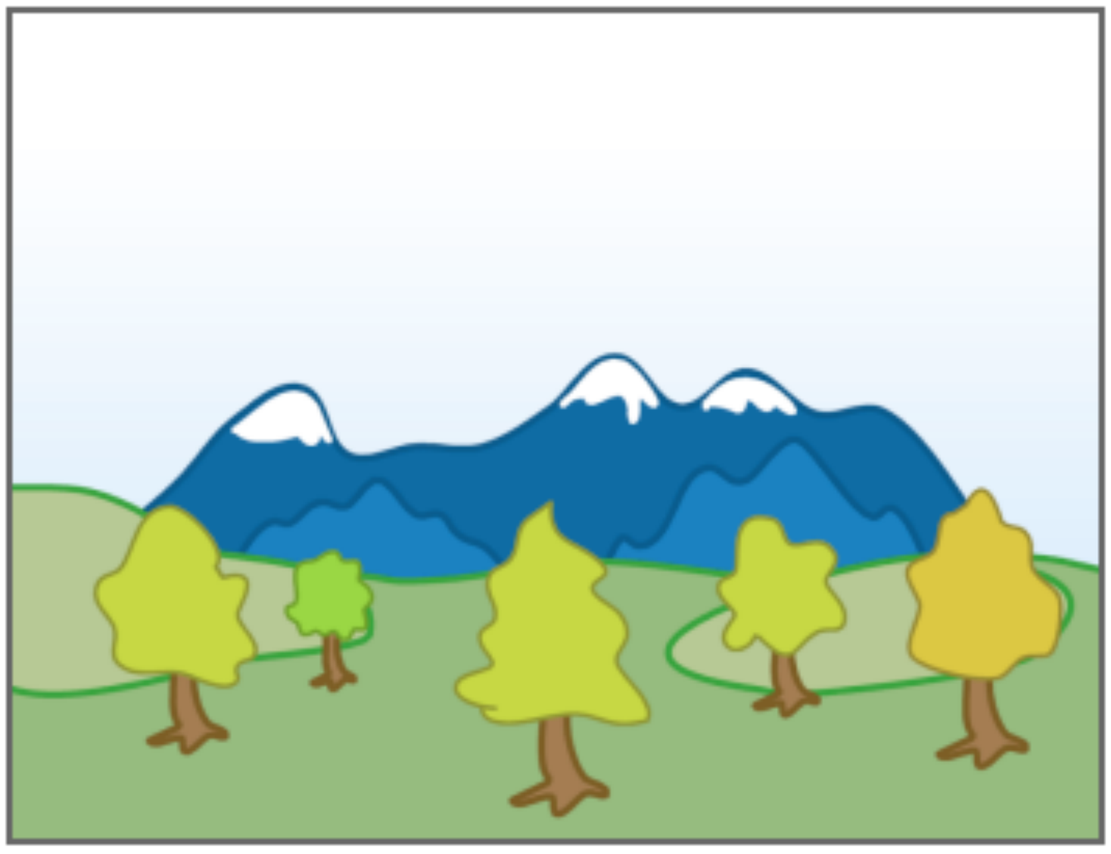

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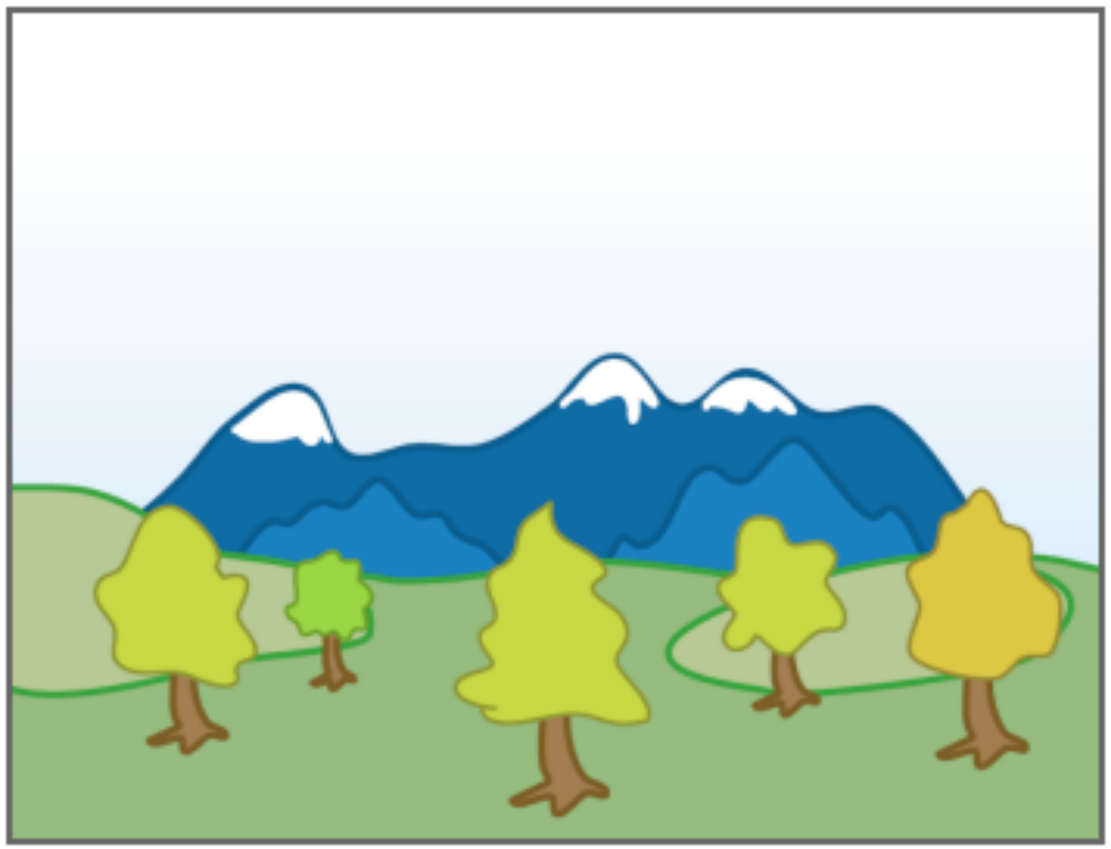

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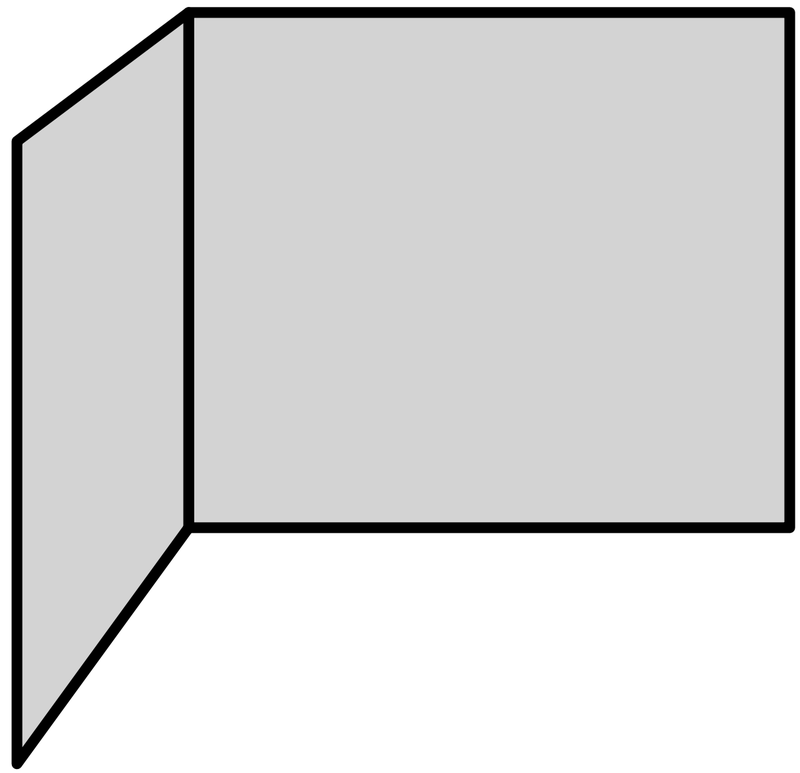

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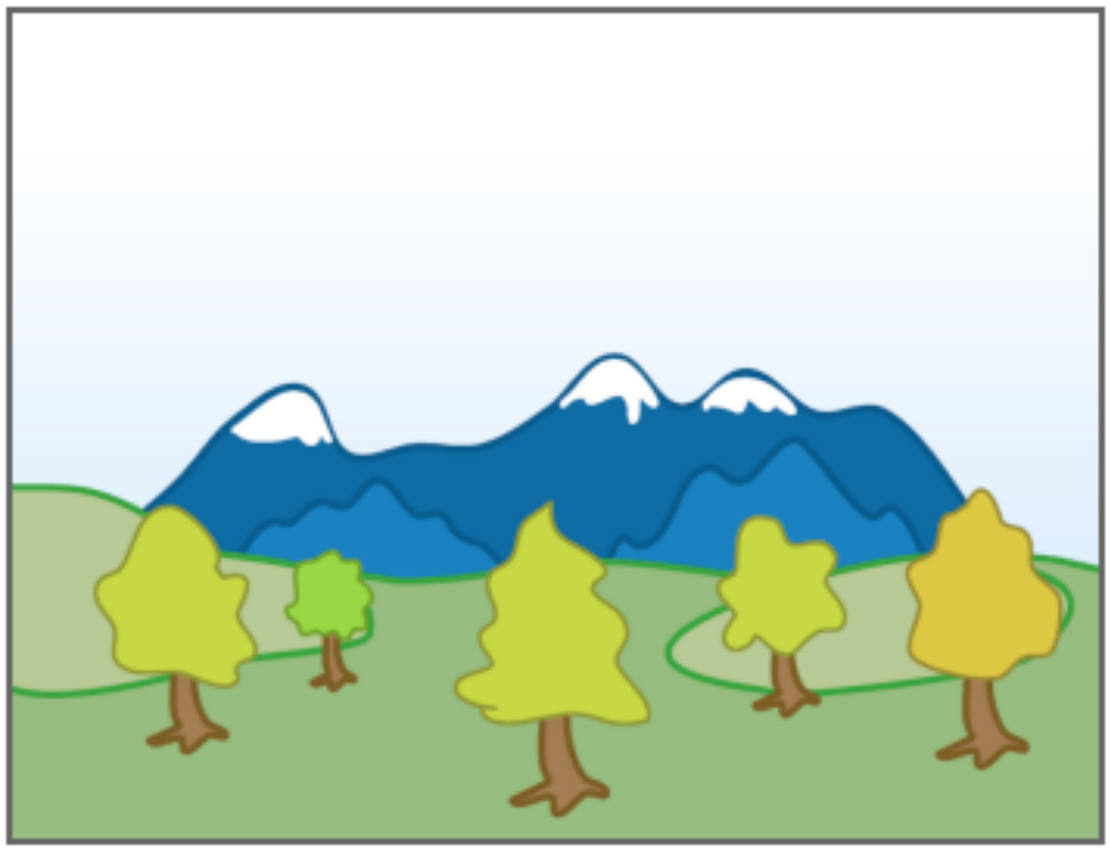

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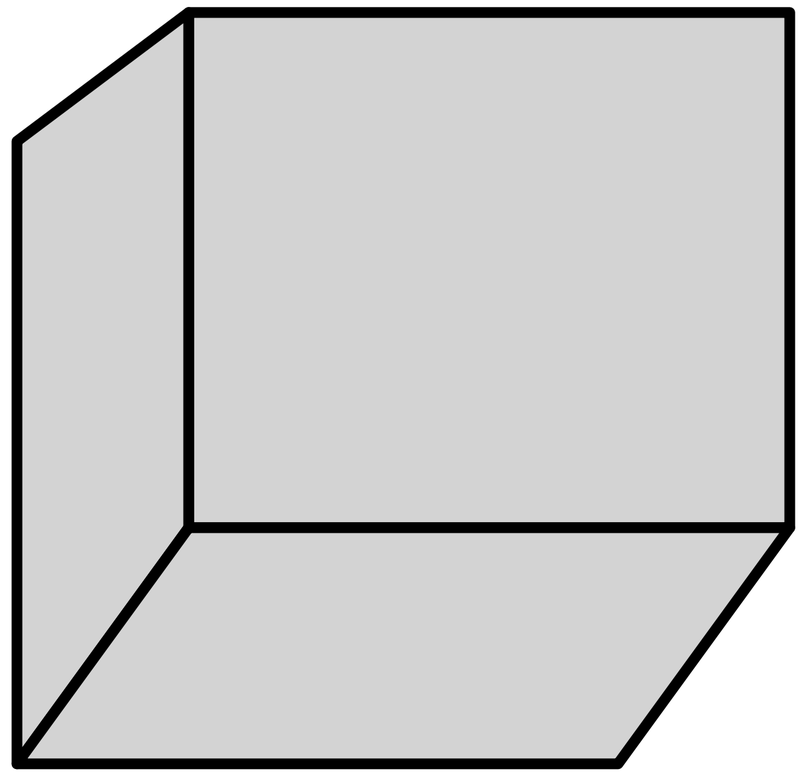

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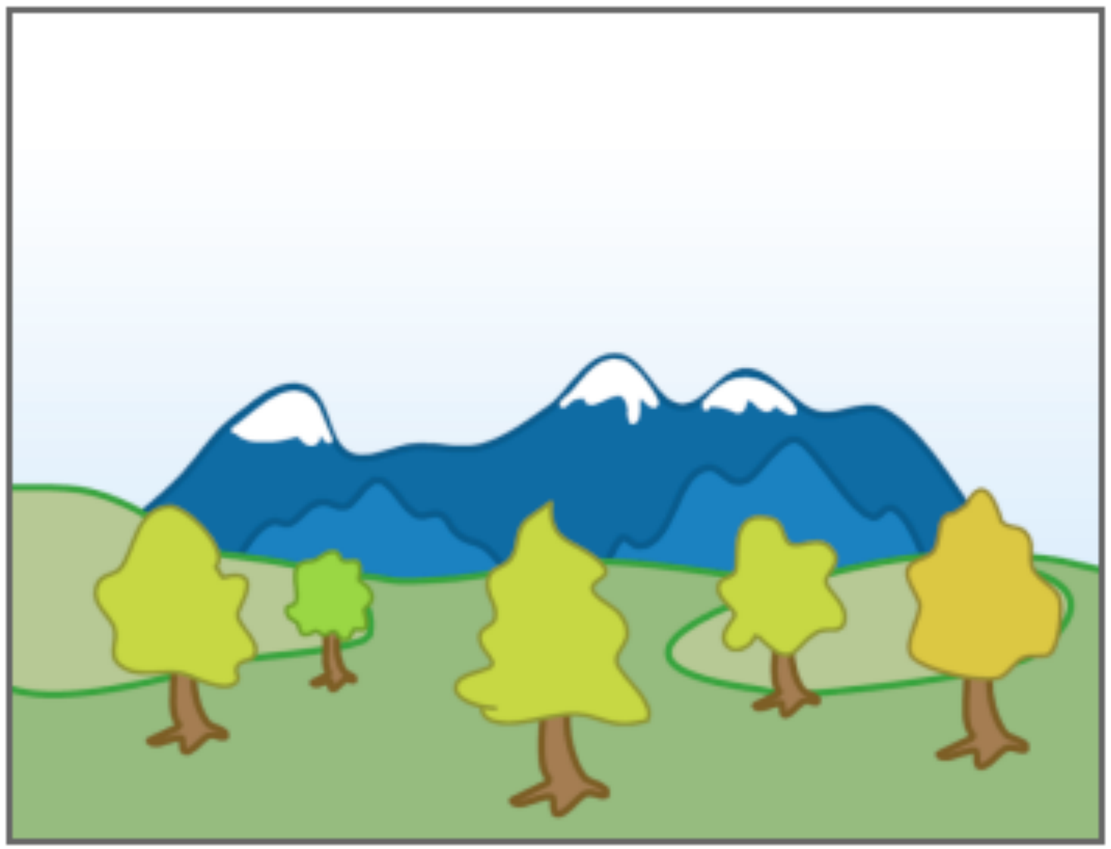

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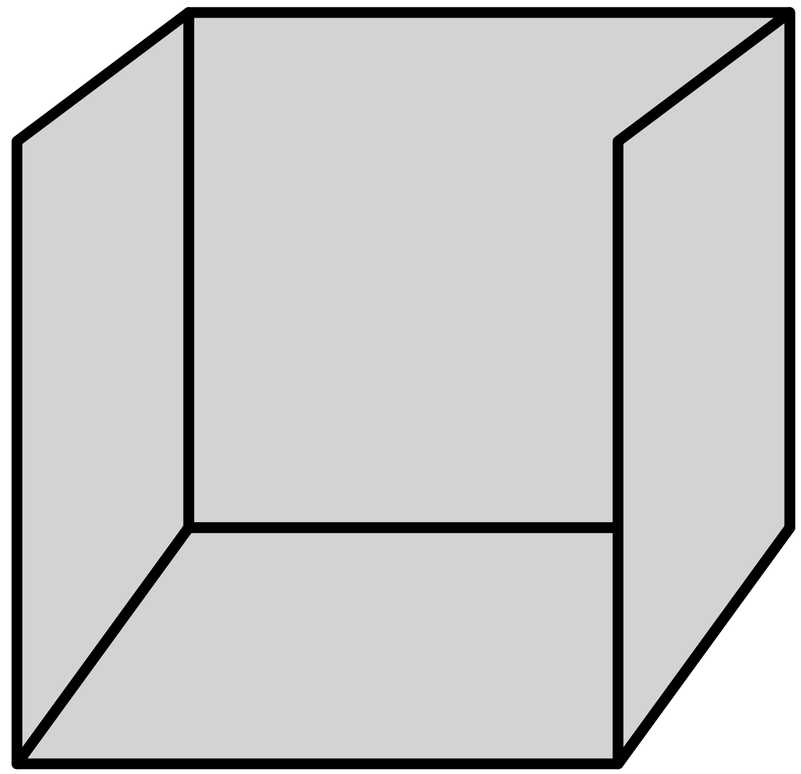

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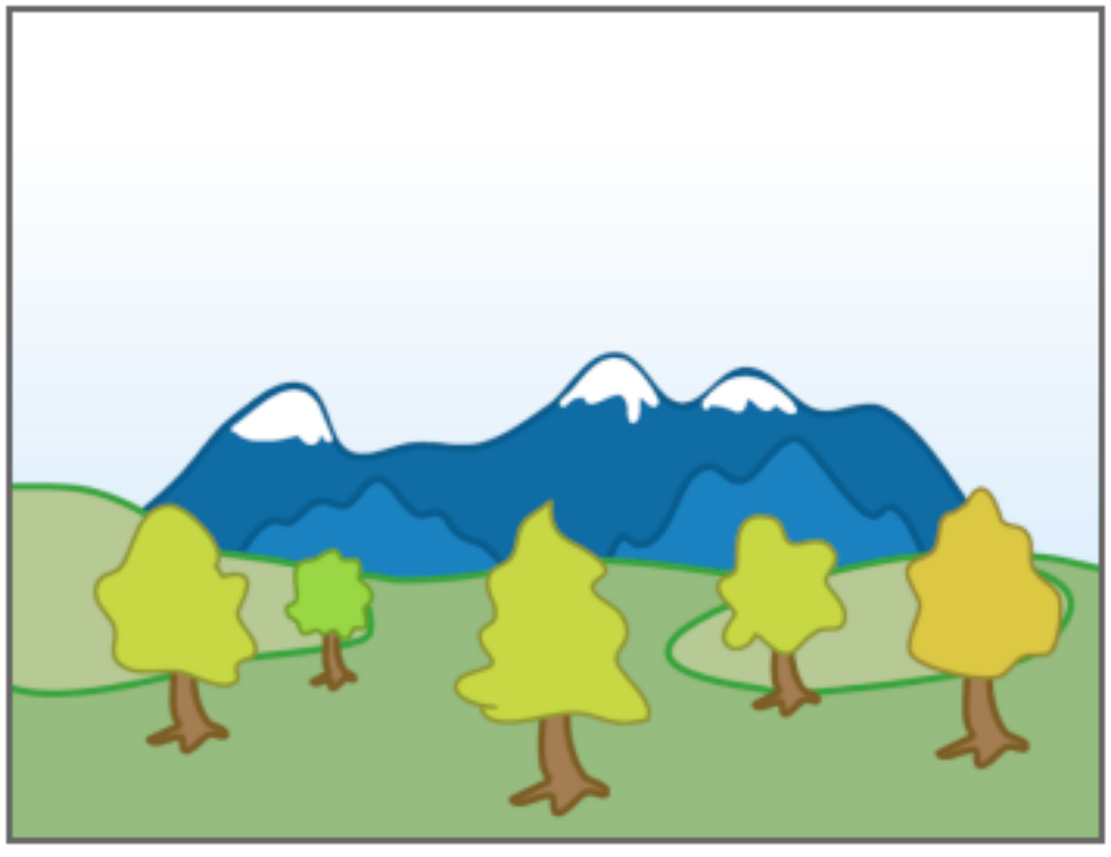

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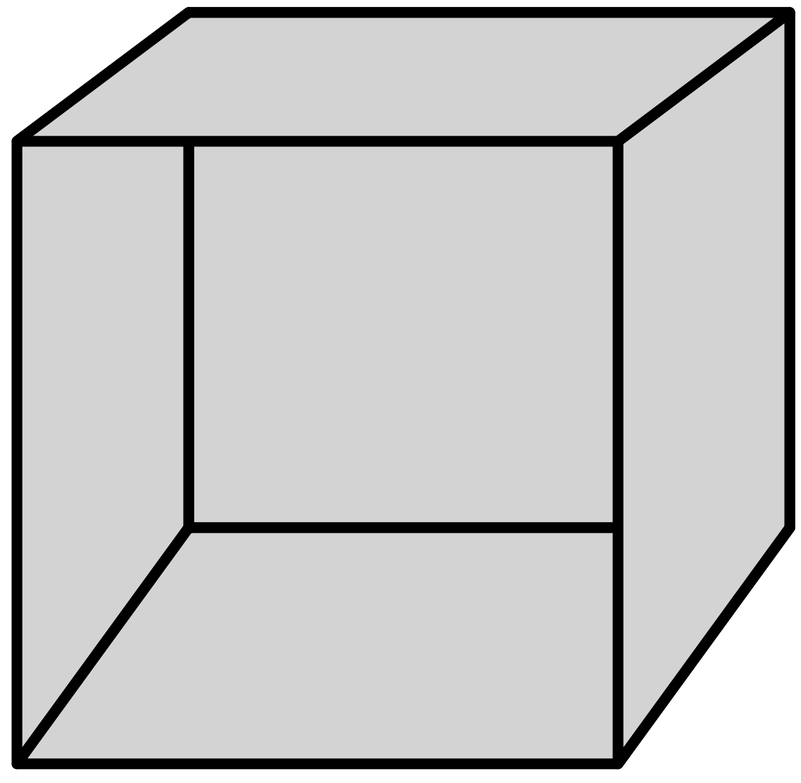

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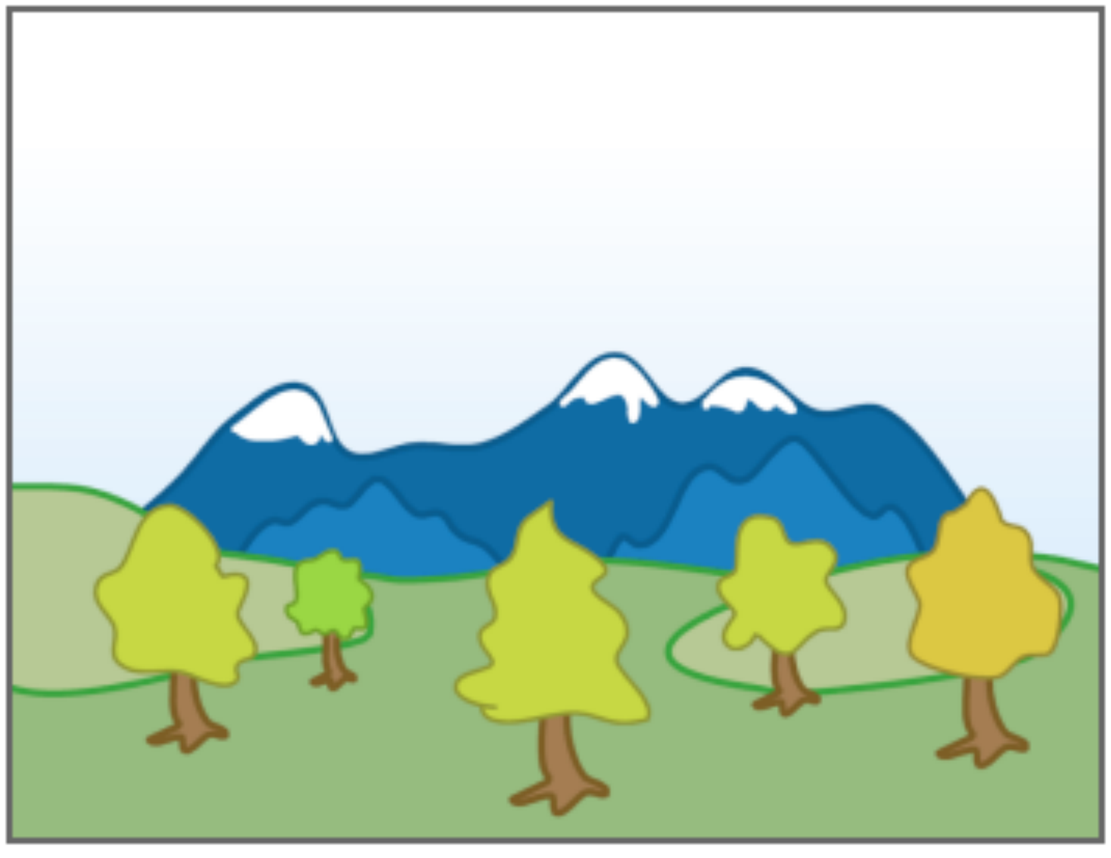

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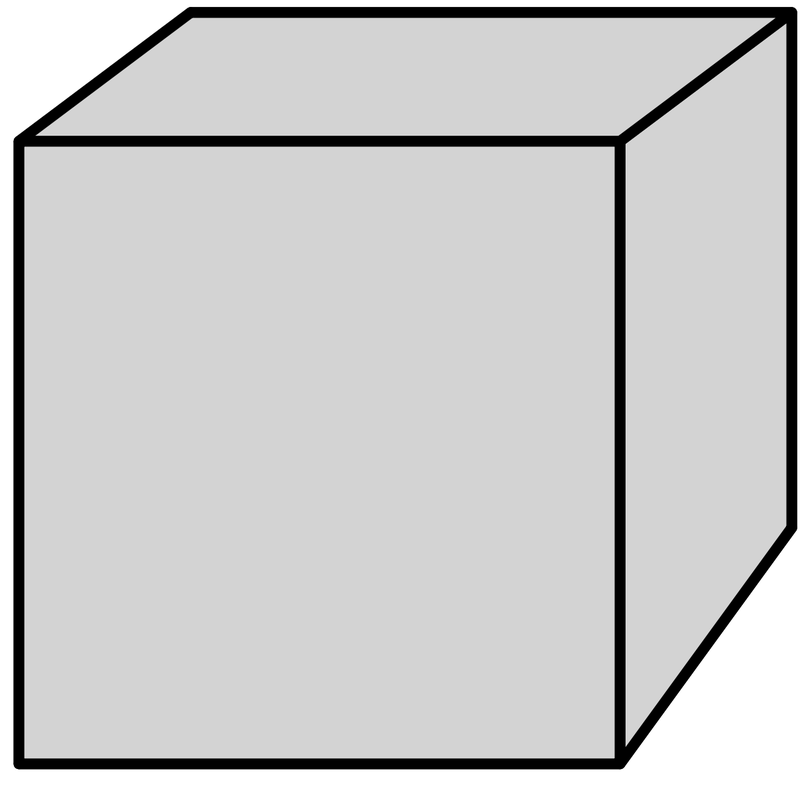
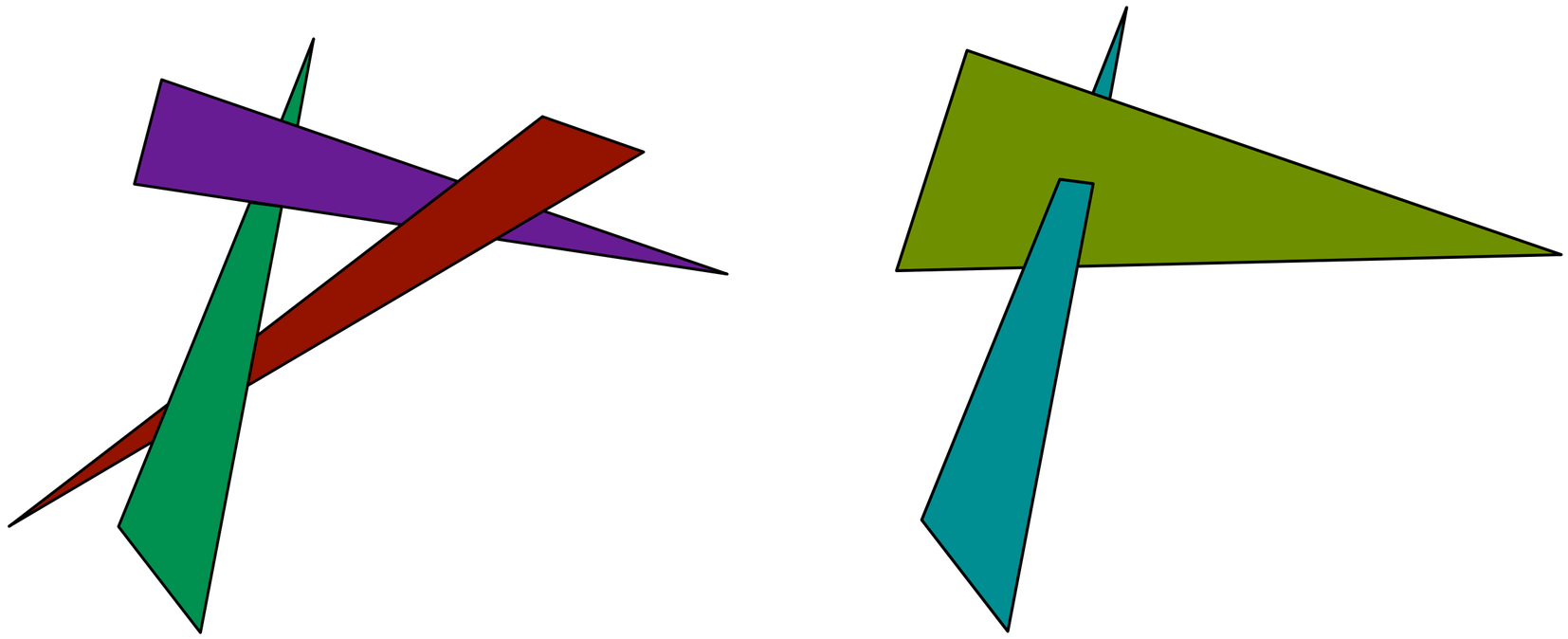


Painter's Algorithm

Requires sorting in depth (O(n log n) for n triangles)

Can have unresolvable depth order



(BSP Trees will provide a way of dealing with both issues.)

Z-Buffer

Z-Buffer

This is the hidden-surface-removal algorithm that eventually won.

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Idea:

Z-Buffer

This is the hidden-surface-removal algorithm that eventually won.

Idea:

- Store current min. z-value for each sample position

Z-Buffer

This is the hidden-surface-removal algorithm that eventually won.

Idea:

- **Store current min. z-value for <u>each</u> sample position**
- **Needs an additional buffer for depth values**

Z-Buffer

This is the hidden-surface-removal algorithm that eventually won.

Idea:

- Store current min. z-value for <u>each</u> sample position
- Needs an additional buffer for depth values
  - framebuffer stores RBG color values

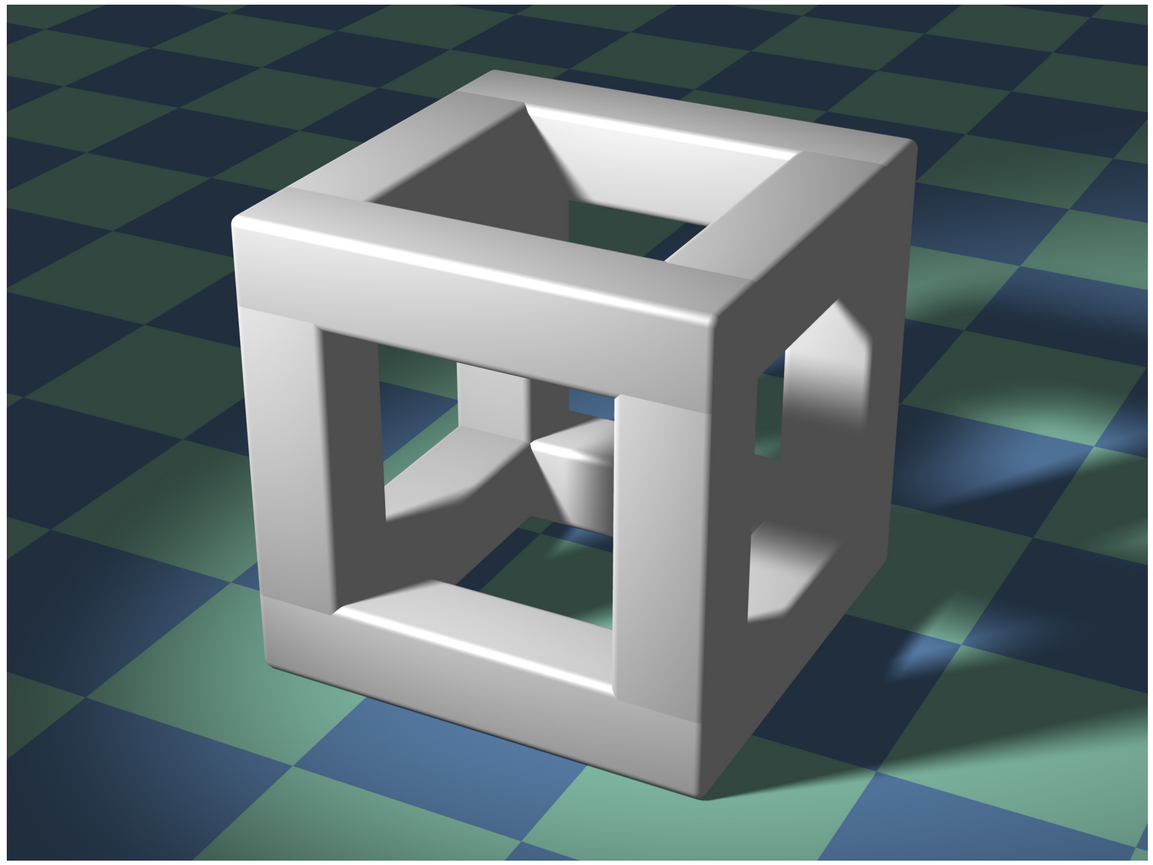
Z-Buffer

This is the hidden-surface-removal algorithm that eventually won.

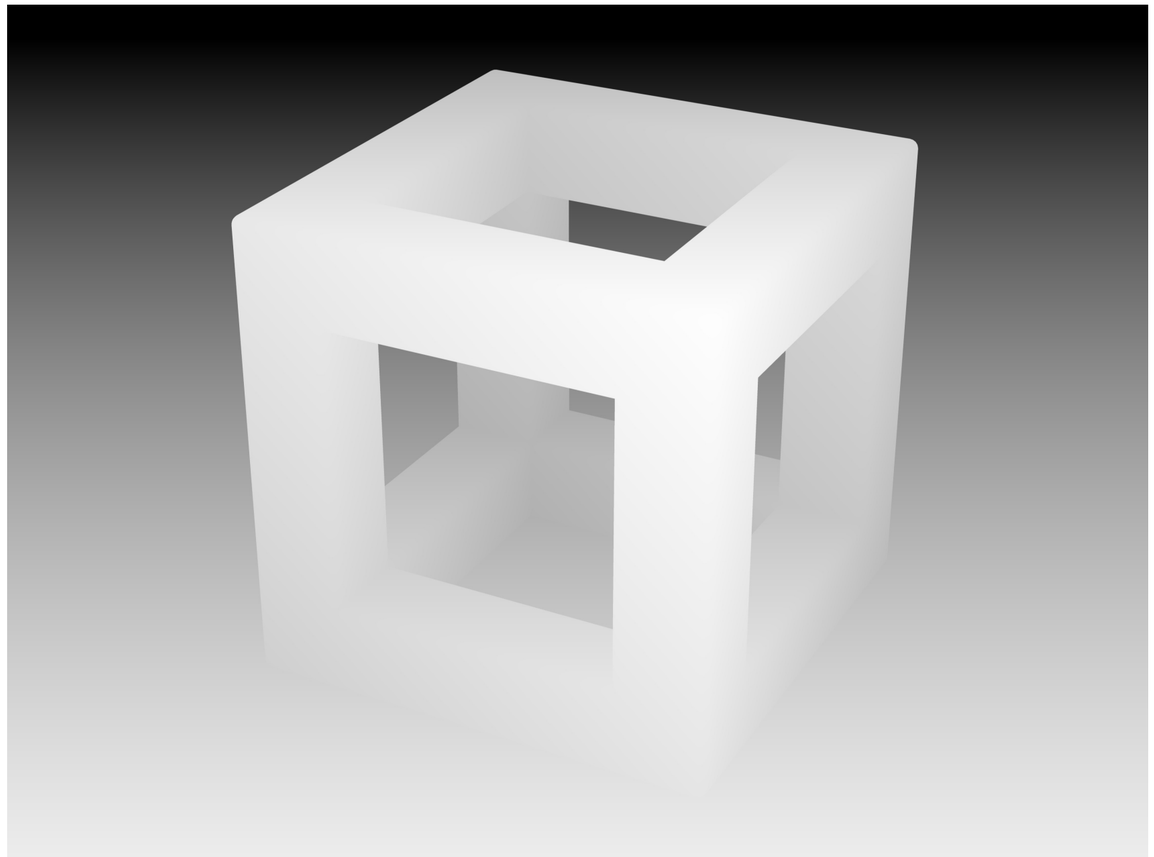
Idea:

- **Store current min. z-value for <u>each</u> sample position**
- **Needs an additional buffer for depth values**
  - **framebuffer stores RBG color values**
  - **depth buffer (z-buffer) stores depth (16 to 32 bits)**

Z-Buffer Example



Rendering



Depth buffer

Image credit: Dominic Alves, flickr.

Z-Buffer Algorithm

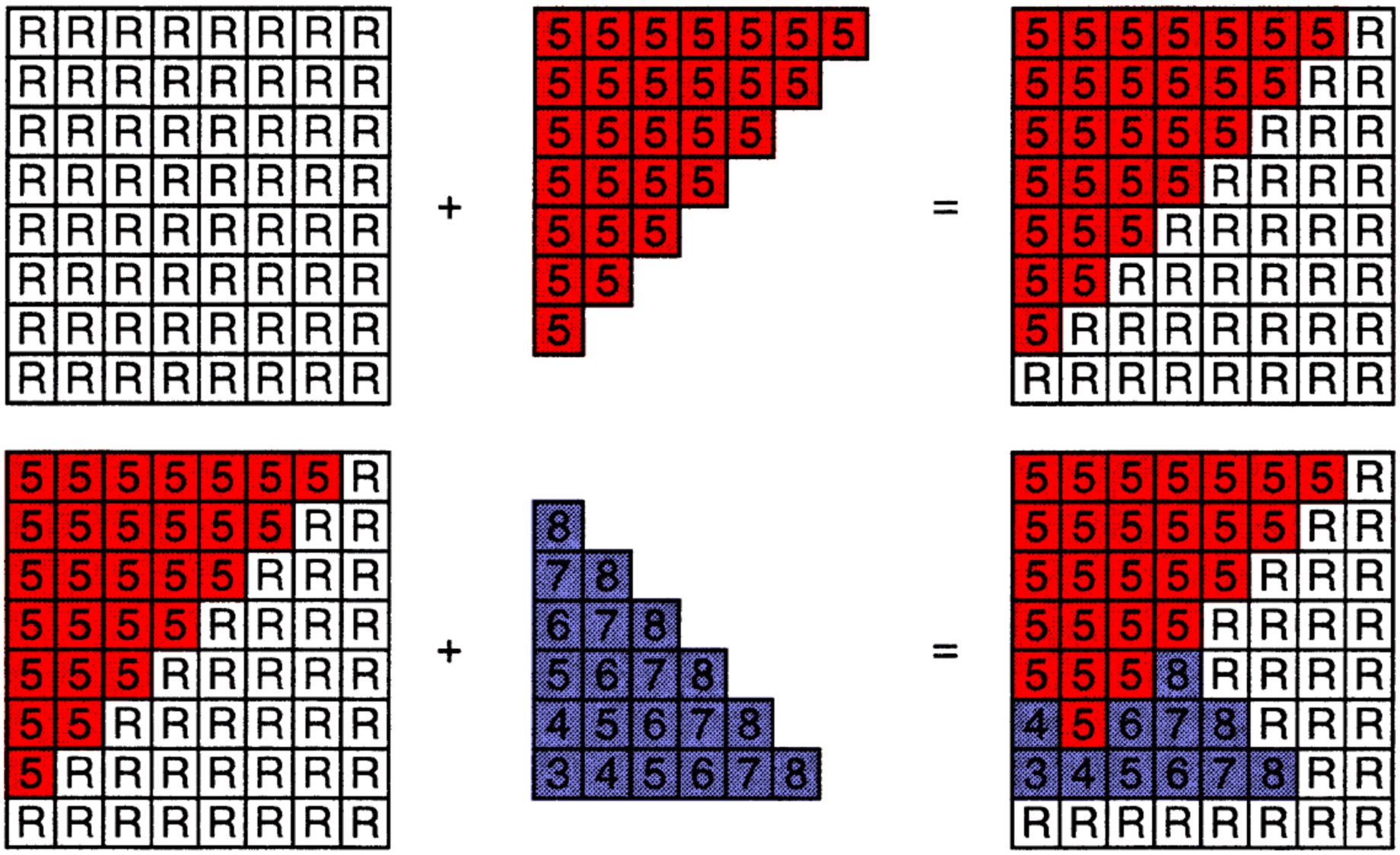
Initialize depth buffer to ∞

During rasterization:

```
for (each triangle T)
    for (each sample (x,y,z) in T)
        if (z < zbuffer[x,y])              // closest sample so far
            framebuffer[x,y] = rgb;        // update (compute) color
            zbuffer[x,y]     = z;          // update z
        else
            ;       // do nothing, this sample is not closest
```

// Careful with Z vs -Z. No point in storing sign bit.

Z-Buffer Algorithm



(In this example positive values are further.)

Z-Buffer Complexity

Complexity

- O(*n*) for *n* triangles
- How can we sort *n* triangles in linear time?

Most important visibility algorithm

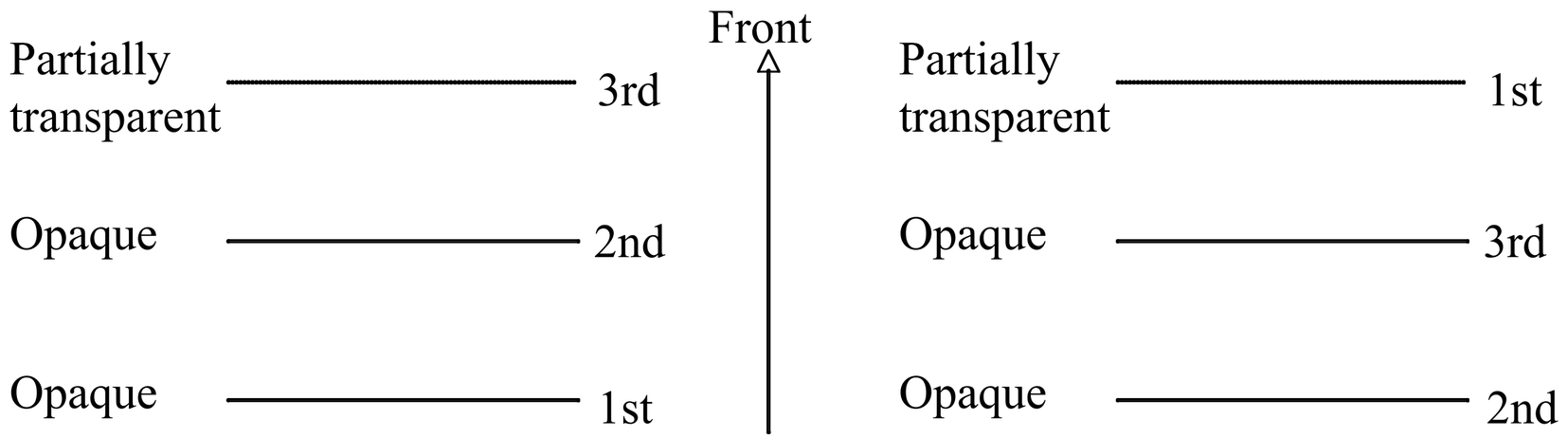
- Implemented in hardware for all GPUs
- Used by OpenGL

Issues

- Overdraw
- Transparency

Z-Buffer and Transparency

Transparency requires partial sorting



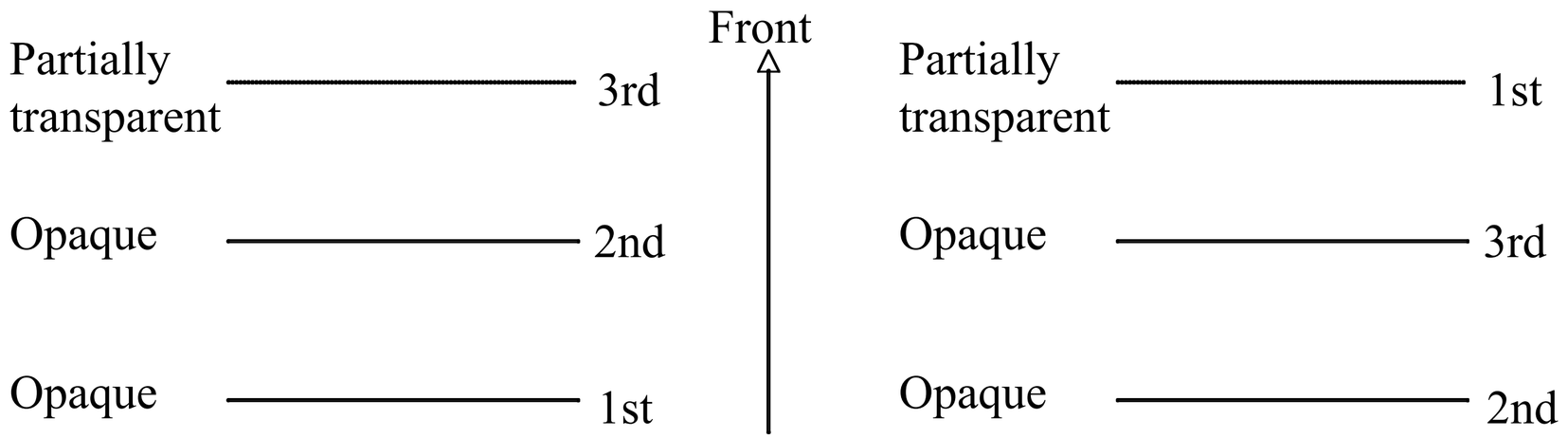
Good Not Good

Common solution:

- **Draw opaque polygons first**
- **Then draw transparent polygons (Ideally in sorted order)**

Z-Buffer and Transparency

Transparency requires partial sorting



Good Not Good

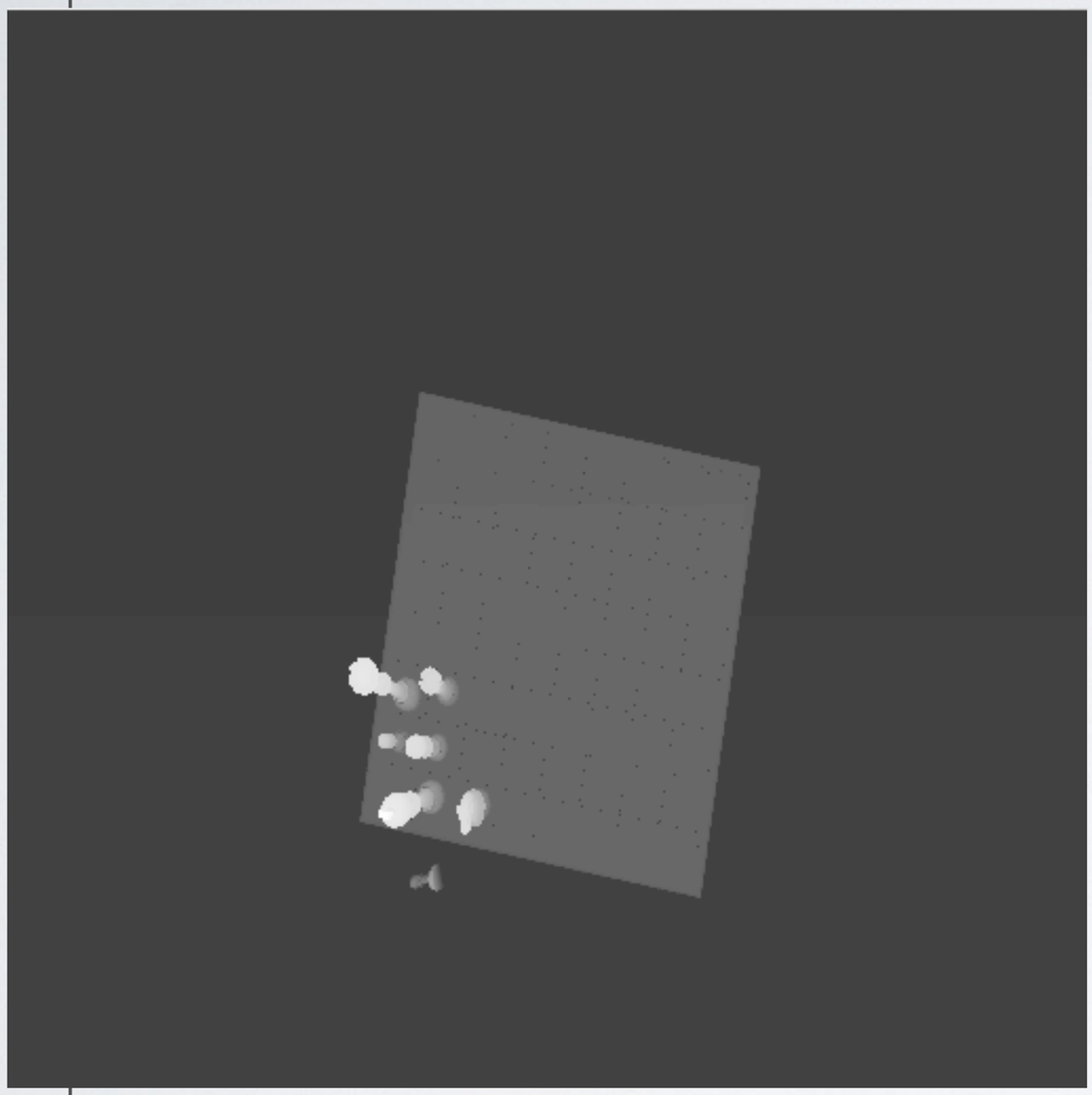
Another solution:

- **Linked list of RGB-Z-α at each pixel (Deep Alpha Buffer)**

Shadow Maps

- Pre-render scene from perspective of light source
  - Only render Z-Buffer (the shadow buffer)
- Render scene from camera perspective
  - Compare with shadow buffer
  - If nearer light, if further shadow

Shadow Maps



Shadow Buffer

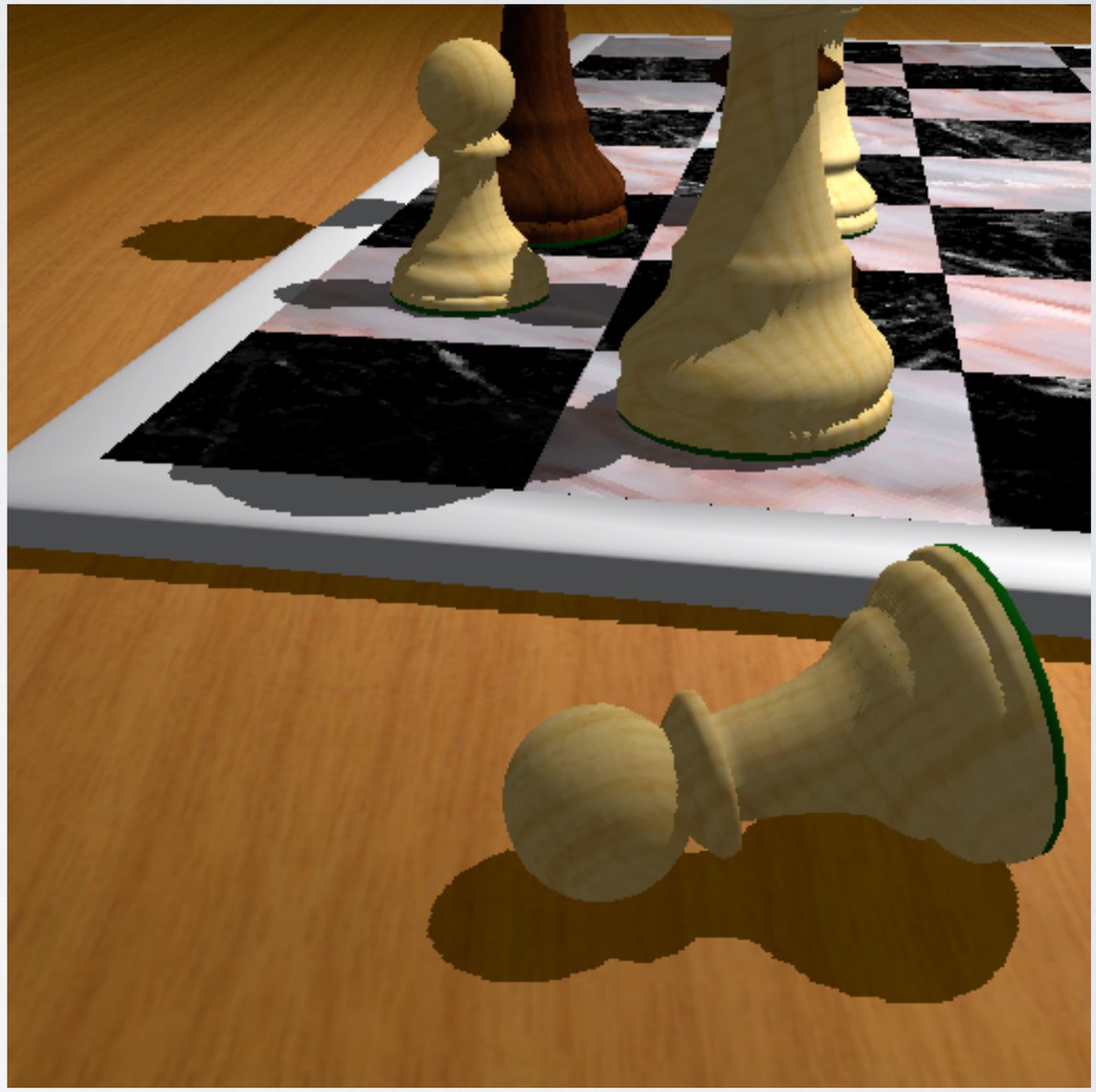


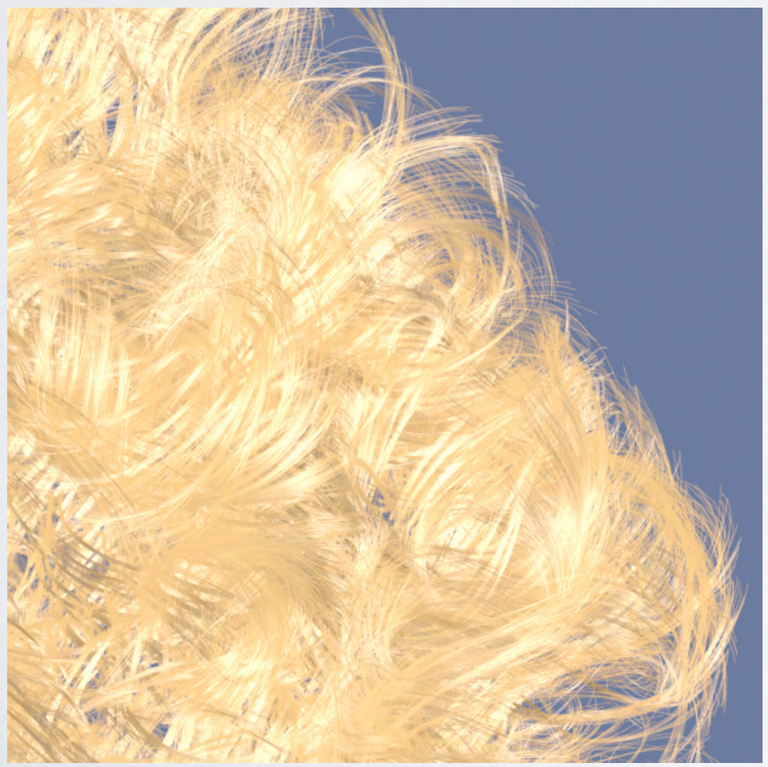
Image w/ Shadows

From Stamminger and Drettakis
SIGGRAPH 2002

Note: These images don't really go together, see the paper...

Deep Shadow Maps

- Some objects only partially occlude light
  - A single shadow value will not work
  - Similar to transparency in Z-Buffer



From
Lokovic and Veach
SIGGRAPH 2000

Attendance

If you are seated in class, go to this form and sign in:

Word of the day: ***visibility***

Lecture number: ***8***

https://tinyurl.com/184fall26

Simple Shading (Blinn-Phong Reflection Model)

Simple Shading vs Realistic Lighting & Materials

What we will cover today

- A local shading model: simple, per-pixel, fast
- Based on perceptual observations, not physics

What we will cover later in the course

- Physics-based lighting and material representations
- Global light transport simulation

Perceptual Observations

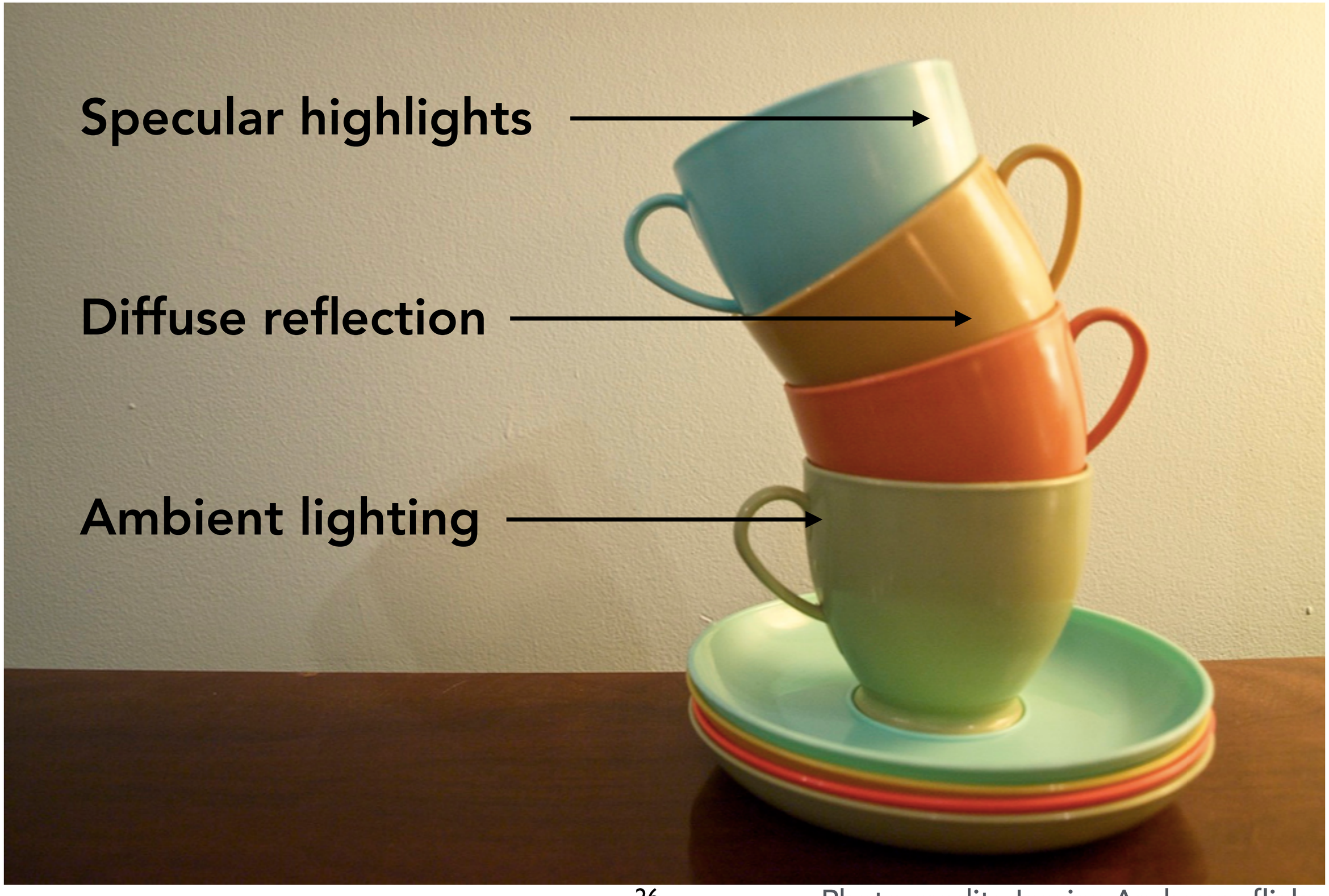


Photo credit: Jessica Andrews, flickr

Local Shading

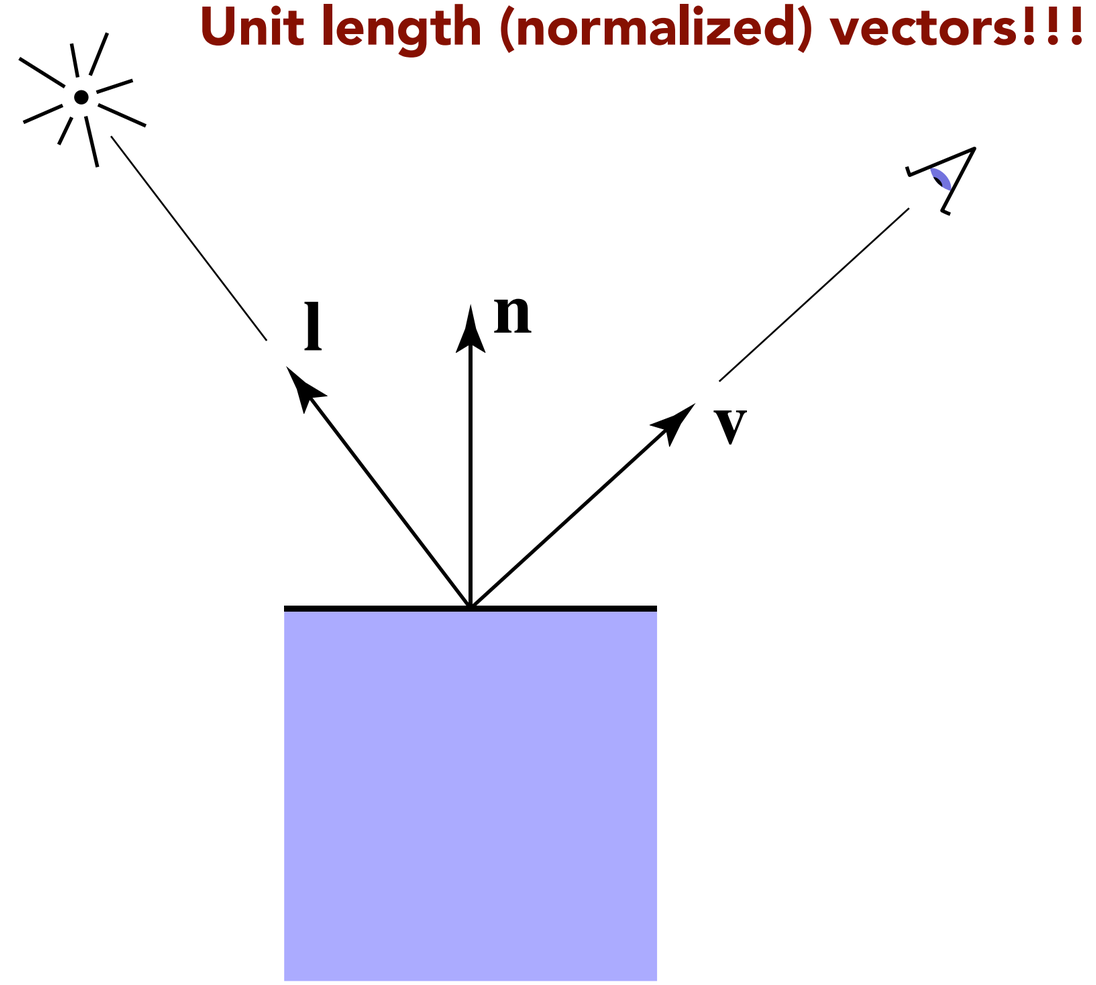
Compute light reflected toward camera

Inputs:

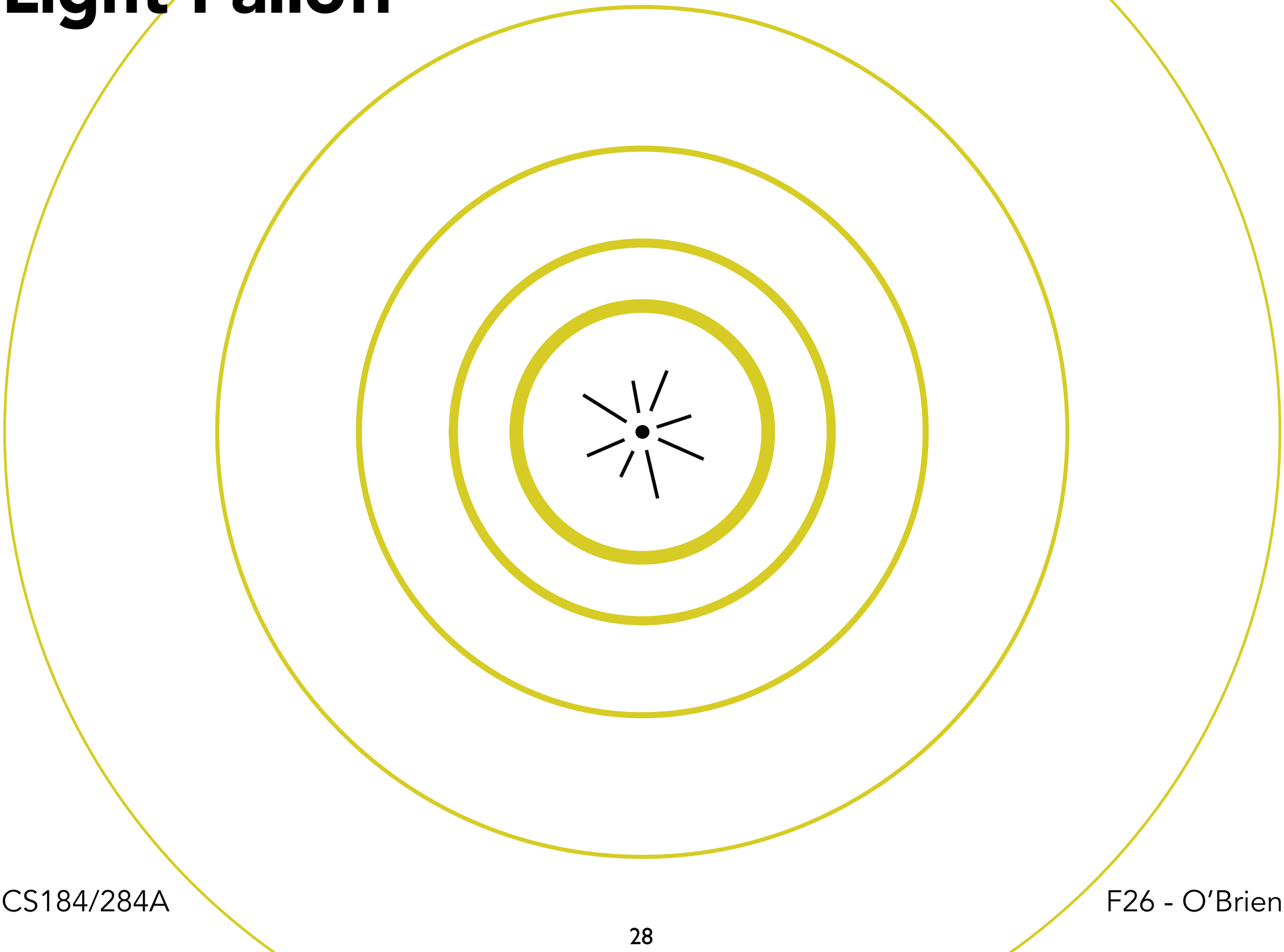
- Viewer direction, v
- Surface normal, n
- Light direction, l
  (for each of many lights)
- Surface parameters
  (color, shininess, …)

No “global” effects.

Unit length (normalized) vectors!!!



Light Falloff



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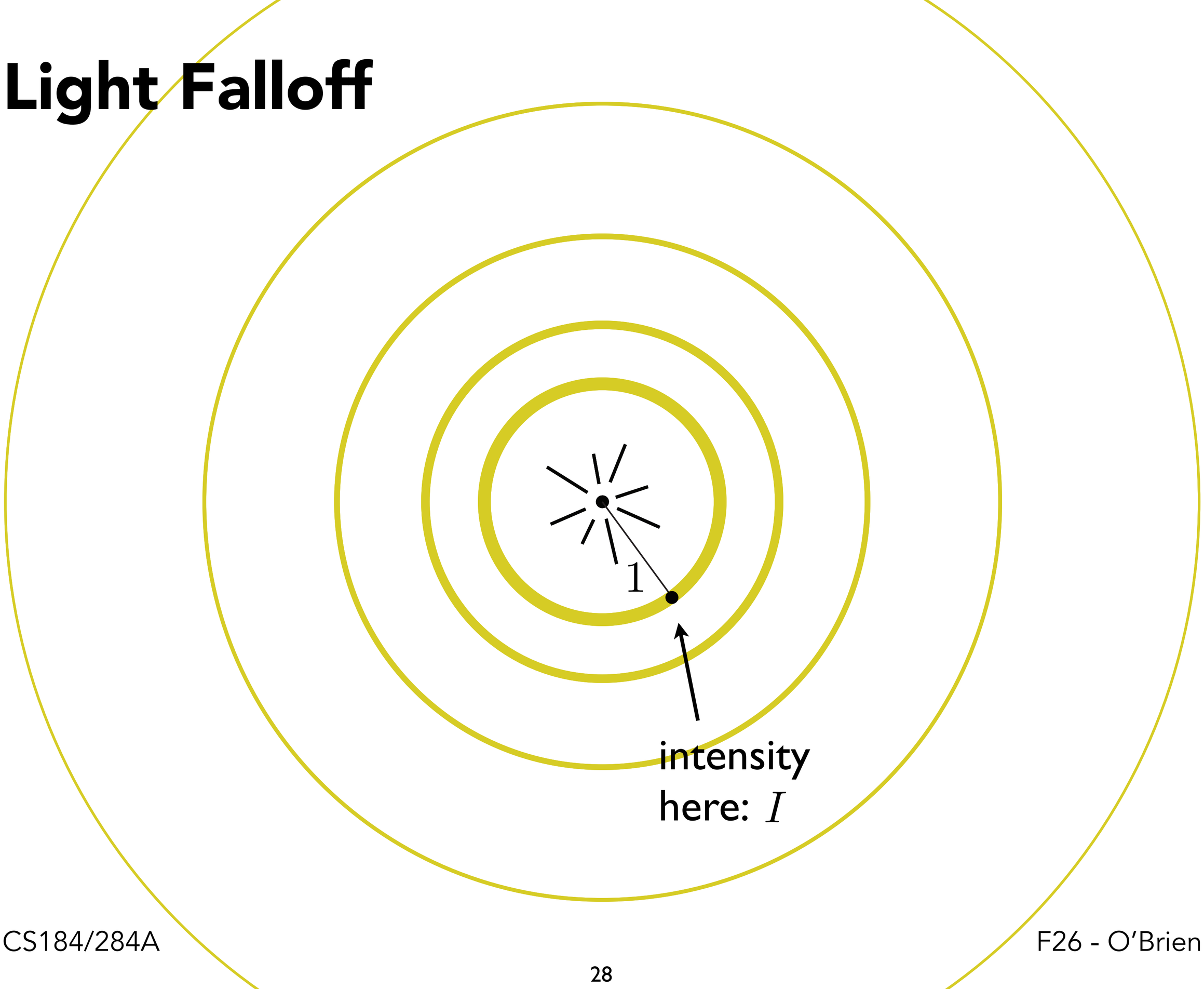

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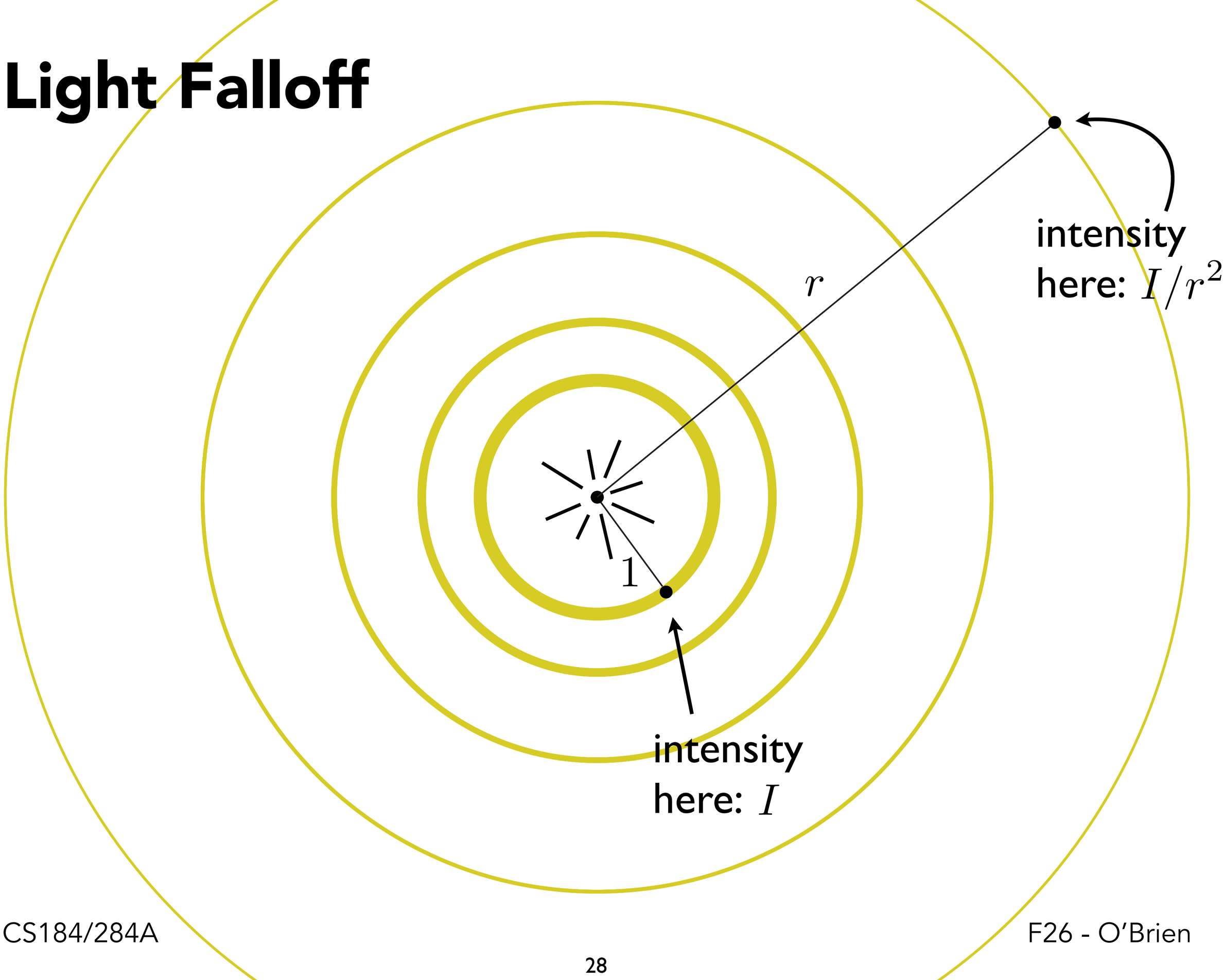


Falloff

Physically correct: $1/r^2$ light intensify falloff

- Tends to look bad with local shading (why?)

Sometimes compromise of $1/r$ used.

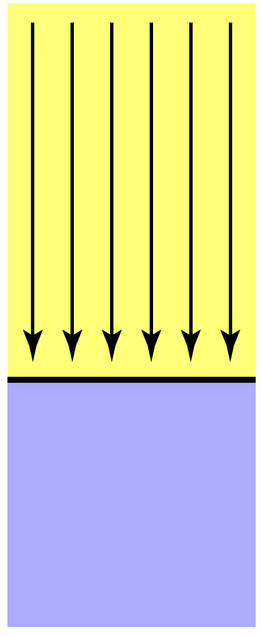
Very important to use $1/r^2$ for correct global illumination methods.

Diffuse Reflection

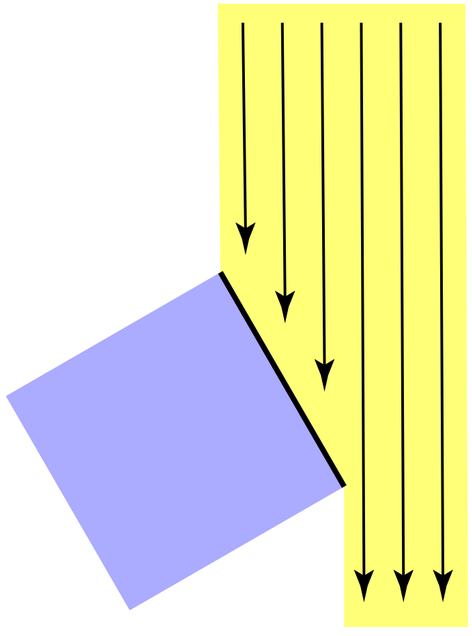
Light is scattered uniformly in all directions

- Surface color is the same for all viewing directions

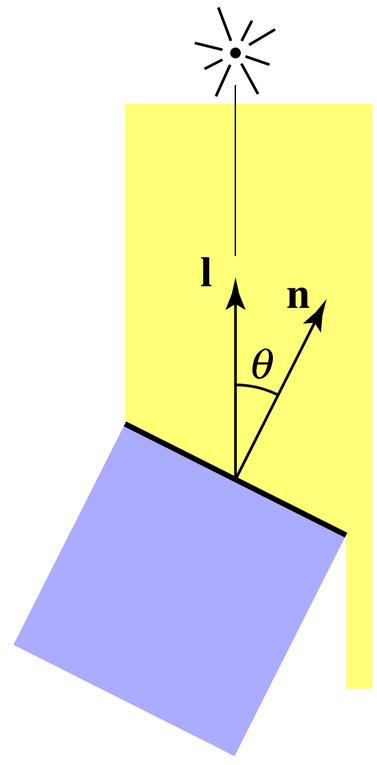
Lambert's cosine law



Top face of cube receives a certain amount of light



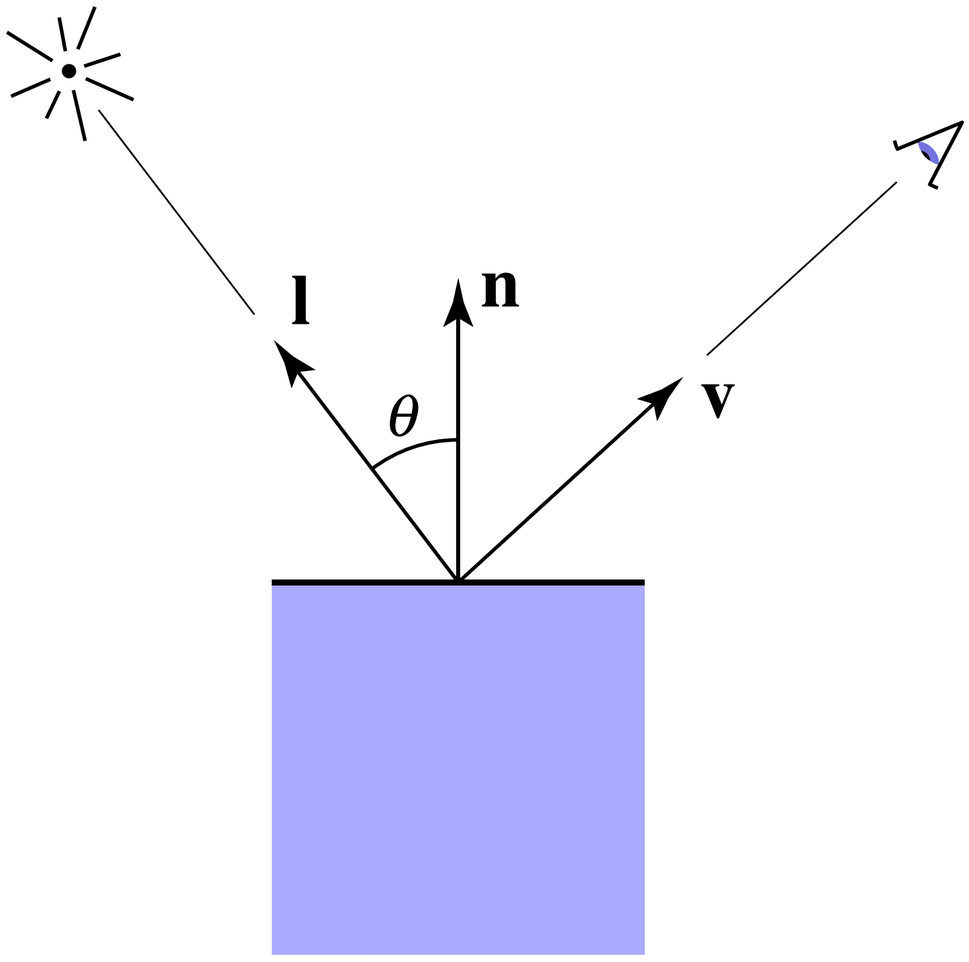
Top face of 60° rotated cube intercepts half the light



In general, light per unit area is proportional to cos θ = **l** • **n**

Lambertian (Diffuse) Shading

Shading independent of view direction



illumination from source

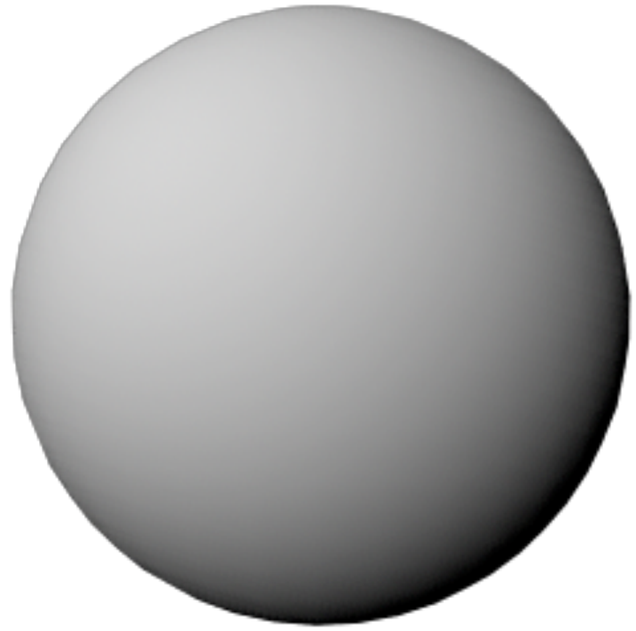
$$L_d = k_d \, (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{l})$$

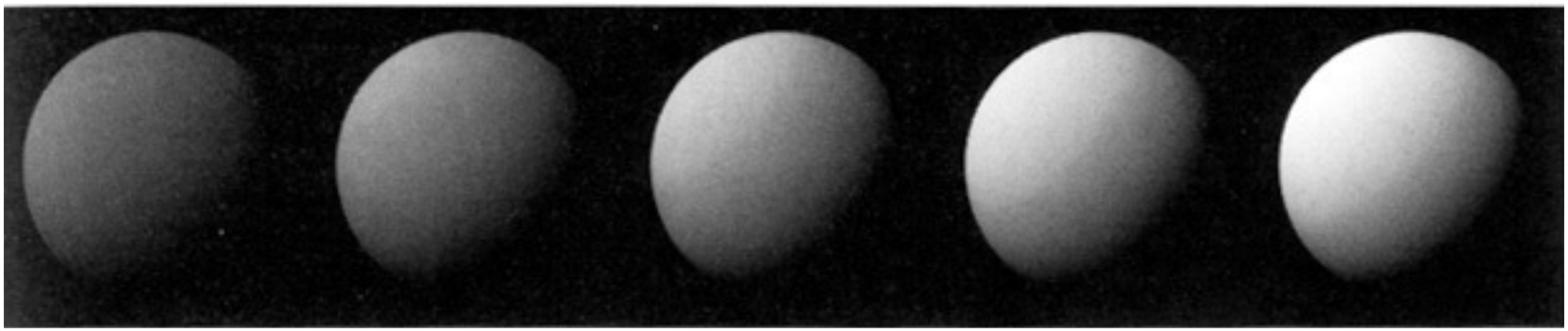
diffuse coefficient

diffusely reflected light

Lambertian (Diffuse) Shading

Produces matte appearance





k_d ⟶

[Foley et al.]

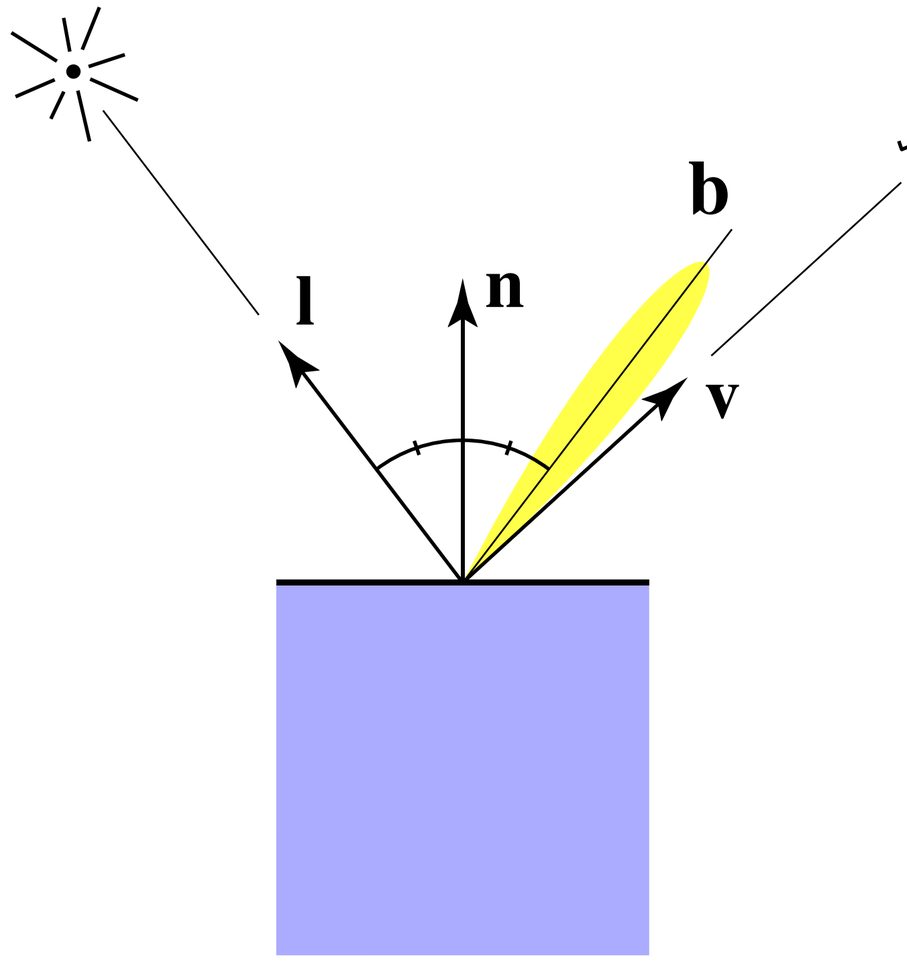
Perceptual Observations

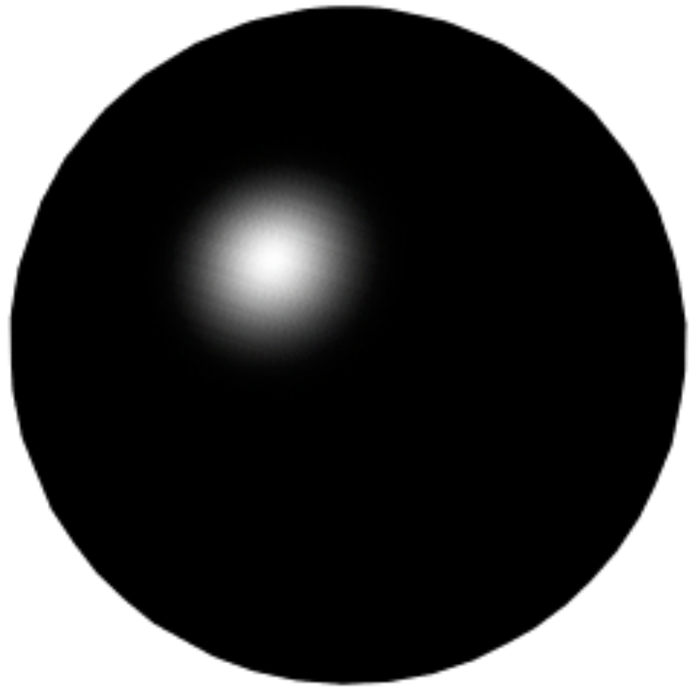
Photo credit: Jessica Andrews, flickr

Specular Shading (Blinn-Phong)

Intensity depends on view direction

- Bright near mirror reflection direction

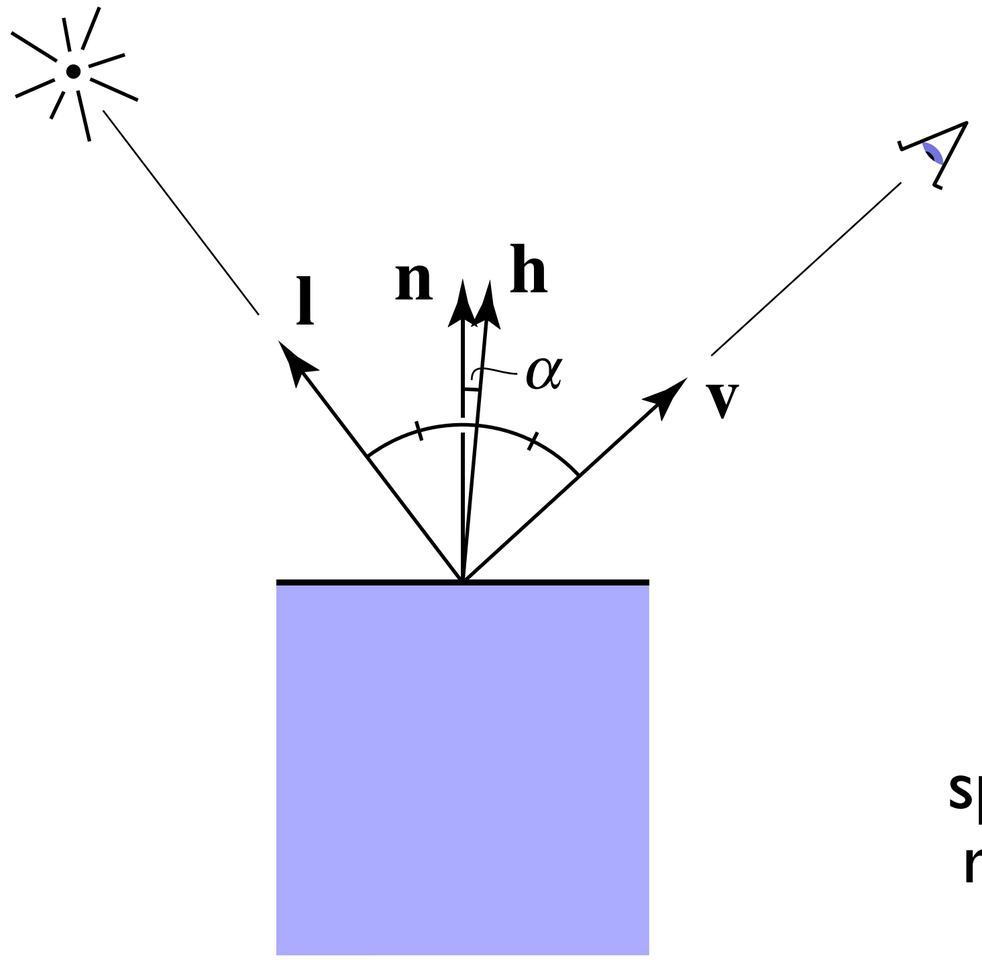




Specular Shading (Blinn-Phong)

Close to mirror direction ⇔ half vector near normal

- Measure "near" by dot product of unit vectors



$$\mathbf{h} = \text{bisector}(\mathbf{v}, \mathbf{l}) = \frac{\mathbf{v} + \mathbf{l}}{\|\mathbf{v} + \mathbf{l}\|}$$

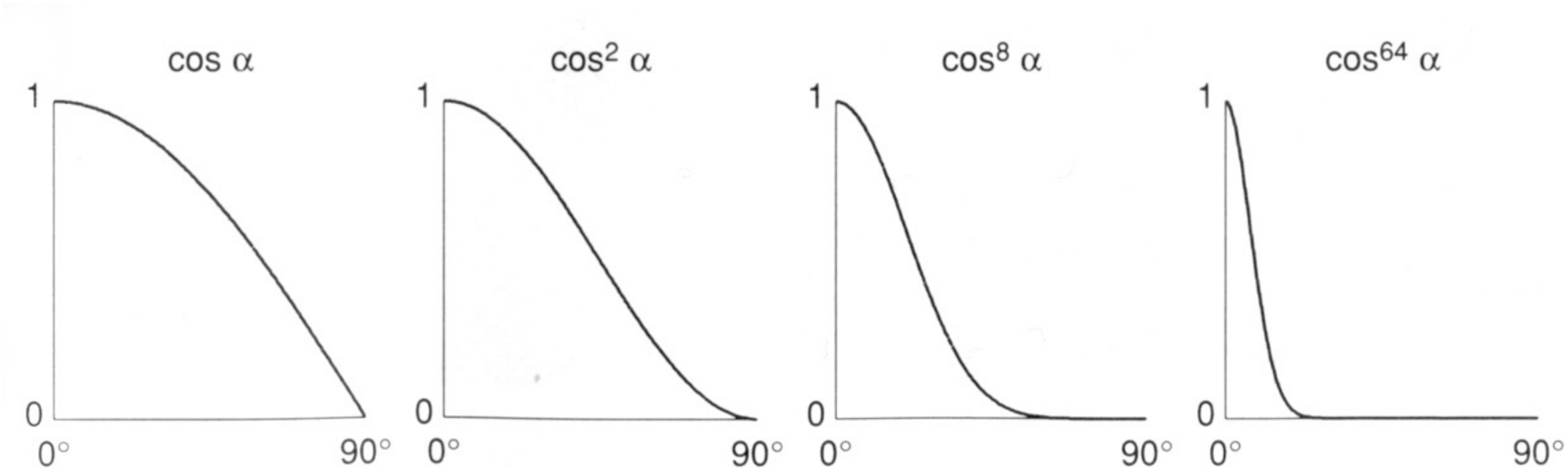
$$L_s = k_s \, (I/r^2) \max(0, \cos\alpha)^p = k_s \, (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{h})^p$$

specularly reflected light

specular coefficient

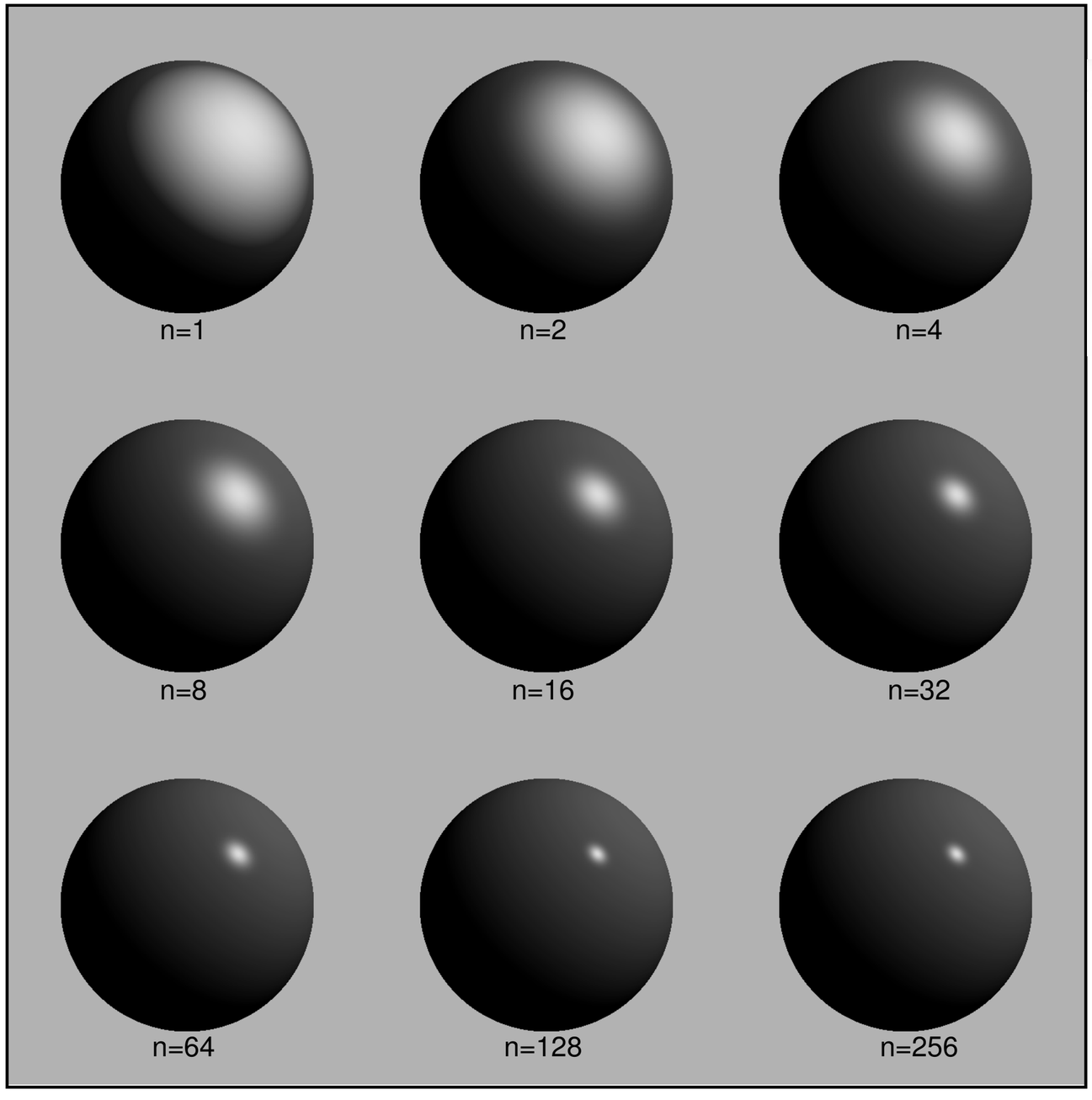
Cosine Power Plots

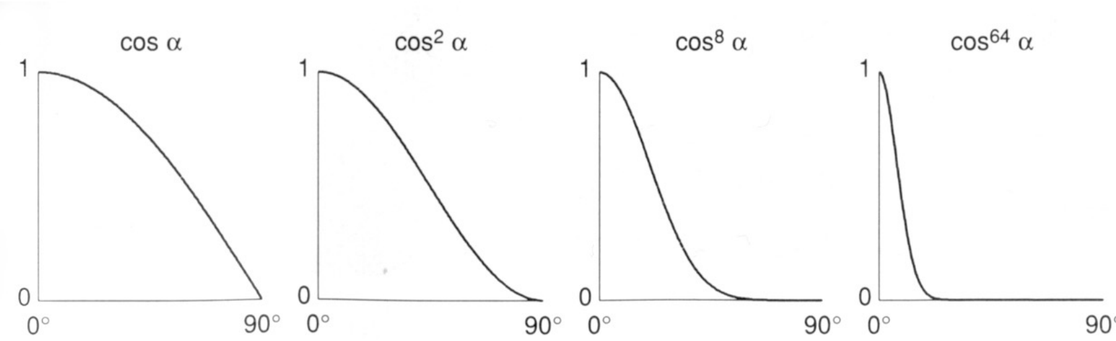
Increasing p narrows the reflection lobe



[Foley et al.]

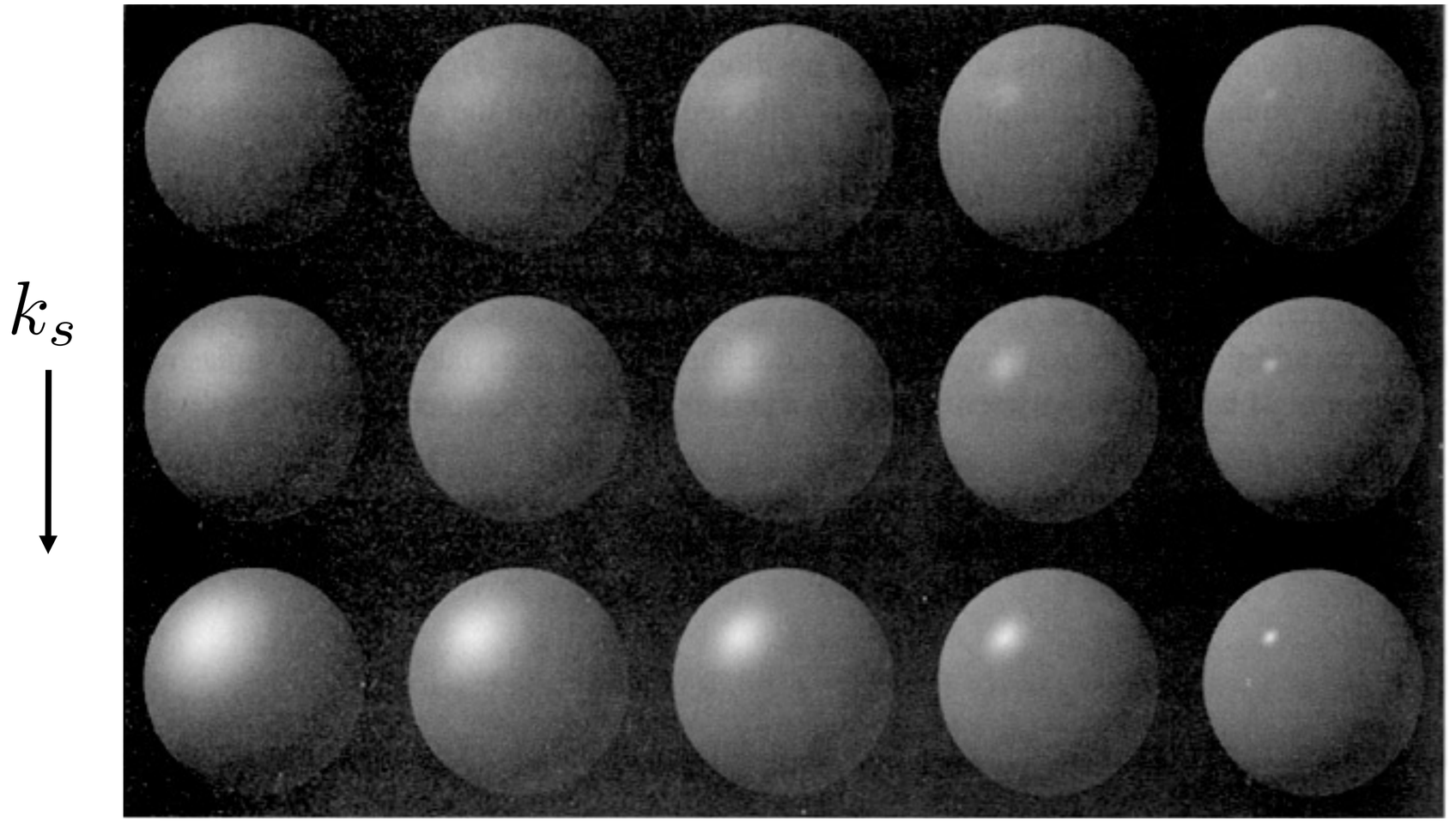
Specular Shading (Blinn-Phong)





Specular Shading (Blinn-Phong)

$$L_s = k_s \left(I/r^2\right) \max(0, \mathbf{n} \cdot \mathbf{h})^p$$



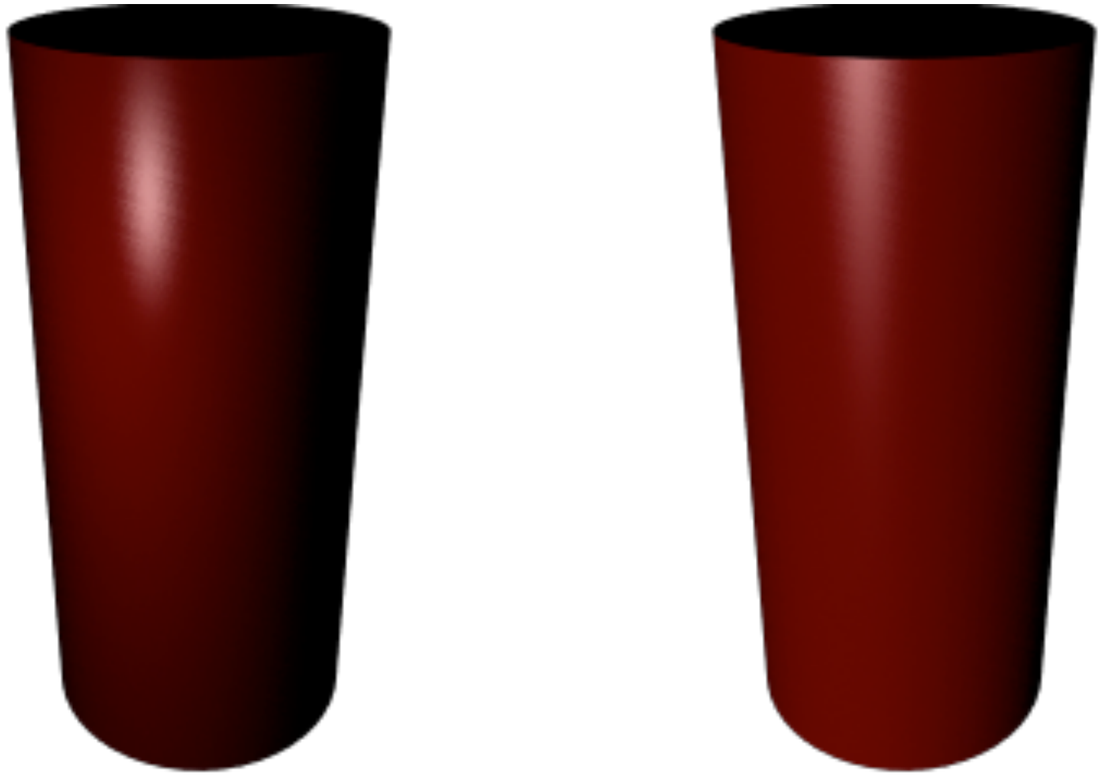
[Foley et al.]

Direction -vs- Point Lights

For a point light, the light direction changes over the surface.

For “distant” light, the direction is constant

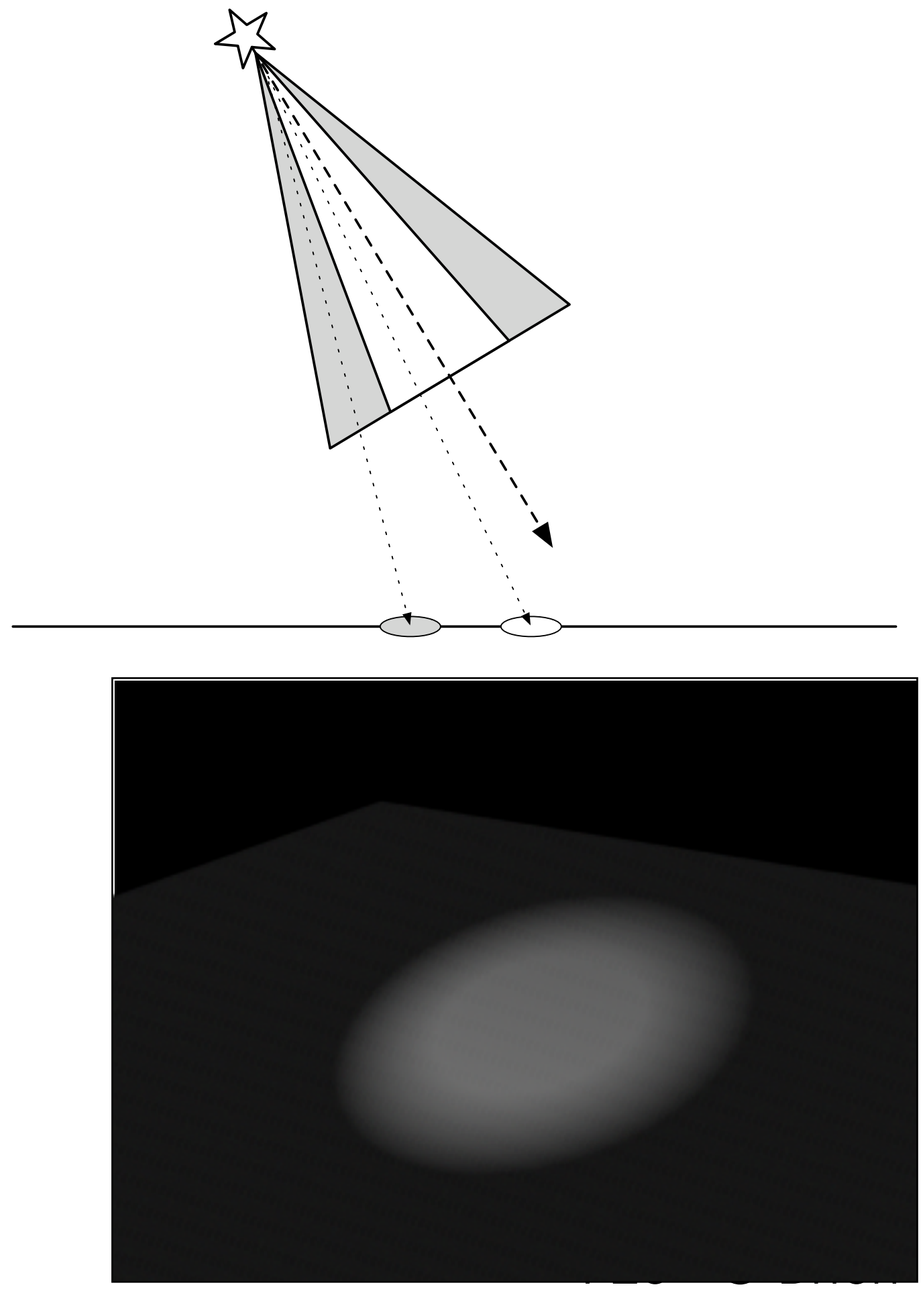
Similar for orthographic/perspective viewer



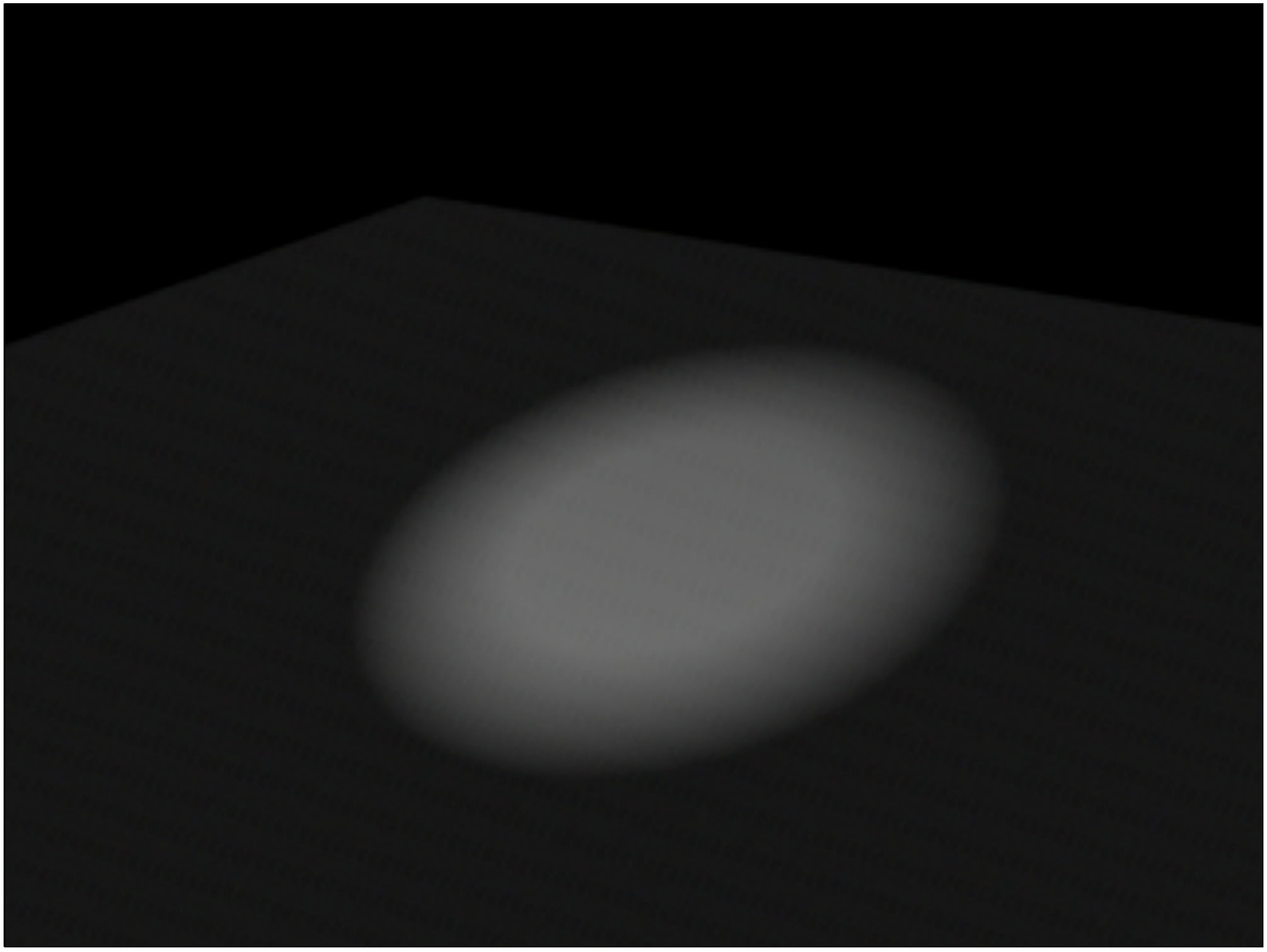
Spot and Other Lights

Other calculations for useful effects

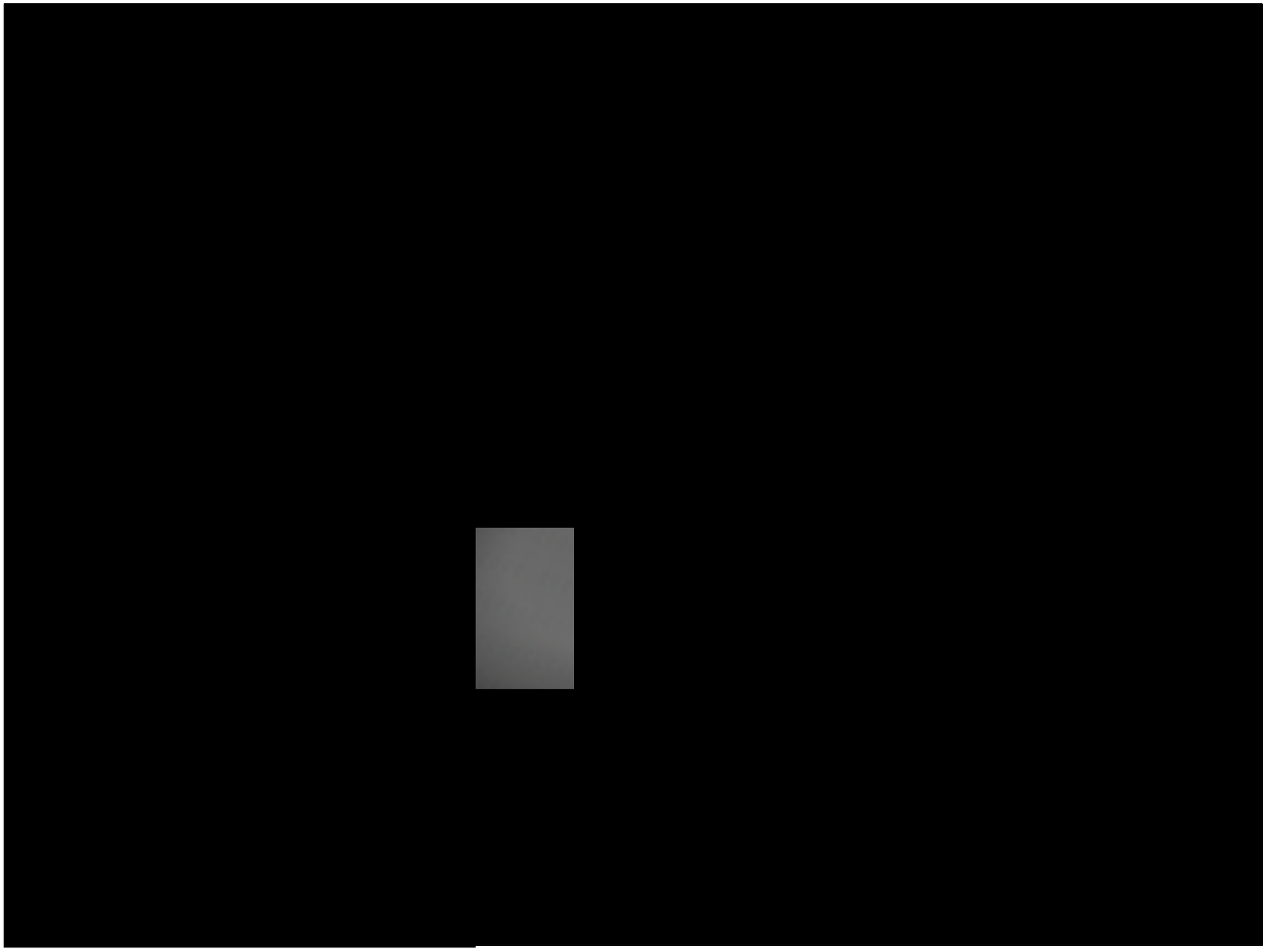
- Spot light
- Only light certain objects
- Negative lights
- ***etc.***



Ugly....



Ugly....



Perceptual Observations

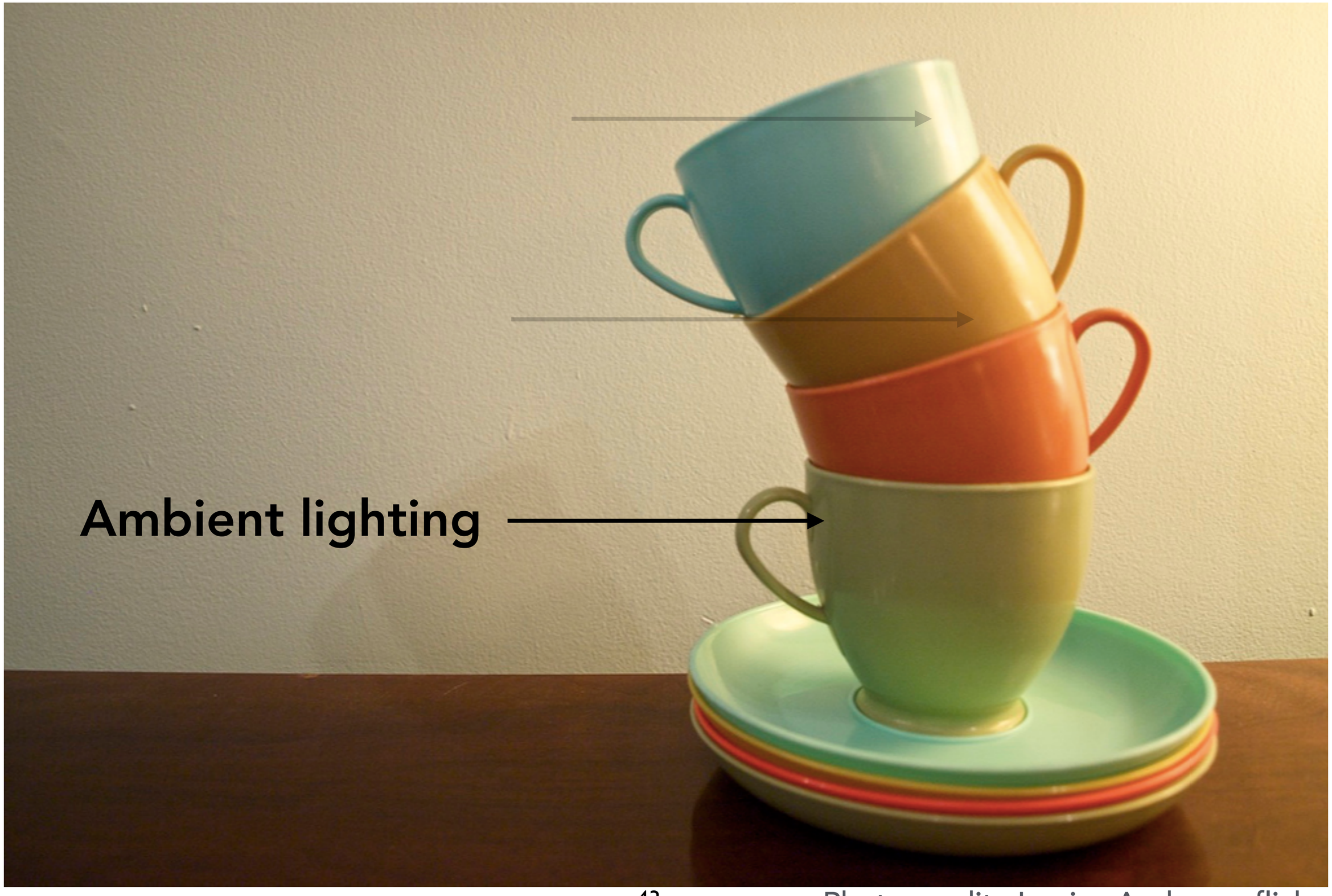
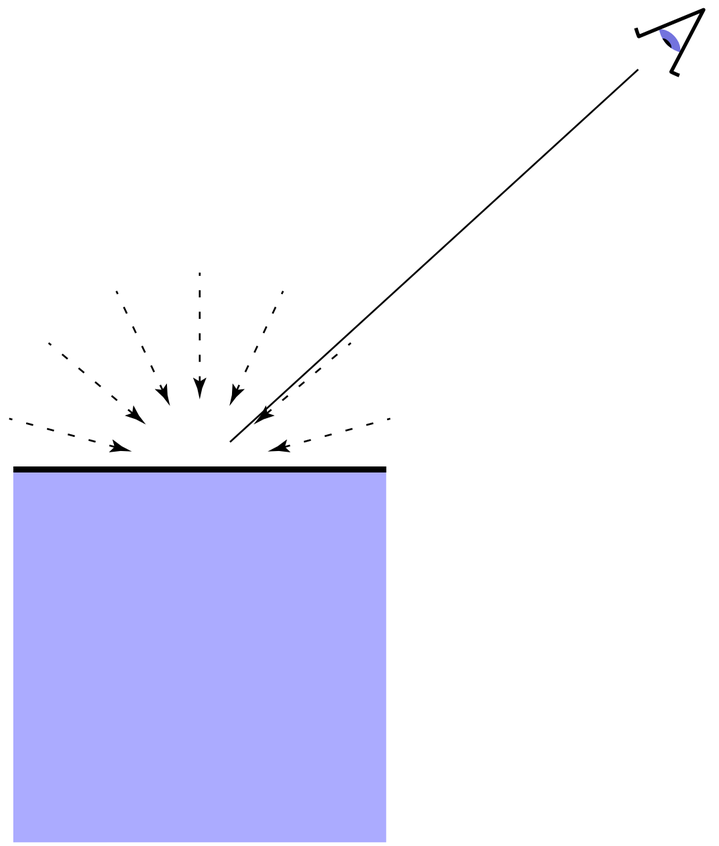


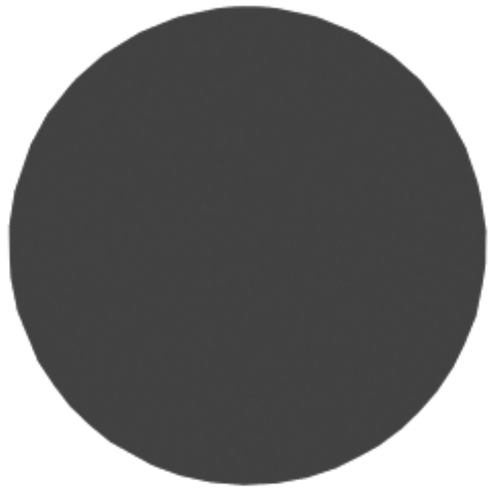
Photo credit: Jessica Andrews, flickr

Ambient Shading

Shading that does not depend on anything

- Add constant color to account for disregarded illumination and fill in black shadows



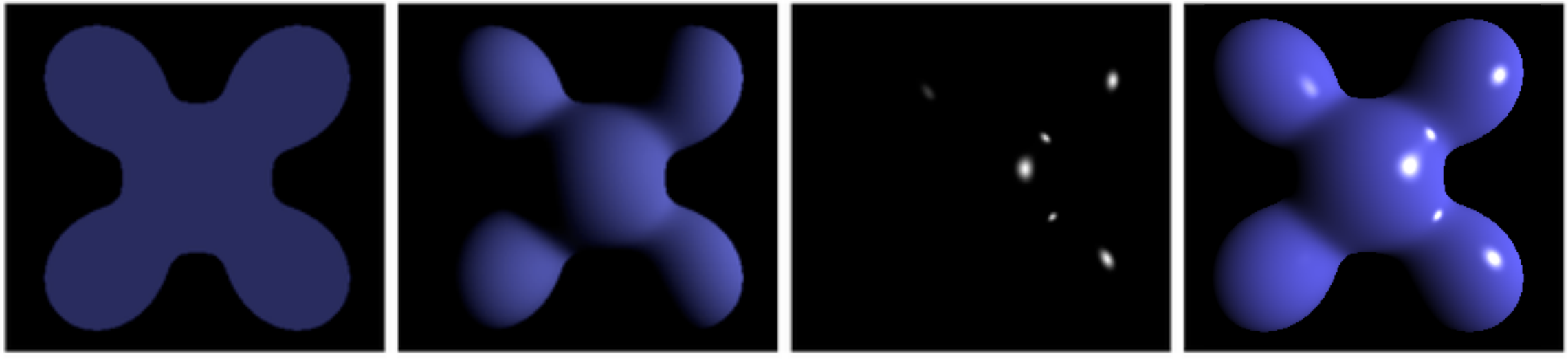


$$L_a = k_a \, I_a$$

reflected ambient light

ambient coefficient

Blinn-Phong Reflection Model



Ambient + Diffuse + Specular = Phong Reflection

$$
\begin{aligned}
L &= L_a + L_d + L_s \\
&= k_a\, I_a + k_d\, (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{l}) + k_s\, (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{h})^p
\end{aligned}
$$

Note: Sum over lights for multiple lights.

Blinn-Phong Reflection Model

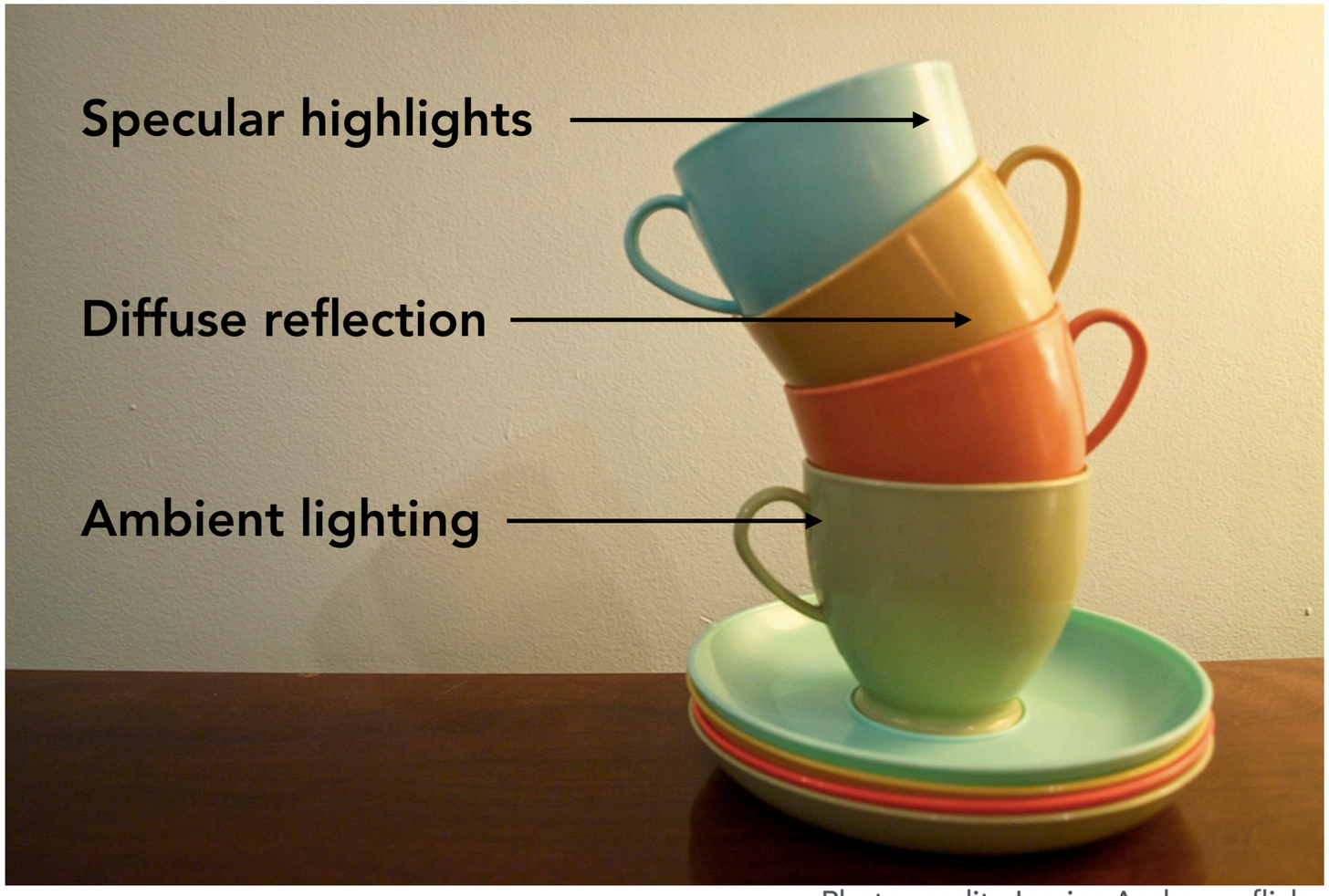


Photo credit: Jessica Andrews, flickr

$$L = L_a + L_d + L_s$$
$$= k_a\, I_a + k_d\, (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{l}) + k_s\, (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{h})^p$$

Ashikhmin-Shirley BRDF

- More realistic specular term (for some materials)
- Anisotropic specularities
- Fresnel behavior (grazing angle highlights)
- Energy preserving diffuse term
- Sum of diffuse and specular terms (as before)

$$\rho(\hat{\mathbf{l}}, \hat{\mathbf{v}}) = \rho_d(\hat{\mathbf{l}}, \hat{\mathbf{v}}) + \rho_s(\hat{\mathbf{l}}, \hat{\mathbf{v}})$$

Michael Ashikhmin and Peter Shirley. 2000. An anisotropic phong BRDF model. J. Graph. Tools 5, 2 (February 2000), 25-32.
https://www.cs.utah.edu/~shirley/papers/jgtbrdf.pdf

Ashikhmin-Shirley BRDF

$$\rho_s(\hat{\mathbf{l}}, \hat{\mathbf{e}}) = \frac{\sqrt{(p_u+1)(p_v+1)}}{8\pi} \frac{(\hat{\mathbf{n}} \cdot \hat{\mathbf{h}})^{p_u \cos^2\phi + p_v \sin^2\phi}}{(\hat{\mathbf{h}} \cdot \hat{\mathbf{e}}) \max\Big((\hat{\mathbf{n}} \cdot \hat{\mathbf{e}}), (\hat{\mathbf{n}} \cdot \hat{\mathbf{l}})\Big)} F(\hat{\mathbf{h}} \cdot \hat{\mathbf{e}})$$

$$F(\hat{\mathbf{h}} \cdot \hat{\mathbf{e}}) = K_s + (1 - K_s)(1 - (\hat{\mathbf{h}} \cdot \hat{\mathbf{e}}))^5$$

$\hat{\mathbf{l}}$	Light direction
$\hat{\mathbf{e}}$	Viewer (eye) direction
p_u, p_v	Specular powers
$\hat{\mathbf{n}}$	Normal
$\hat{\mathbf{h}}$	Half angle
K_s	Specular coefficient (color)
$\hat{\mathbf{u}}, \hat{\mathbf{v}}$	Parametric directions

Ashikhmin-Shirley BRDF

$$\rho_s(\hat{\mathbf{l}}, \hat{\mathbf{e}}) = \frac{\sqrt{(p_u+1)(p_v+1)}}{8\pi} \frac{(\hat{\mathbf{n}} \cdot \hat{\mathbf{h}})^{\frac{p_u(\hat{\mathbf{h}}\cdot\hat{\mathbf{u}})^2 + p_u(\hat{\mathbf{h}}\cdot\hat{\mathbf{v}})^2}{1-(\hat{\mathbf{h}}\cdot\hat{\mathbf{n}})^2}}}{(\hat{\mathbf{h}} \cdot \hat{\mathbf{e}}) \max\left((\hat{\mathbf{n}} \cdot \hat{\mathbf{e}}), (\hat{\mathbf{n}} \cdot \hat{\mathbf{l}})\right)} F(\hat{\mathbf{h}} \cdot \hat{\mathbf{e}})$$

$$F(\hat{\mathbf{h}} \cdot \hat{\mathbf{e}}) = K_s + (1 - K_s)(1 - (\hat{\mathbf{h}} \cdot \hat{\mathbf{e}}))^5$$

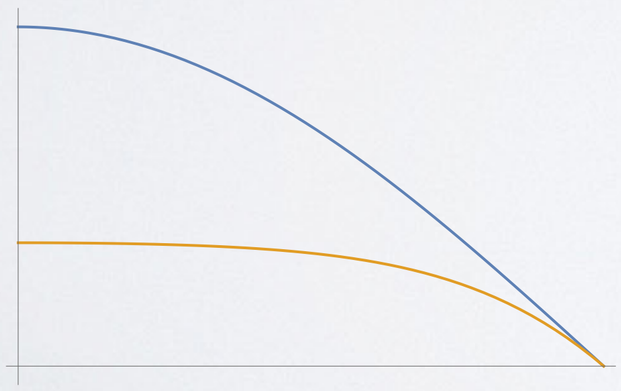
Approximate Fresnel function

$\hat{\mathbf{l}}$	Light direction
$\hat{\mathbf{e}}$	Viewer (eye) direction
p_u, p_v	Specular powers
$\hat{\mathbf{n}}$	Normal
$\hat{\mathbf{h}}$	Half angle
K_s	Specular coefficient (color)
$\hat{\mathbf{u}}, \hat{\mathbf{v}}$	Parametric directions

Ashikhmin-Shirley BRDF

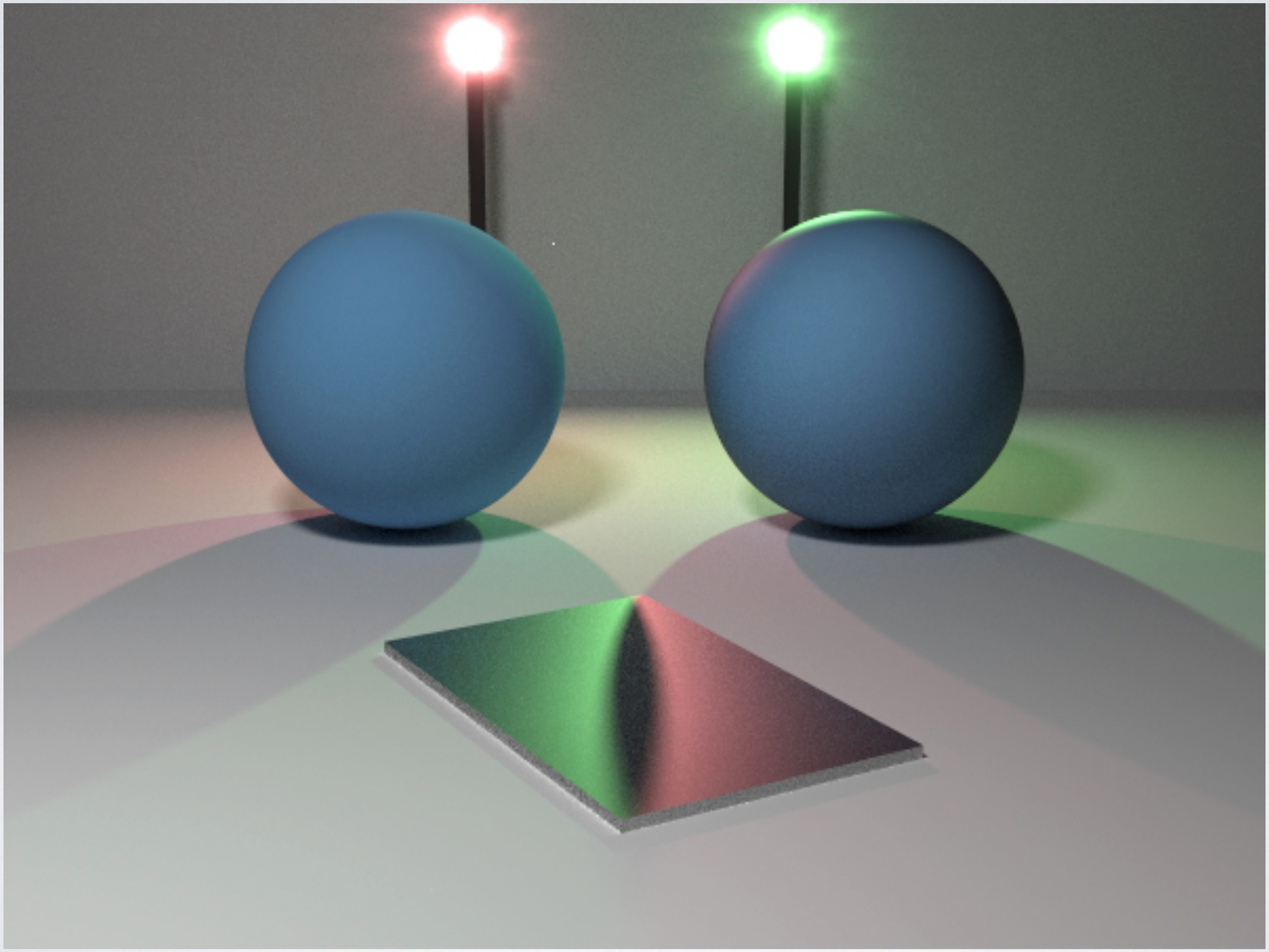
$$\rho_d(\hat{\mathbf{l}}, \hat{\mathbf{e}}) = \frac{28K_d}{23\pi}(1 - K_s)\left(1 - \left(1 - \frac{\hat{\mathbf{n}} \cdot \hat{\mathbf{e}}}{2}\right)^5\right)\left(1 - \left(1 - \frac{\hat{\mathbf{n}} \cdot \hat{\mathbf{l}}}{2}\right)^5\right)$$

Note: The Phong diffuse term (Lambertian) is independent of view. But this term accounts for unavailable light due to specular/ Fresnel reflection.

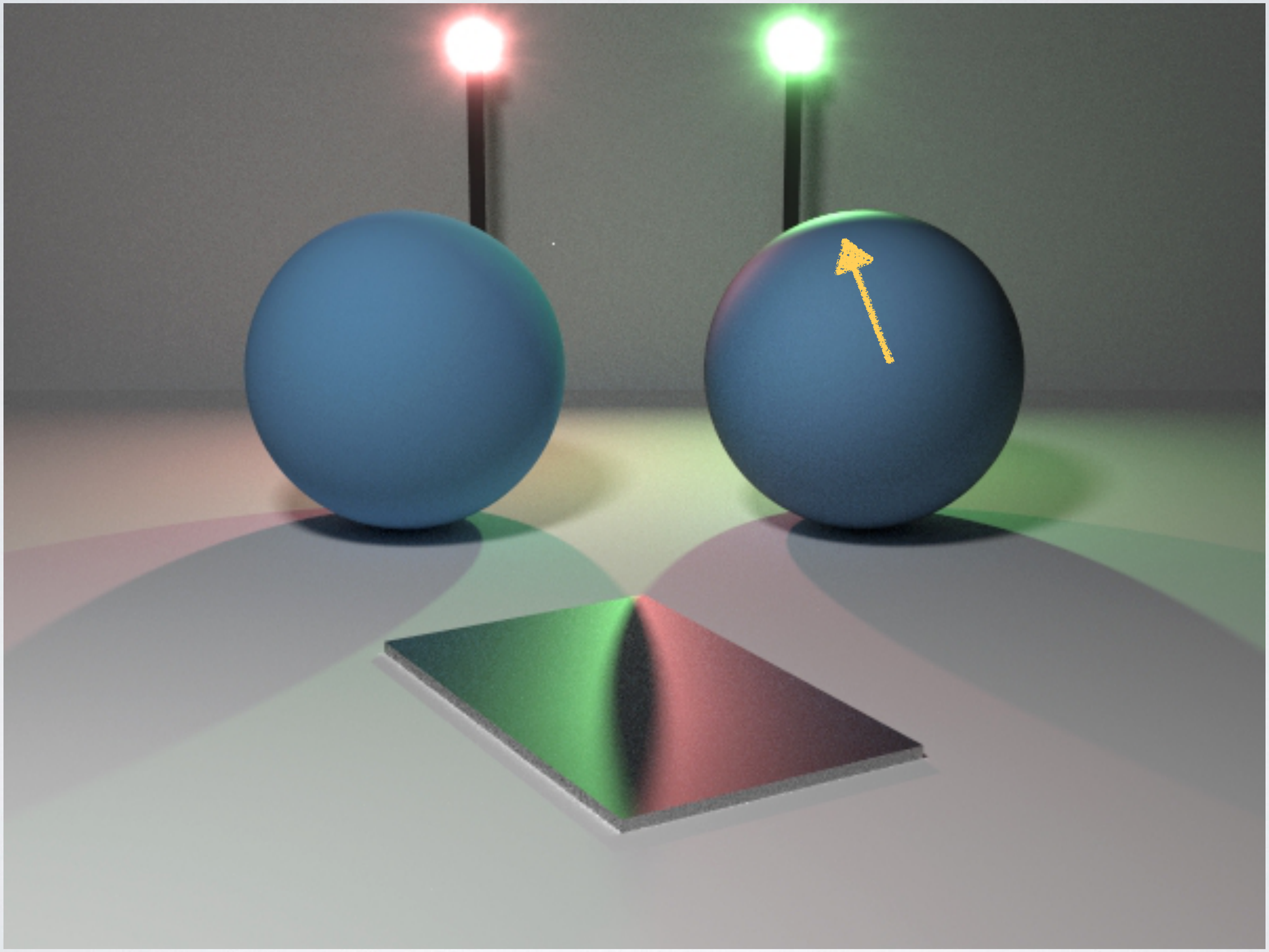


$\hat{\mathbf{l}}$	Light direction
$\hat{\mathbf{e}}$	Viewer (eye) direction
p_u, p_v	Specular powers
$\hat{\mathbf{n}}$	Normal
$\hat{\mathbf{h}}$	Half angle
K_s	Specular coefficient (color)
$\hat{\mathbf{u}}, \hat{\mathbf{v}}$	Parametric directions

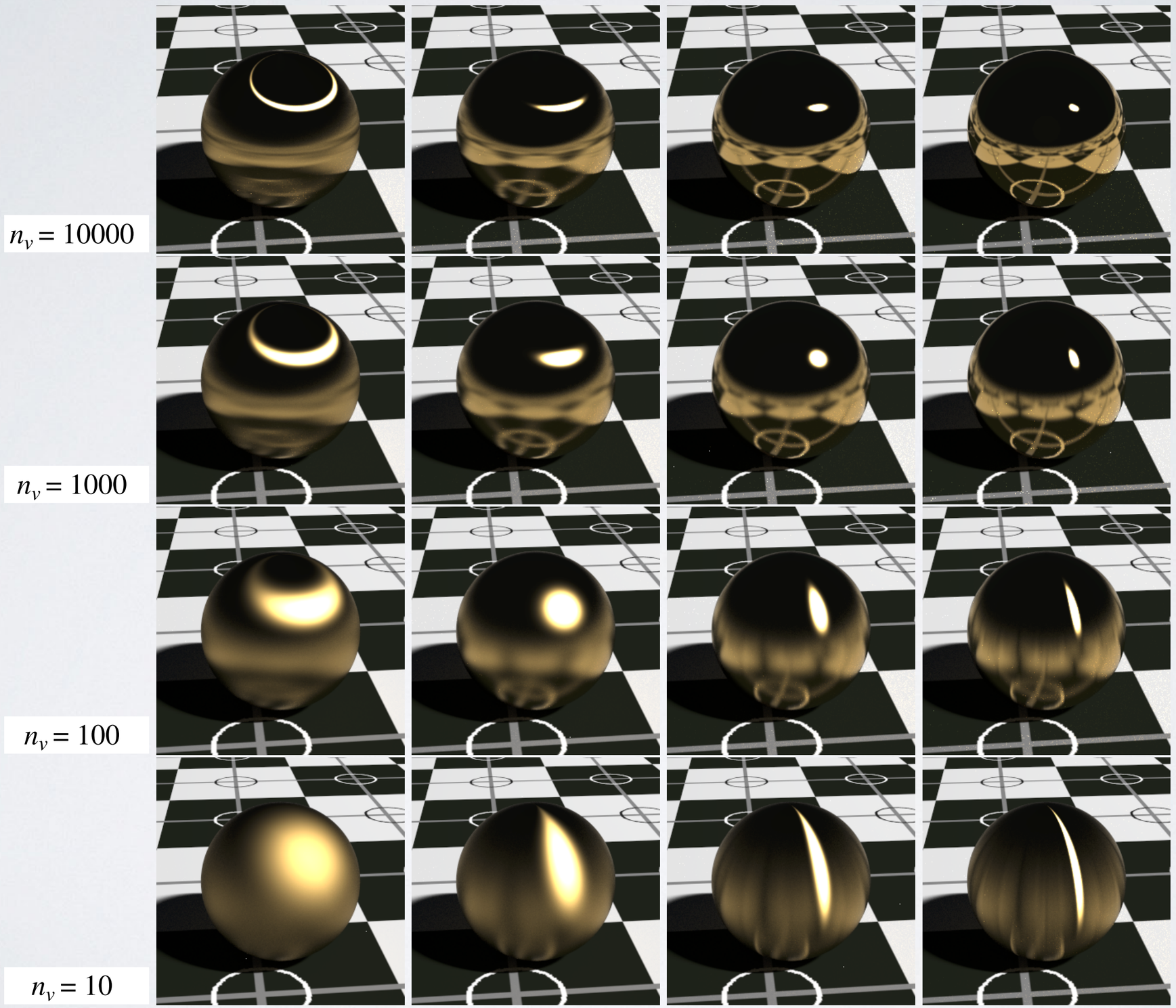
Ashikhmin-Shirley BRDF



Ashikhmin-Shirley BRDF



Ashikhmin-Shirley BRDF



BRDF

- The **B**i-directional **R**eflectance **D**istribution **F**unction
- Given
  - Surface material
  - Incoming light direction
  - Direction of viewer
  - Orientation of surface
- Return:
  - fraction of light that reaches the viewer
- We'll worry about physical units later...

$$\rho = \rho(\theta_V, \theta_L)$$

$$= \rho(\mathbf{v}, \mathbf{l}, \mathbf{n})$$

Phong shading is one example of a computed BRDF.

$$L = L_a + L_d + L_s$$

$$= k_a\, I_a + k_d\, (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{l}) + k_s\, (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{h})^p$$

Beyond BRDFs

The BRDF model does not capture everything

- e.g. Subsurface scattering (BSSRDF)

Images from Jensen ***et. al, SIGGRAPH 2001***

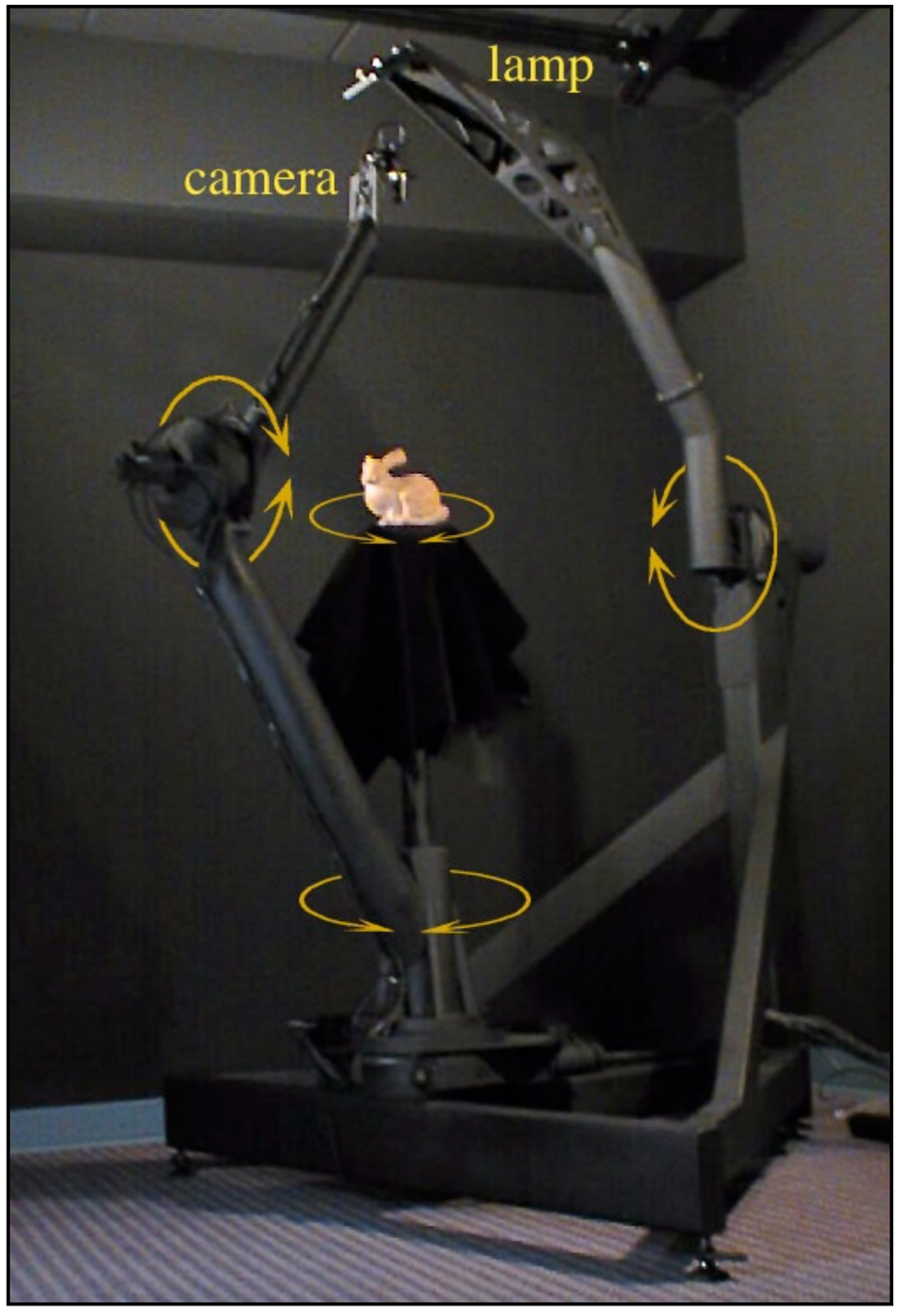
Beyond BRDFs

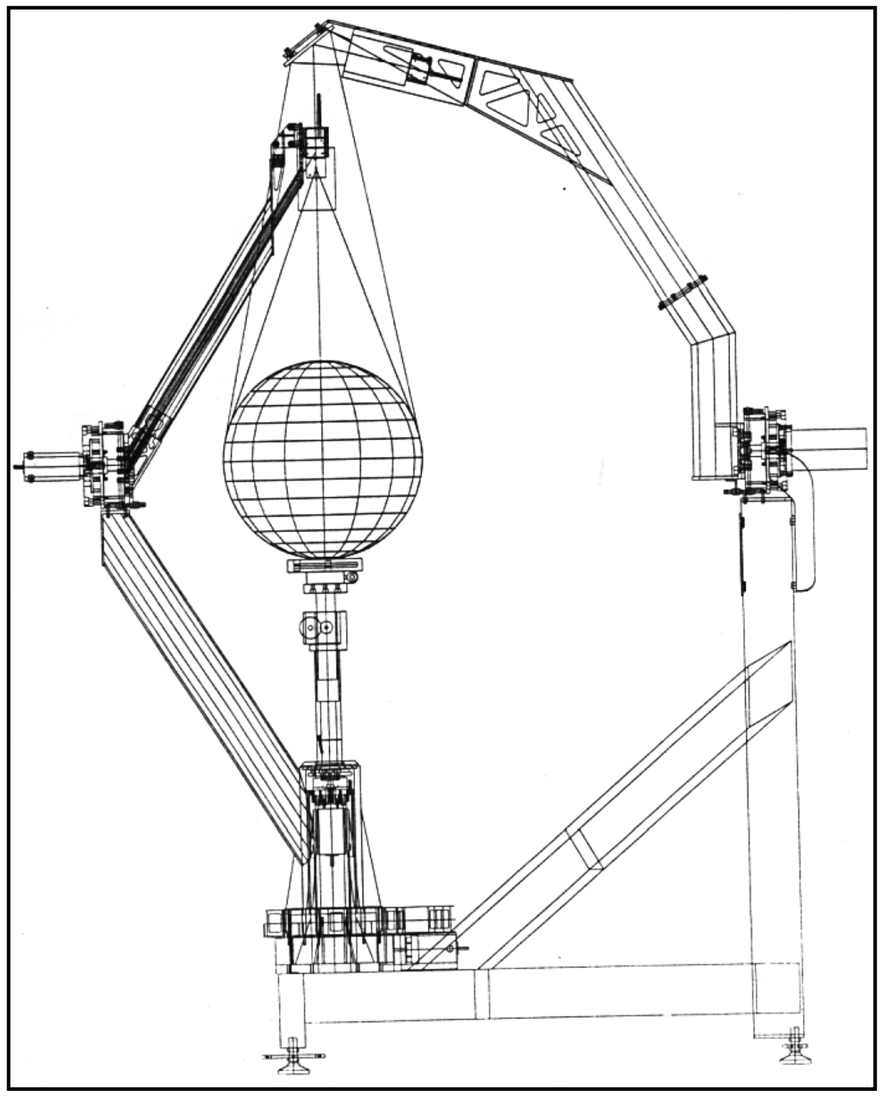
The BRDF model does not capture everything

- e.g. Inter-frequency interactions

$\rho = \rho(\theta_V, \theta_L, \boldsymbol{\lambda}_{\text{in}}, \boldsymbol{\lambda}_{\text{out}})$ This version would work....

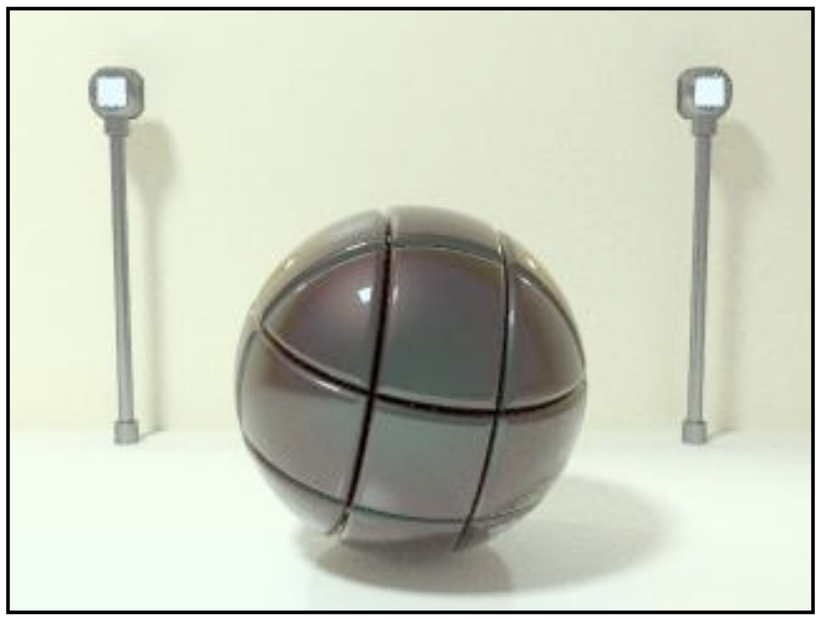
Measuring BRDF

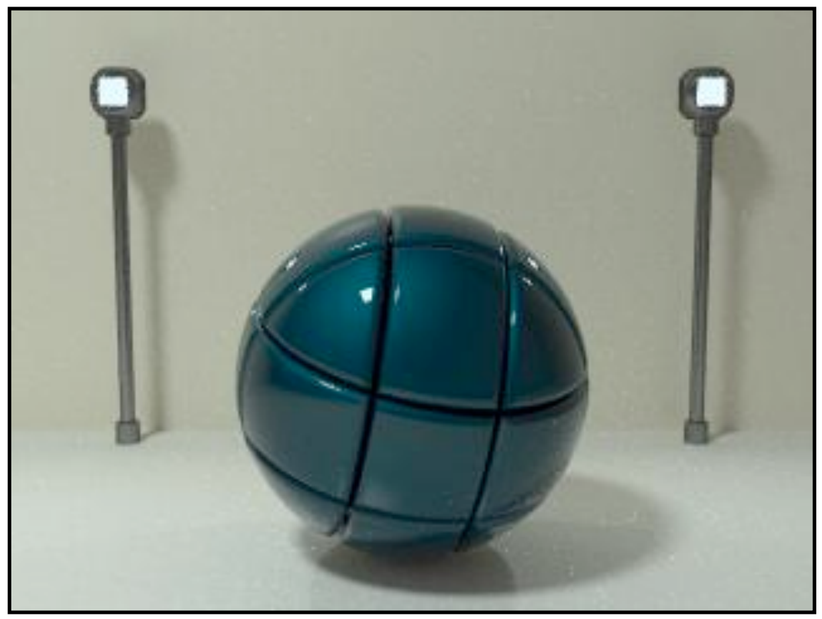




Images from Marc Levoy

Measured BRDFs





BRDFs for automotive paint

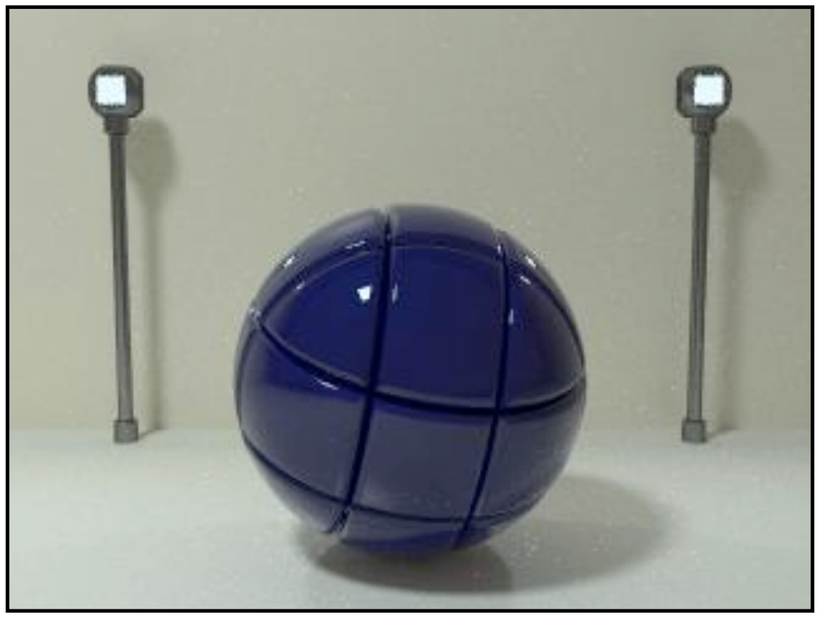
Measured BRDFs

BRDFs for aerosol spray paint

Measured BRDFs

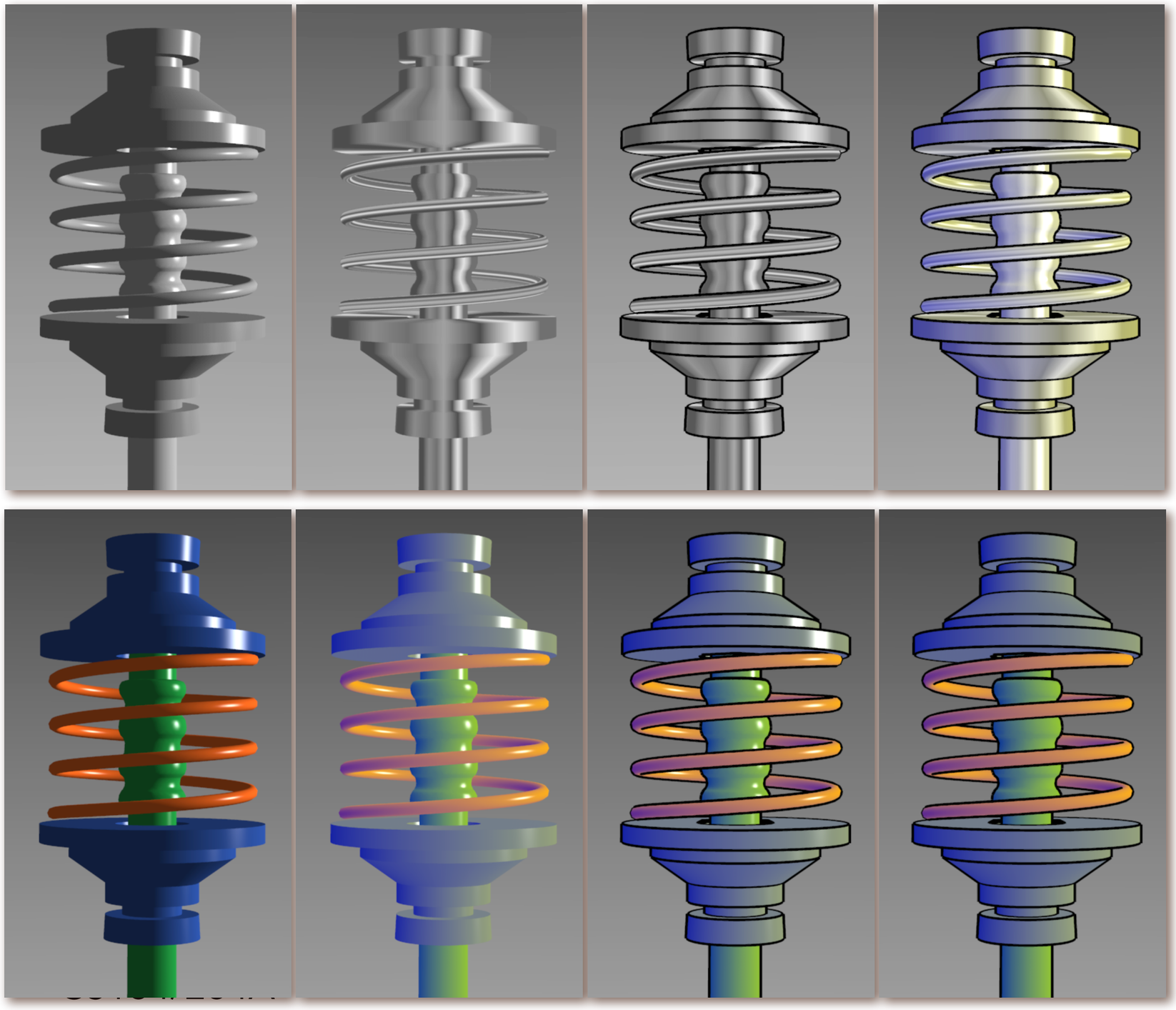
BRDFs for house paint

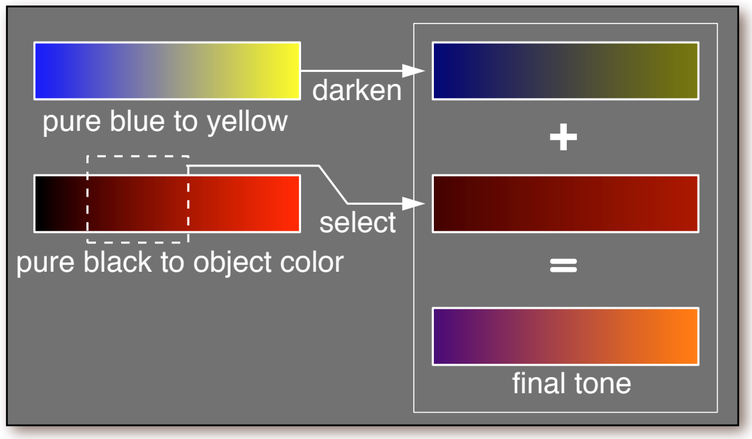
Measured BRDFs



BRDFs for lucite sheet

Other Color Effects





Images from Gooch *et. al, 1998*

Shading Triangle Meshes

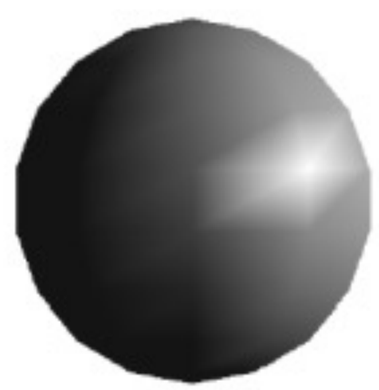
Shading Frequency: Triangle, Vertex or Pixel

Shade each triangle (flat shading)

- Triangle face is flat — one normal vector
- Not good for smooth surfaces

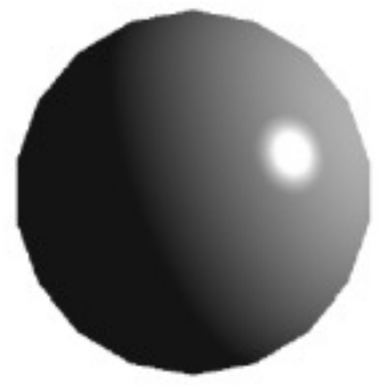
Shade each vertex (“Gouraud” shading)

- Interpolate colors from vertices across triangle
- Each vertex has a normal vector

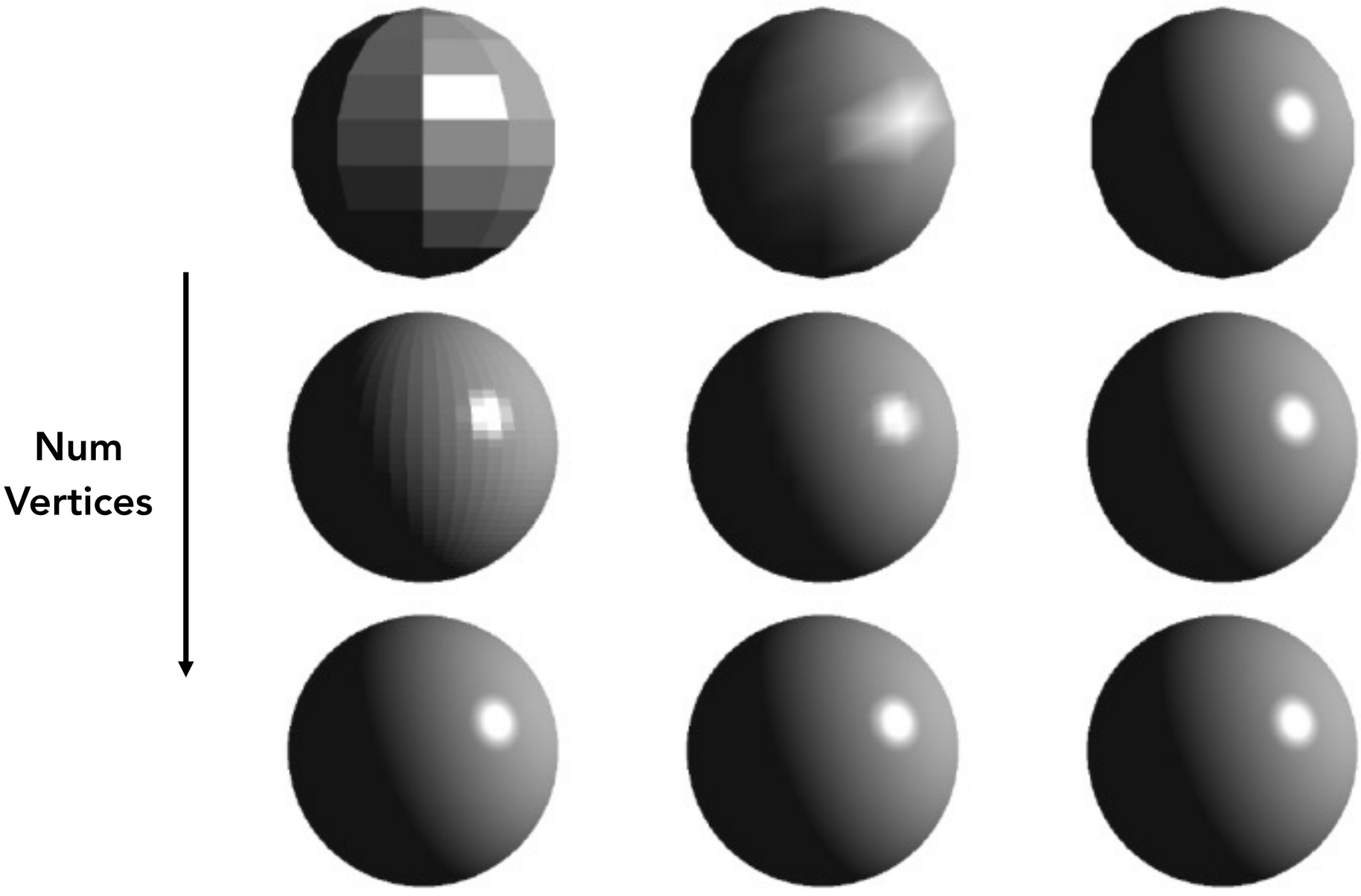


Shade each pixel (“Phong” shading)

- Interpolate normal vectors across each triangle
- Compute full shading model at each pixel



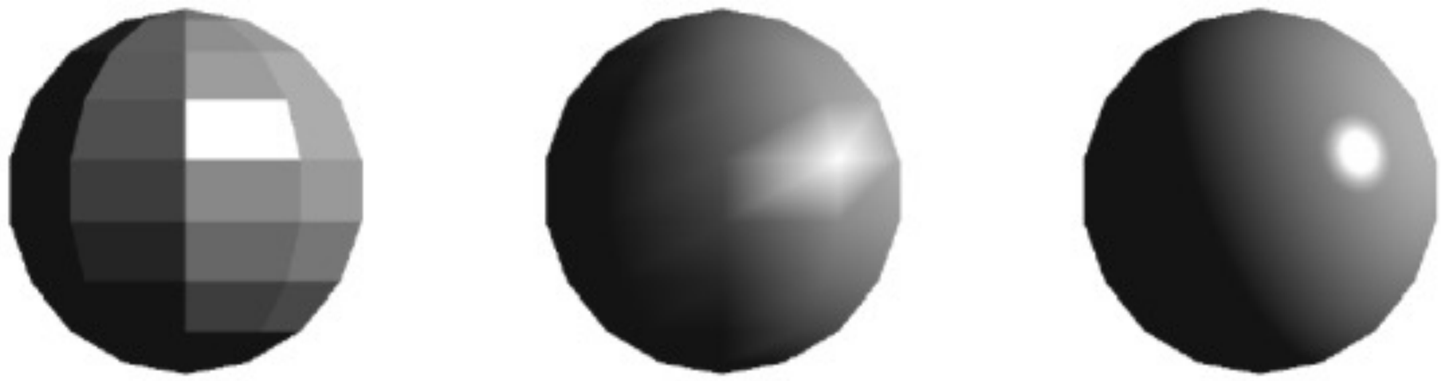
Shading Frequency: Face, Vertex or Pixel



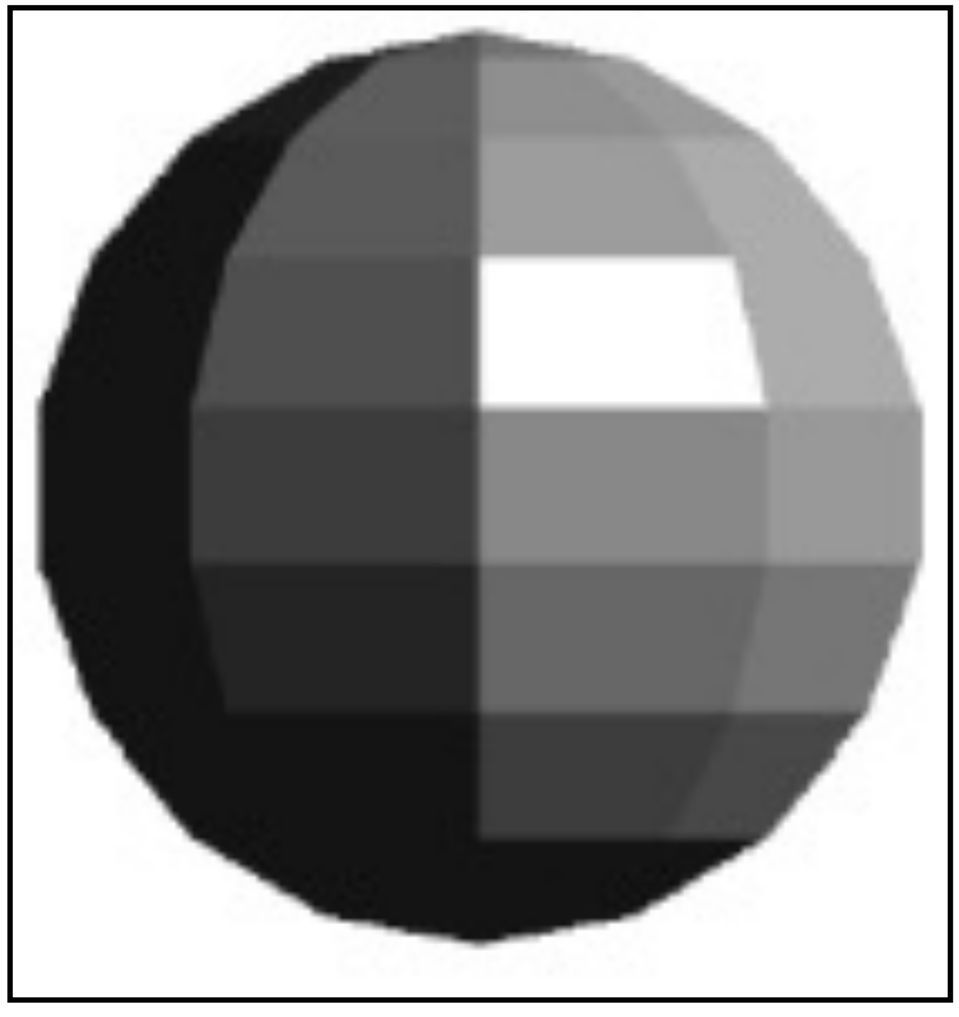
Shading freq. :	Face	Vertex	Pixel
Shading type :	Flat	Gouraud	Phong (*)

 Image credit: Happyman, http://cg2010studio.com/

Shading Frequency: Face, Vertex or Pixel



Note *apparent* edge enhancement



Note specular "taking over" entire face.

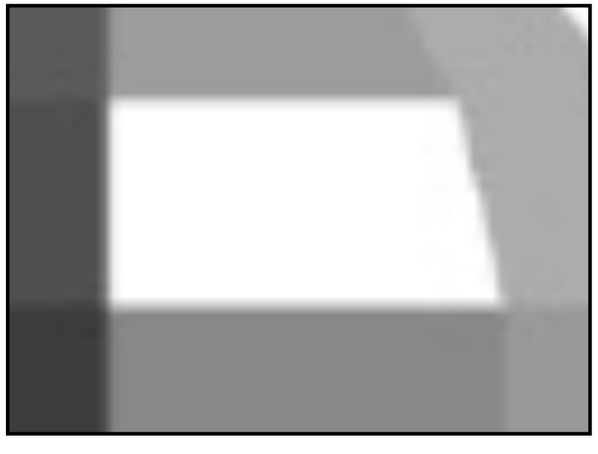
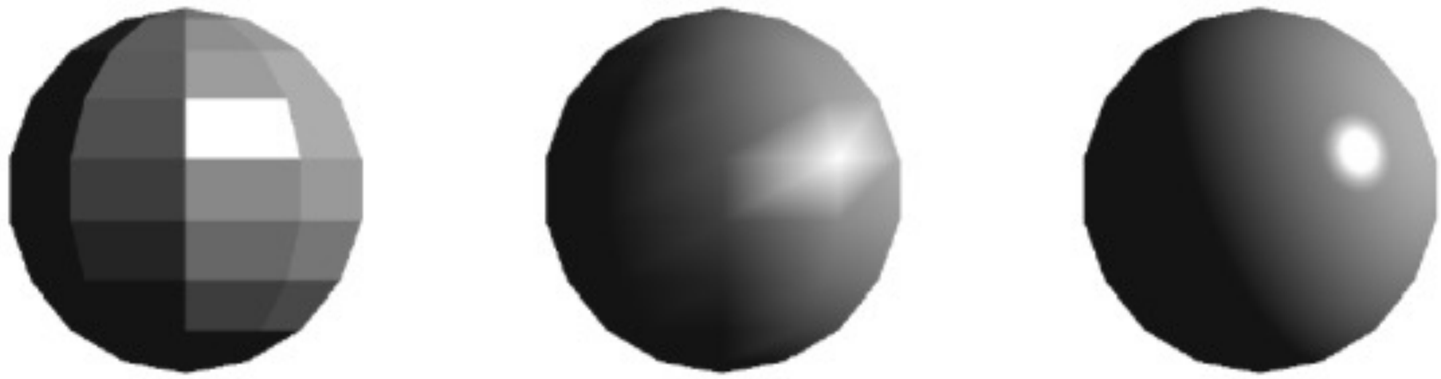
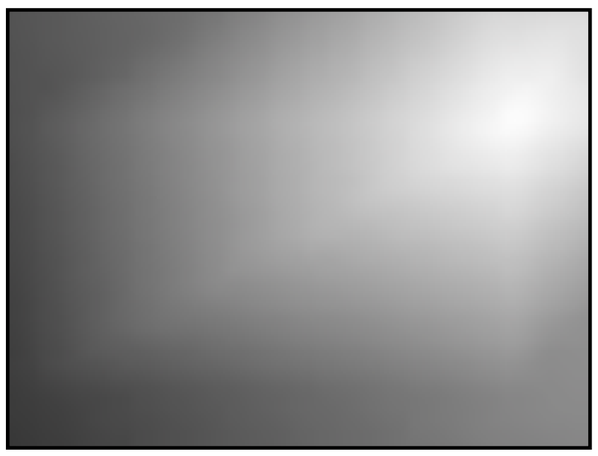


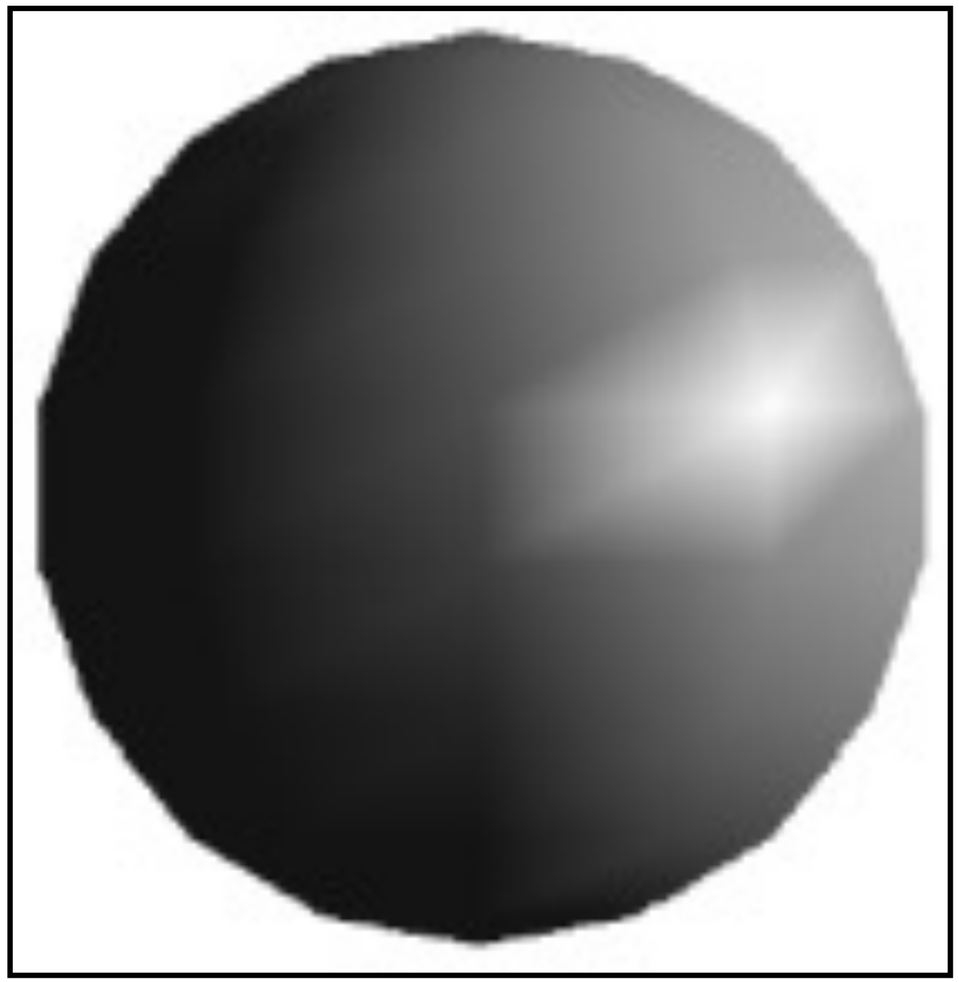
 Image credit: Happyman, http://cg2010studio.com/

Shading Frequency: Face, Vertex or Pixel



Note C1 interpolation artifacts





Note specular spread out over several triangles

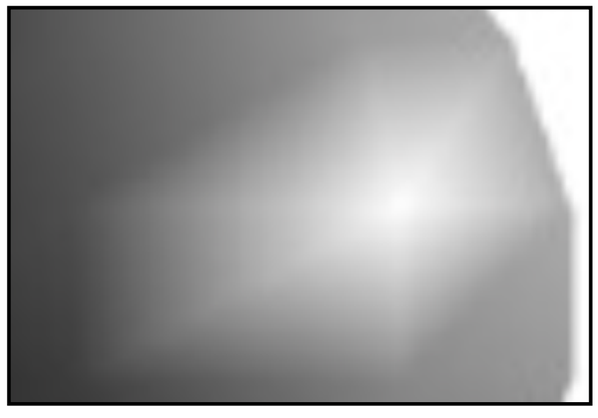
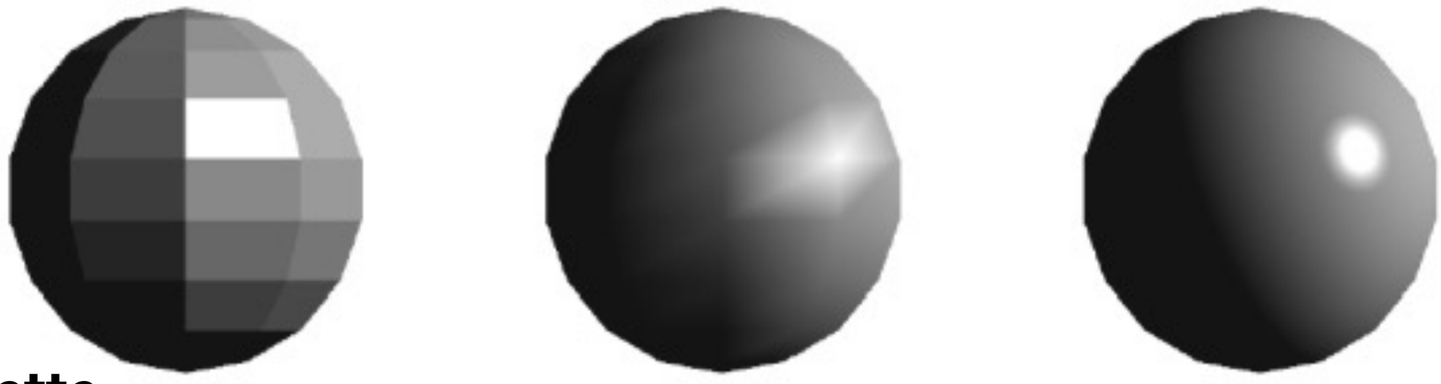


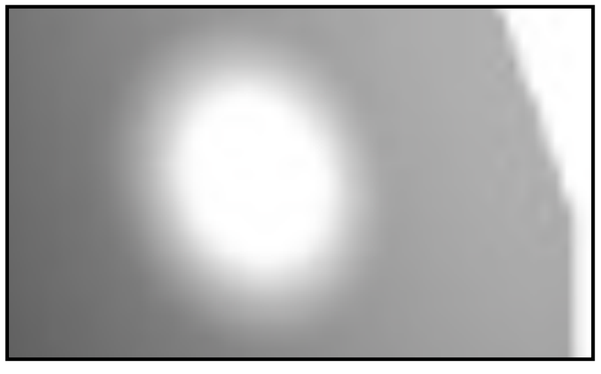
 Image credit: Happyman, http://cg2010studio.com/

Shading Frequency: Face, Vertex or Pixel



Smooth shading, faceted silhouette

Rounded specular highlight



Note clipping in this example where center of specularity appears flat.

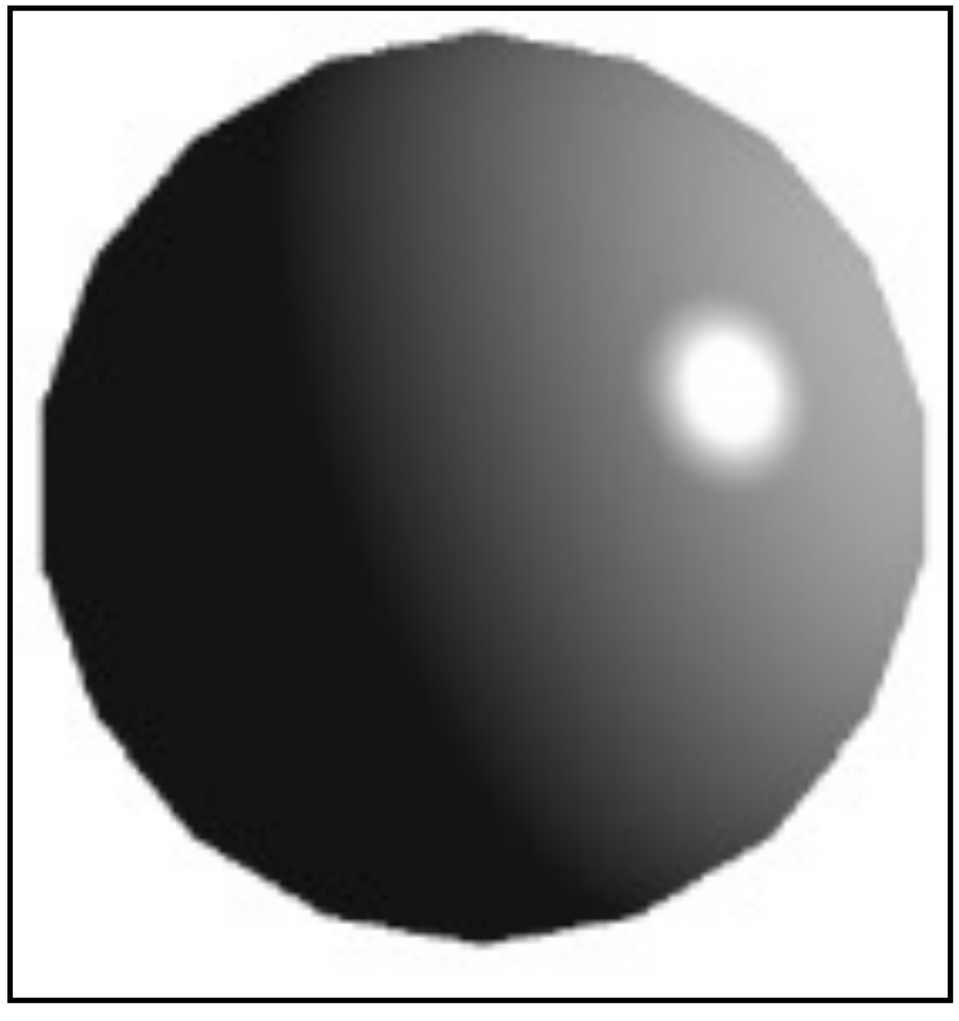
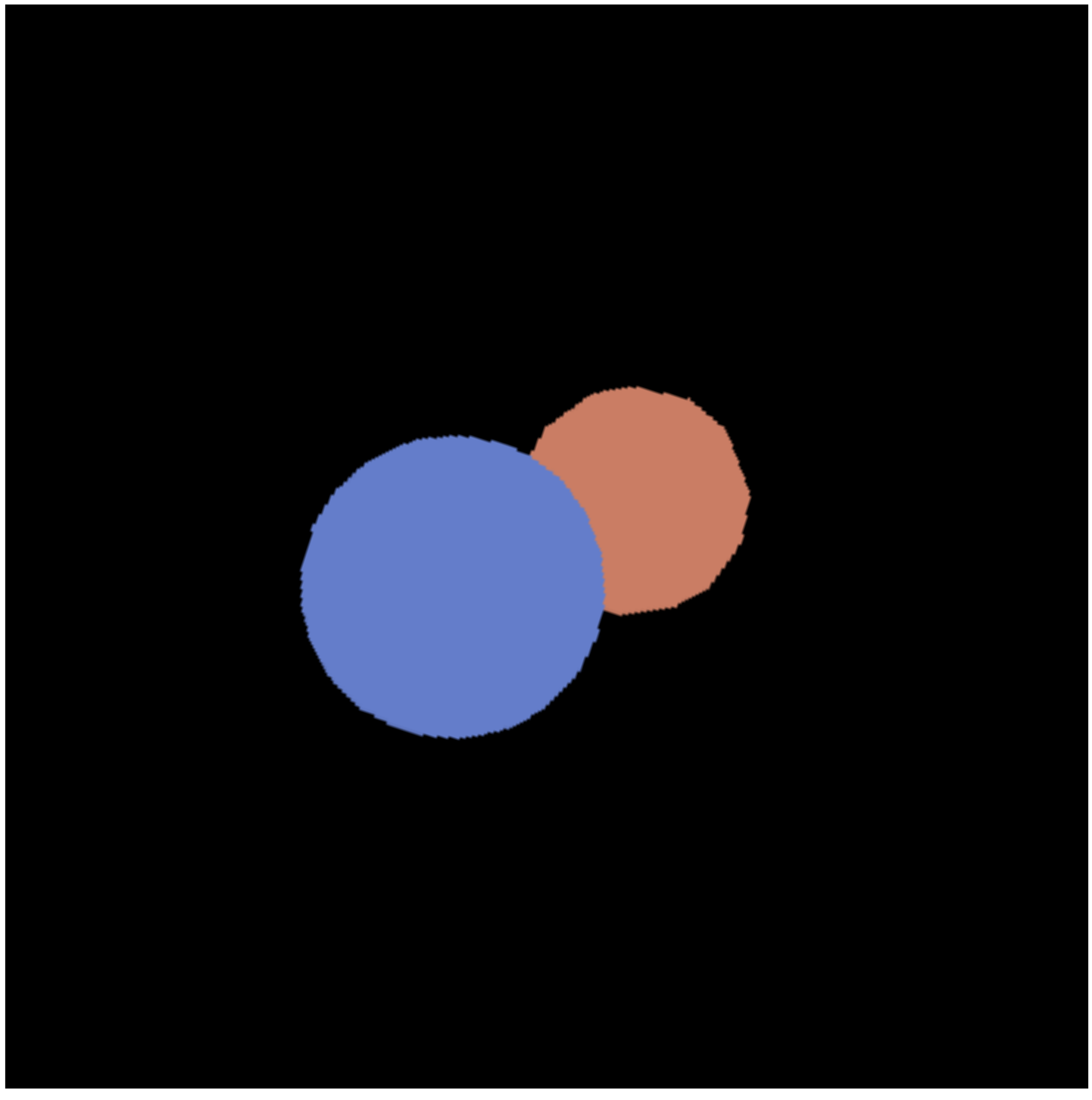


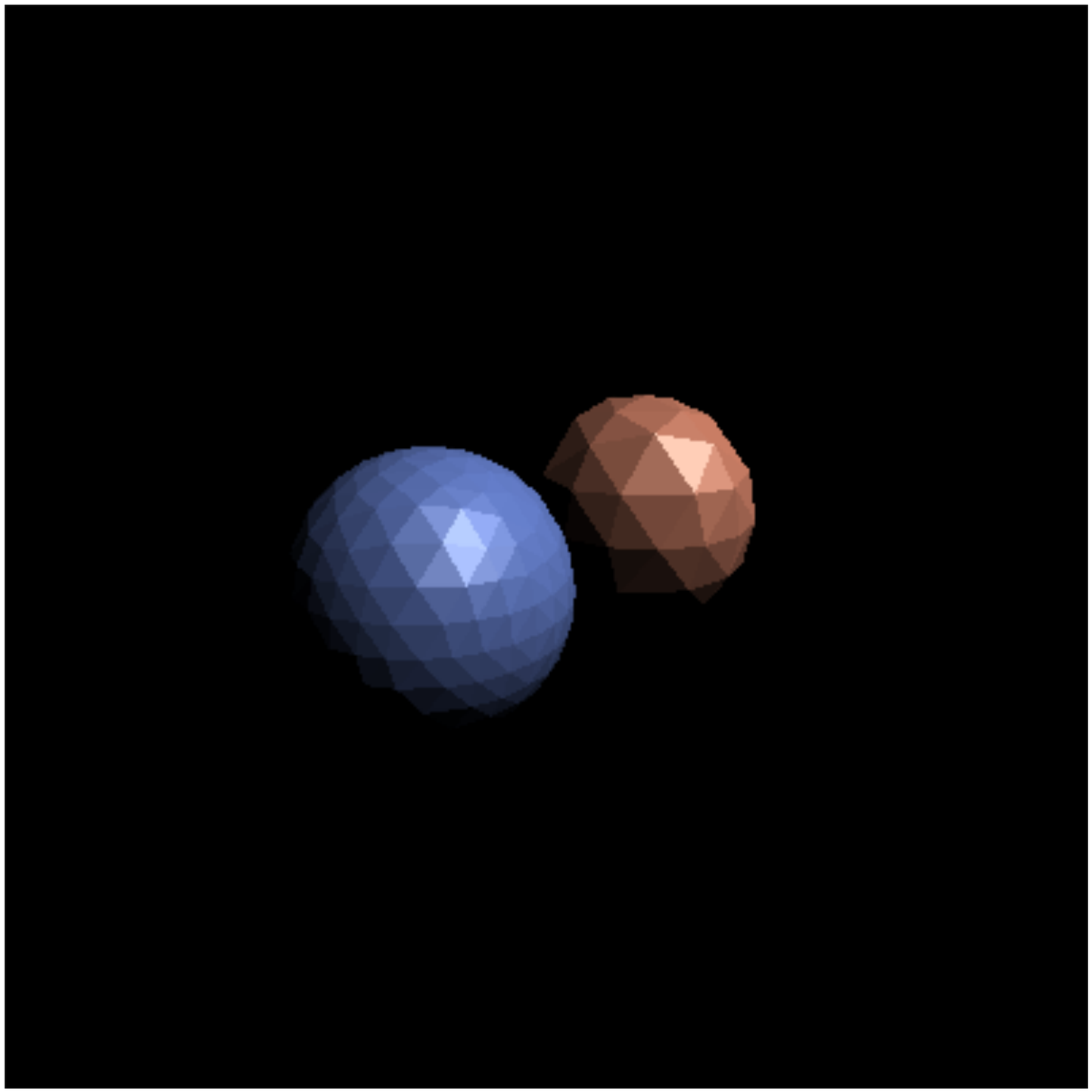
 Image credit: Happyman, http://cg2010studio.com/

Result of Z-Buffer Pipeline

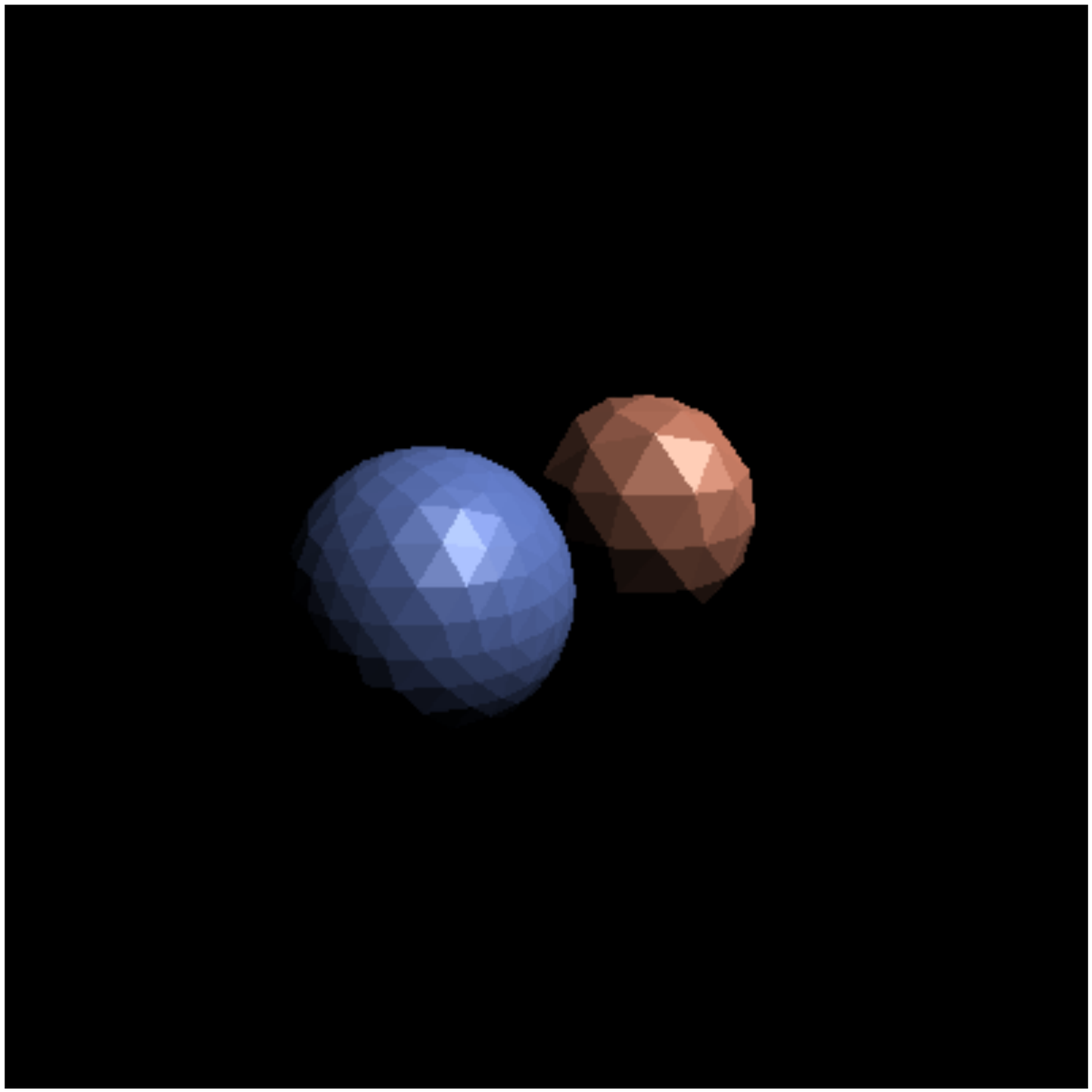


Note: This image does not match next slide, but it's close enough to convey the idea.

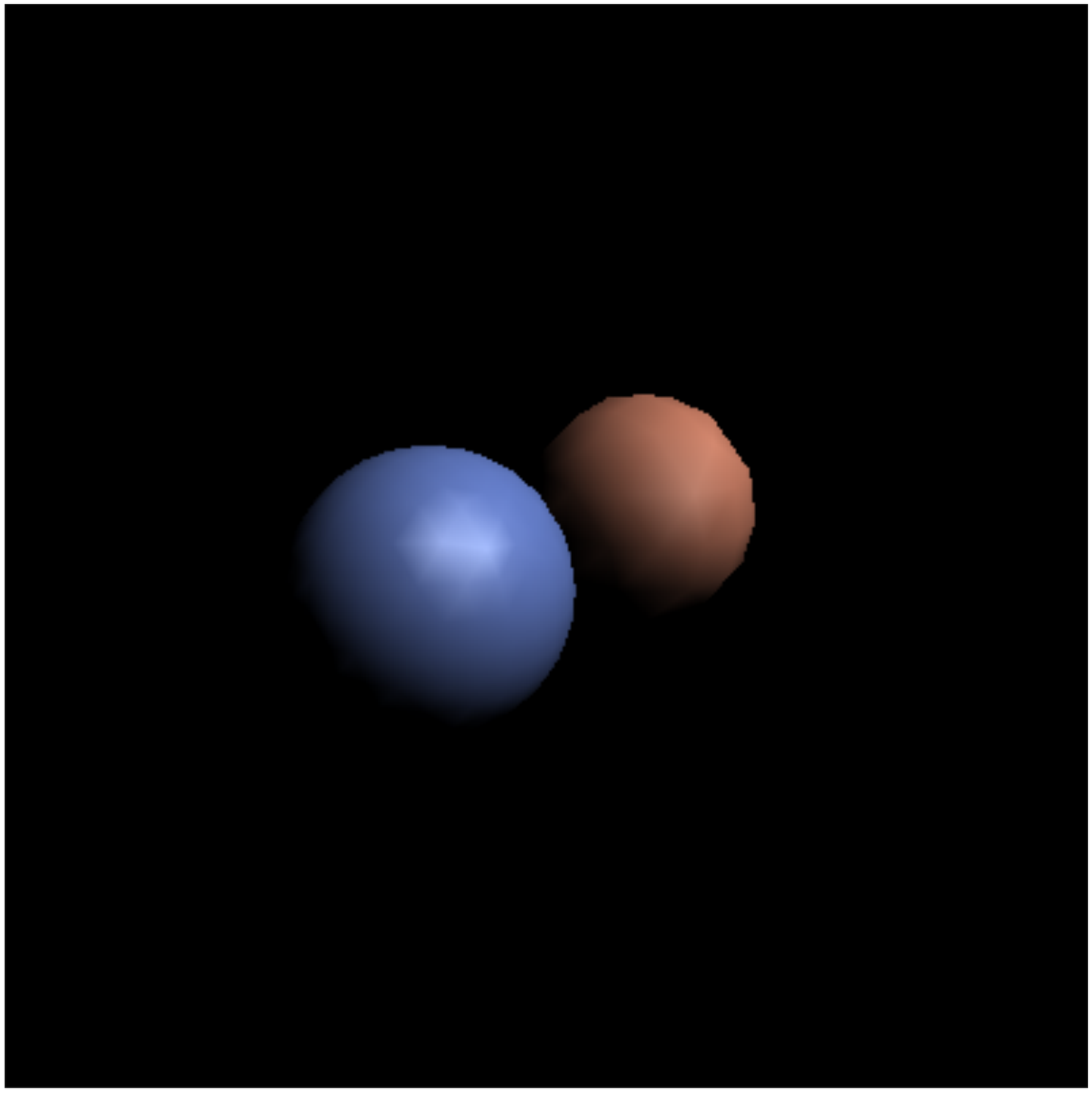
Result of Flat-Shading Pipeline



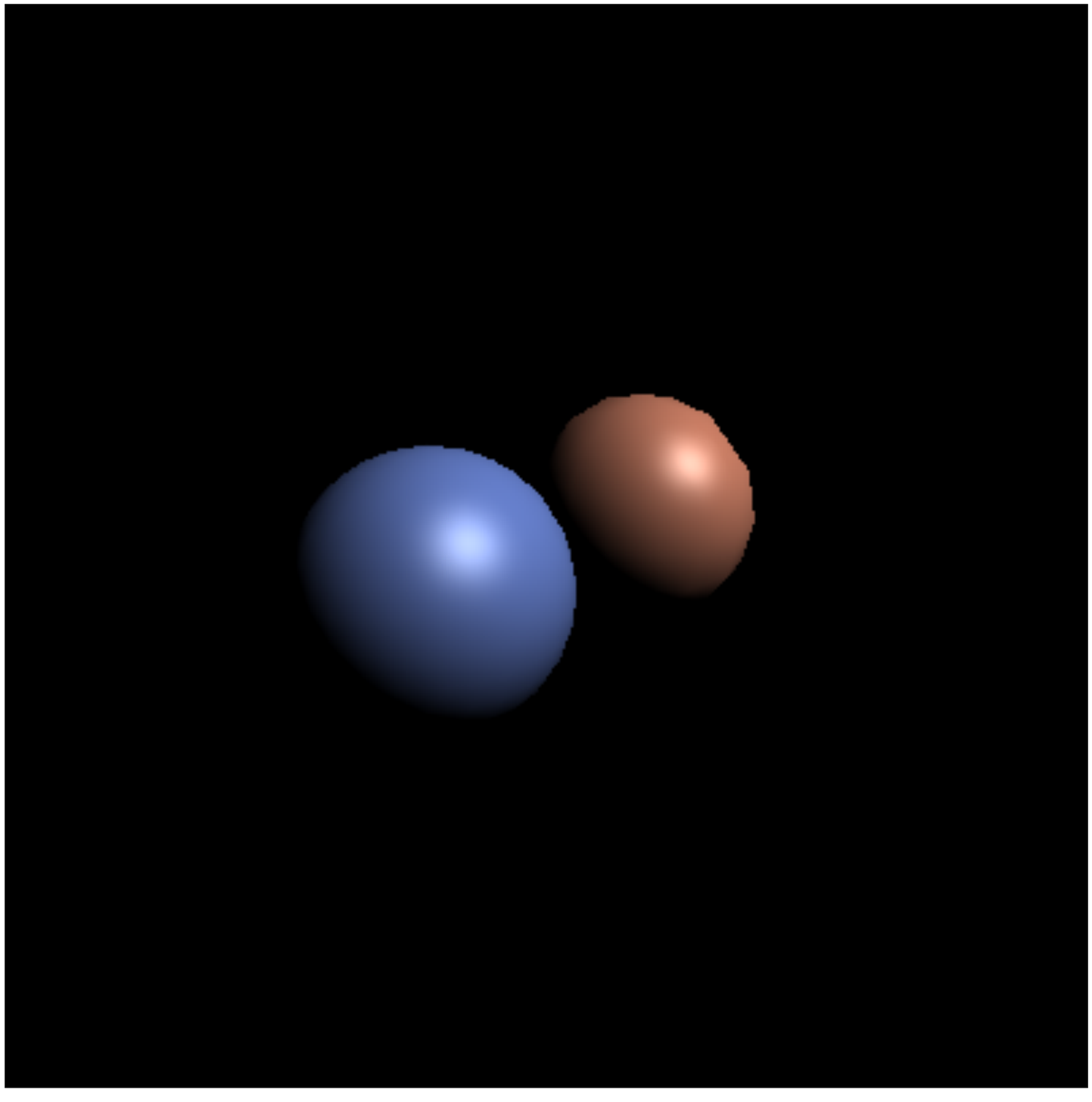
Result of Flat-Shading Pipeline



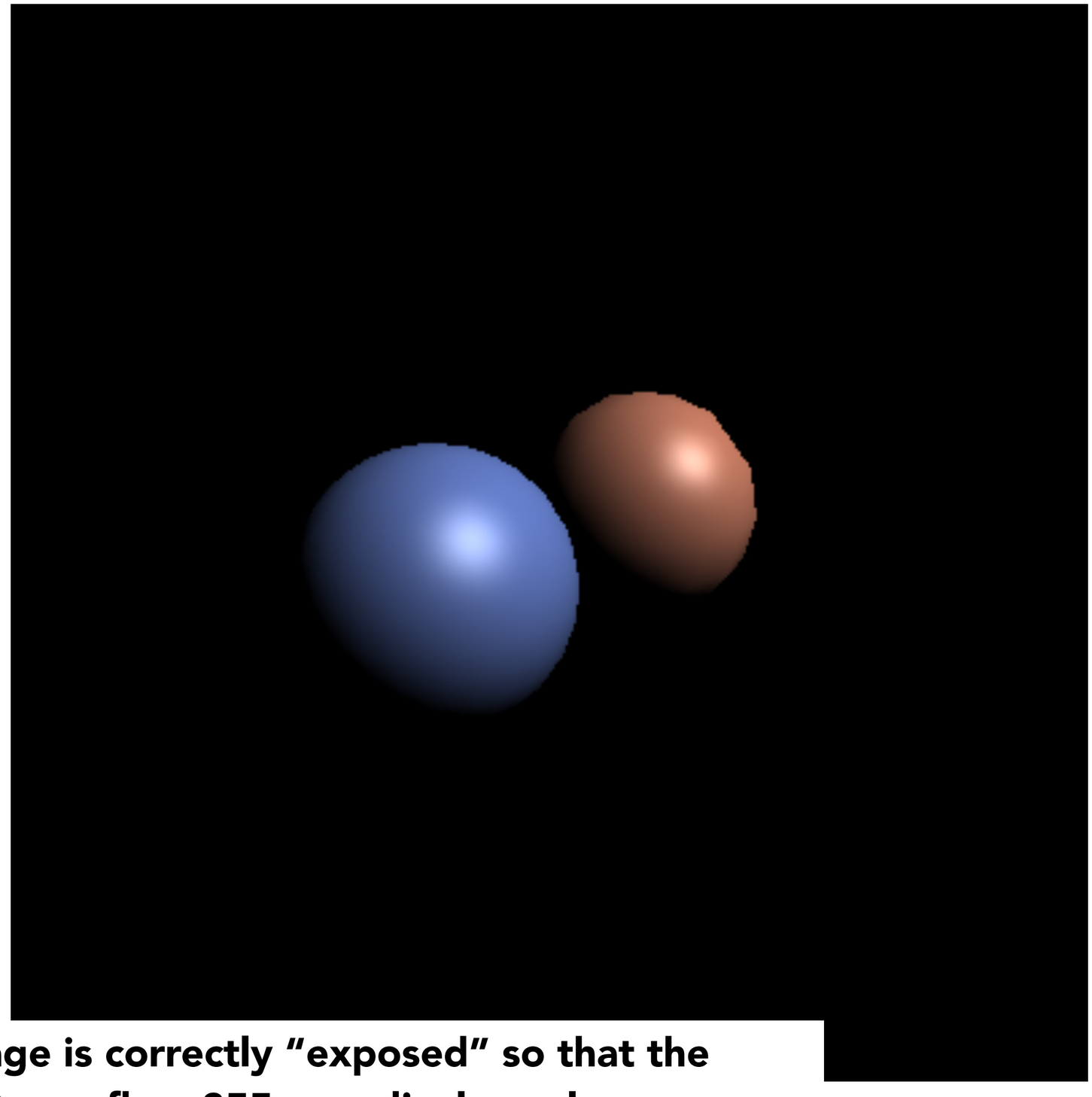
Result of Gouraud Shading Pipeline



Result of Per-Pixel Shading Pipeline



Result of Per-Pixel Shading Pipeline



Note that this image is correctly “exposed” so that the specularities don’t overflow 255 max display value.

Defining Per-Vertex Normal Vectors

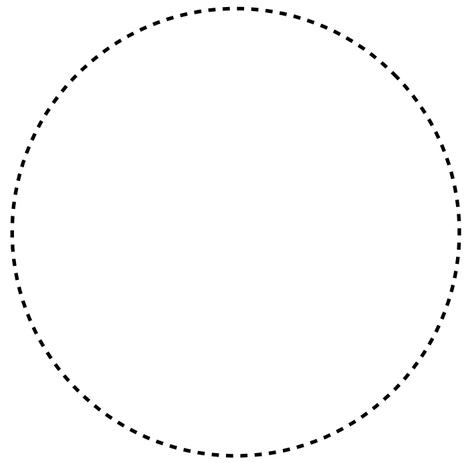
$$\mathbf{N}_v = \frac{\sum_i w_i \mathbf{N_i}}{\sum_i w_i}$$

What weights to use?

Defining Per-Vertex Normal Vectors

Best to get vertex normals from the underlying geometry

- e.g. consider a sphere



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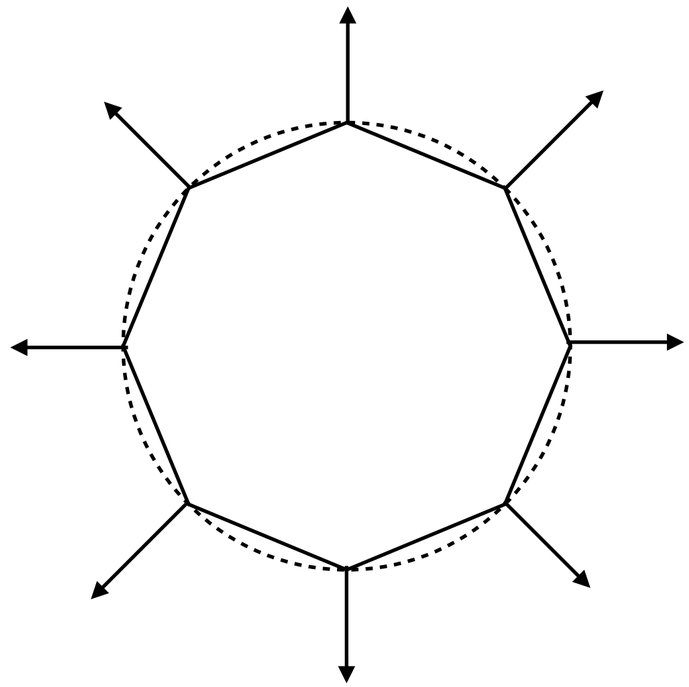


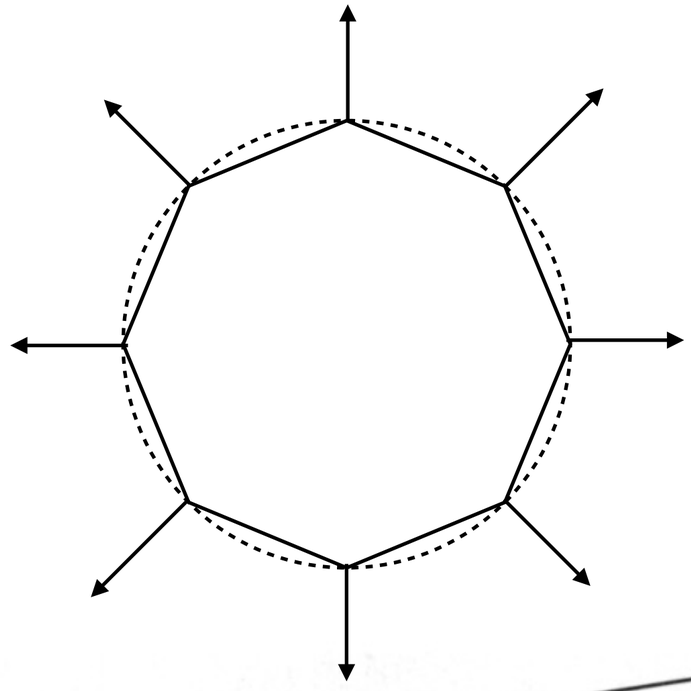
$$\mathbf{N}_v = \frac{\sum_i w_i \mathbf{N_i}}{\sum_i w_i}$$

What weights to use?

Defining Per-Vertex Normal Vectors

Best to get vertex normals from the underlying geometry

- e.g. consider a sphere

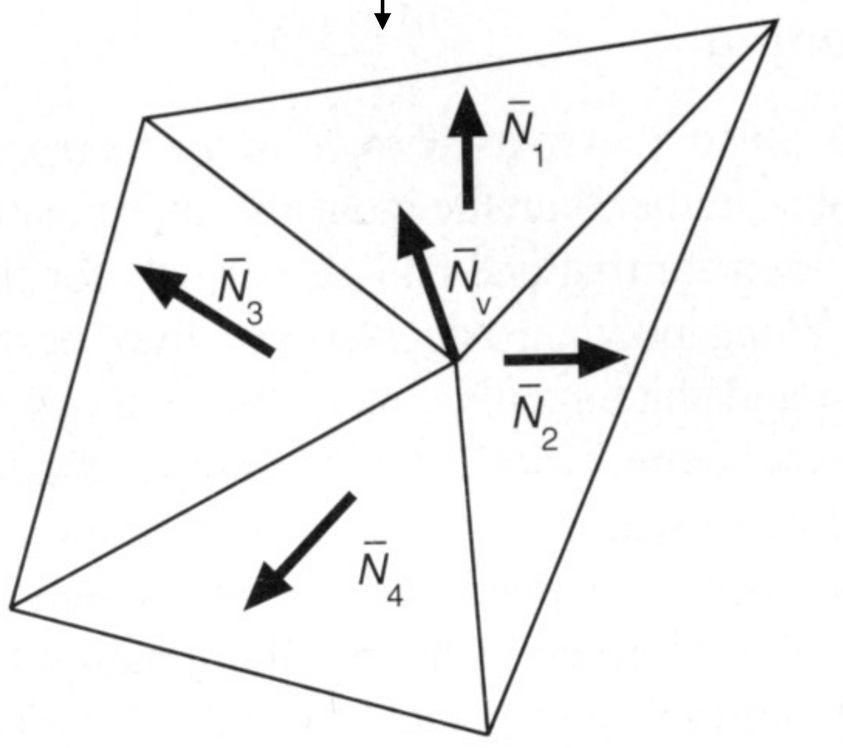


Otherwise have to infer vertex normals from triangle faces

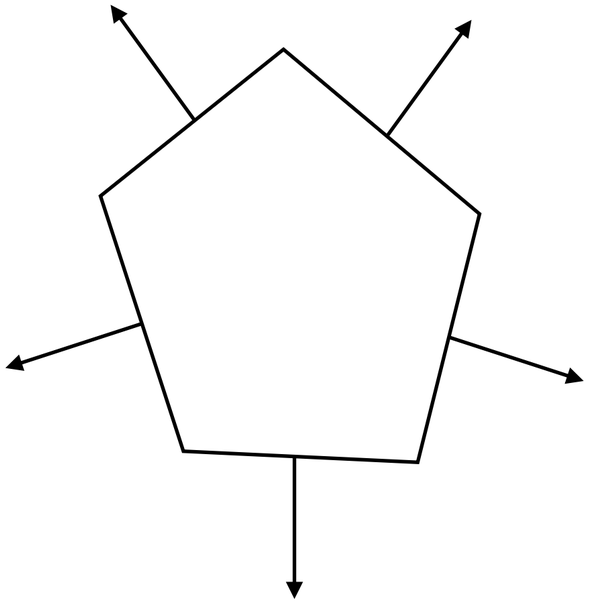
- Simple scheme: average surrounding face normals

$$\mathbf{N}_v = \frac{\sum_i w_i \mathbf{N_i}}{\sum_i w_i}$$

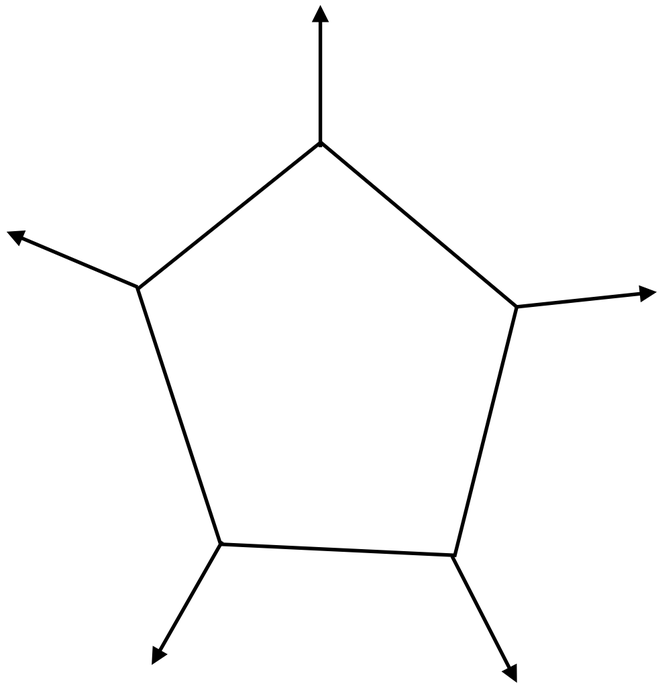
What weights to use?



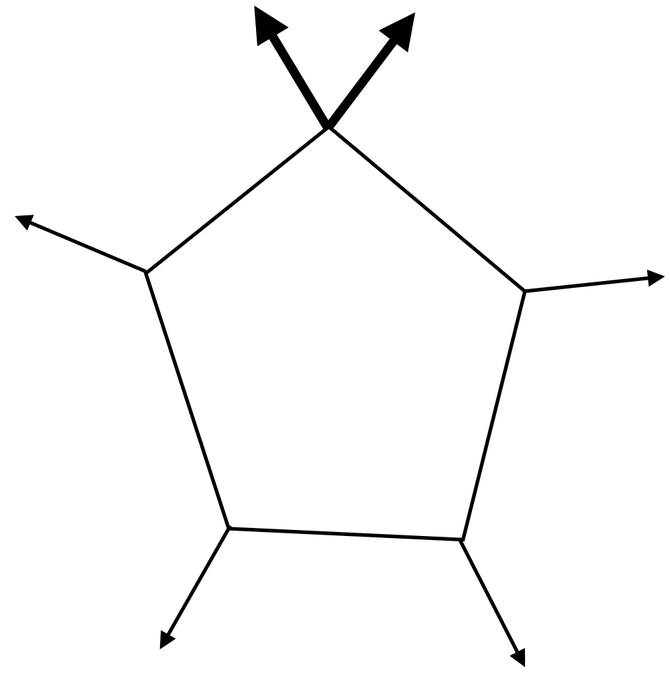
Where are the normals?



Per-Face normals

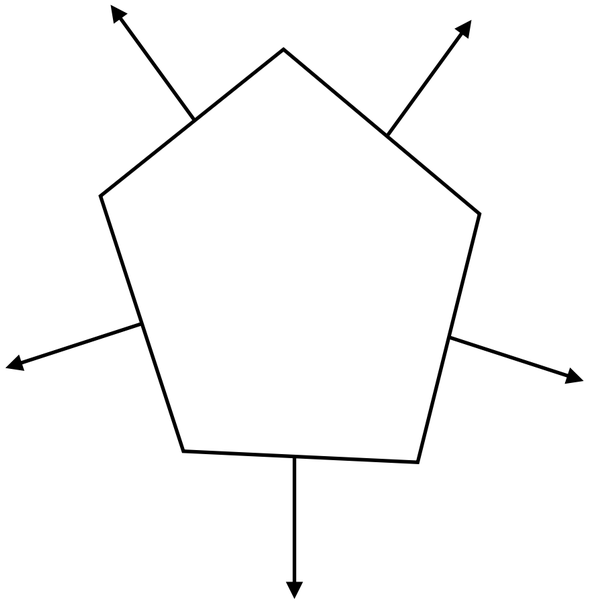


Per-Vertex normals



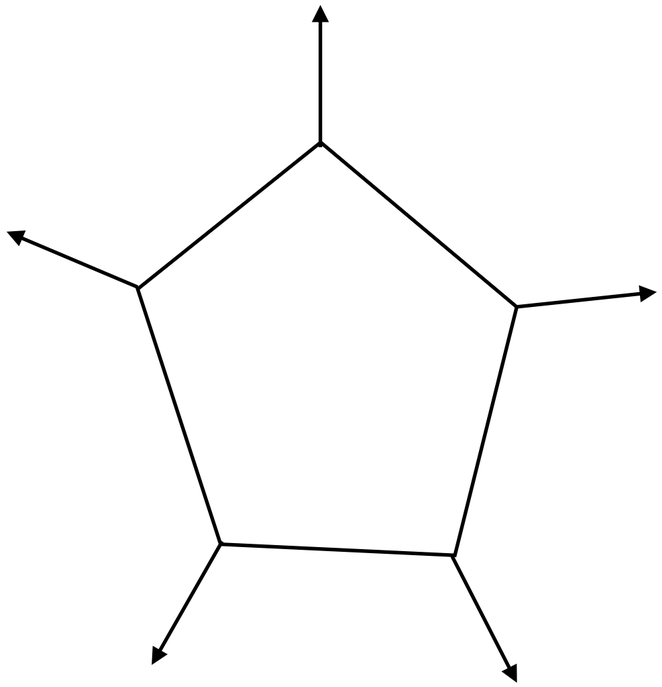
Per-Vertex-Per-Face normals (the bold pair)

Where are the normals?



Per-Face normals

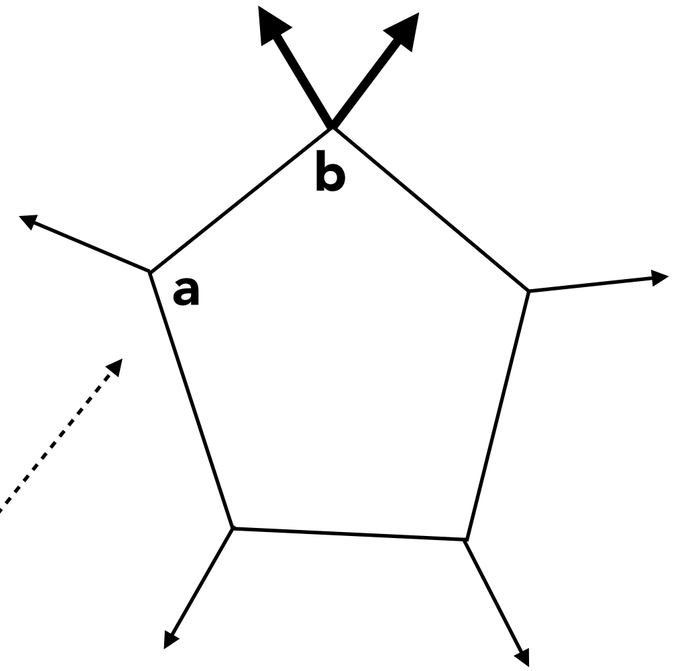
All sharp edges/vertices



Per-Vertex normals

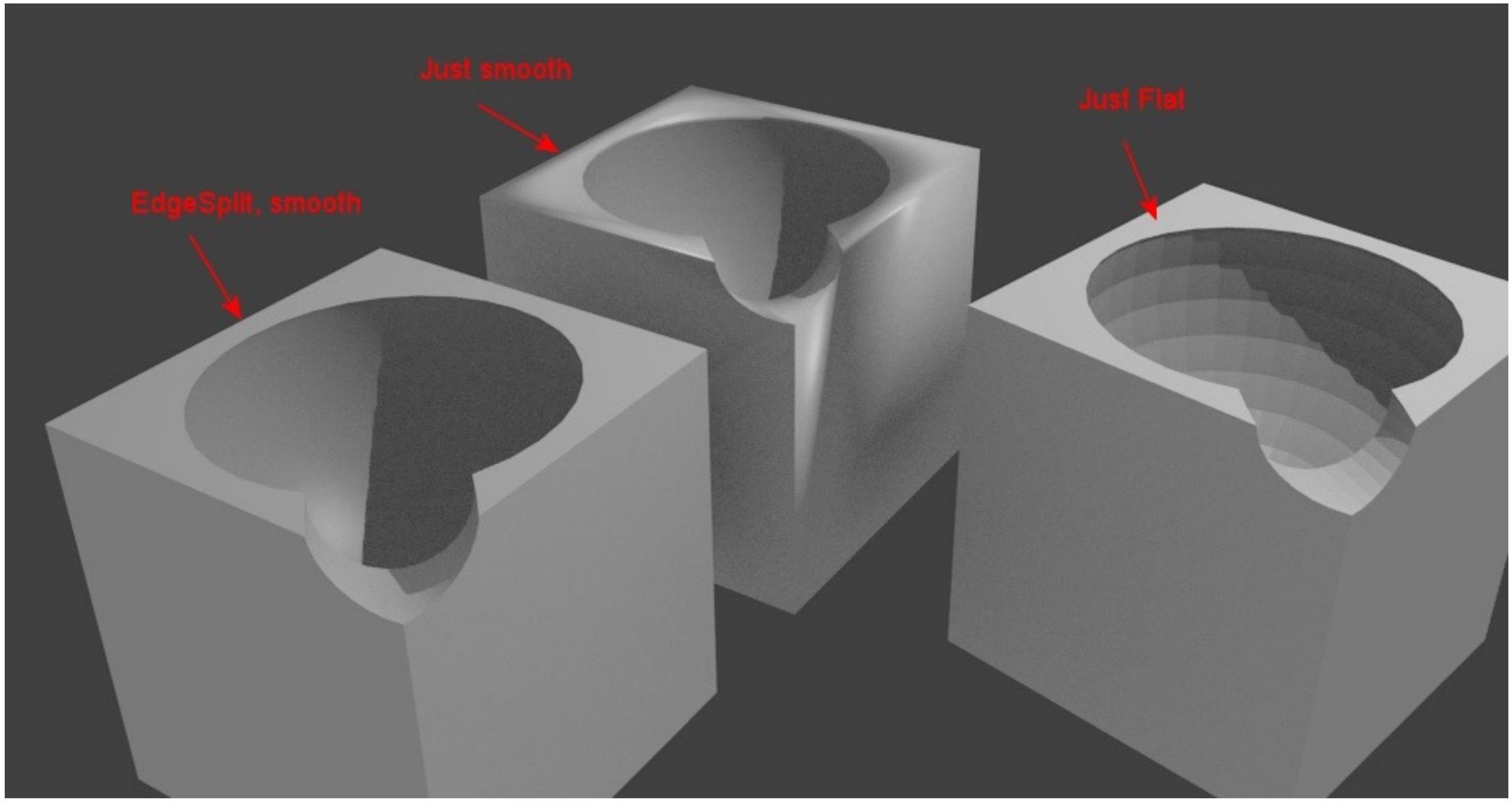
All smooth edges/vertices

a) is a smooth edge/vertex
b) is a sharp edge/vertex



Per-Vertex-Per-Face normals (the bold pair)

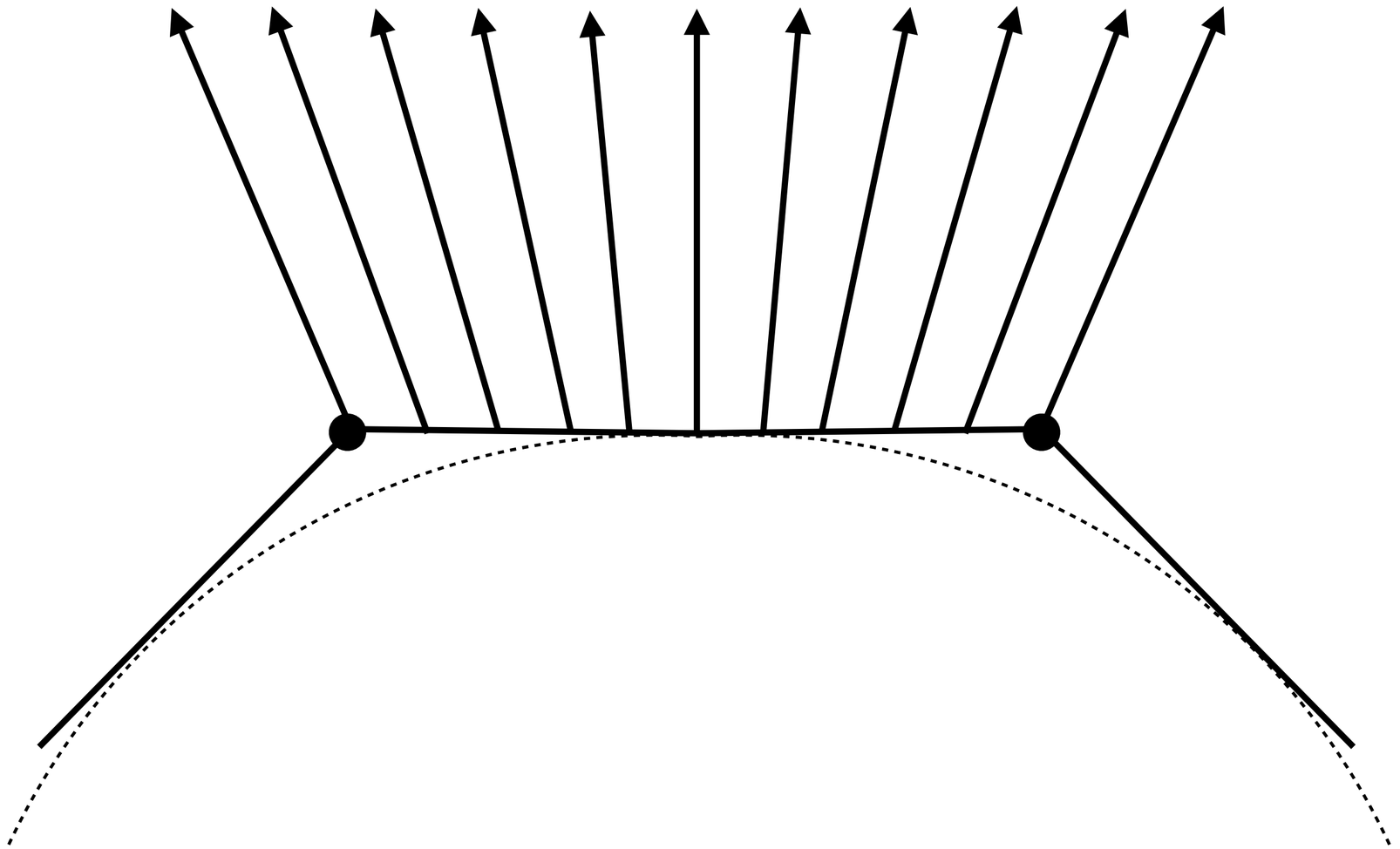
Smooth Shading



From blender.stackexchange.com

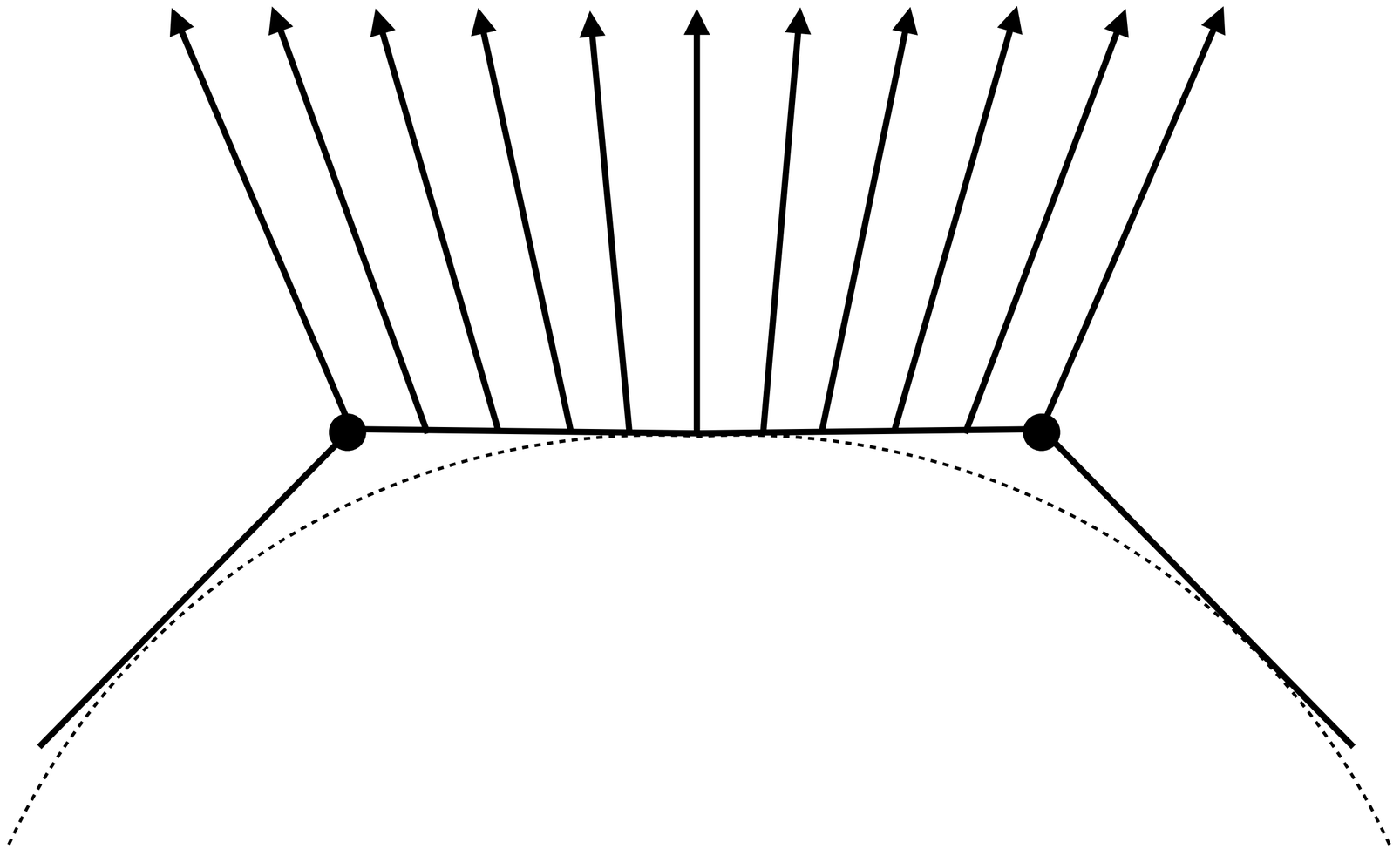
Defining Per-Pixel Normal Vectors

Barycentric interpolation of vertex normals



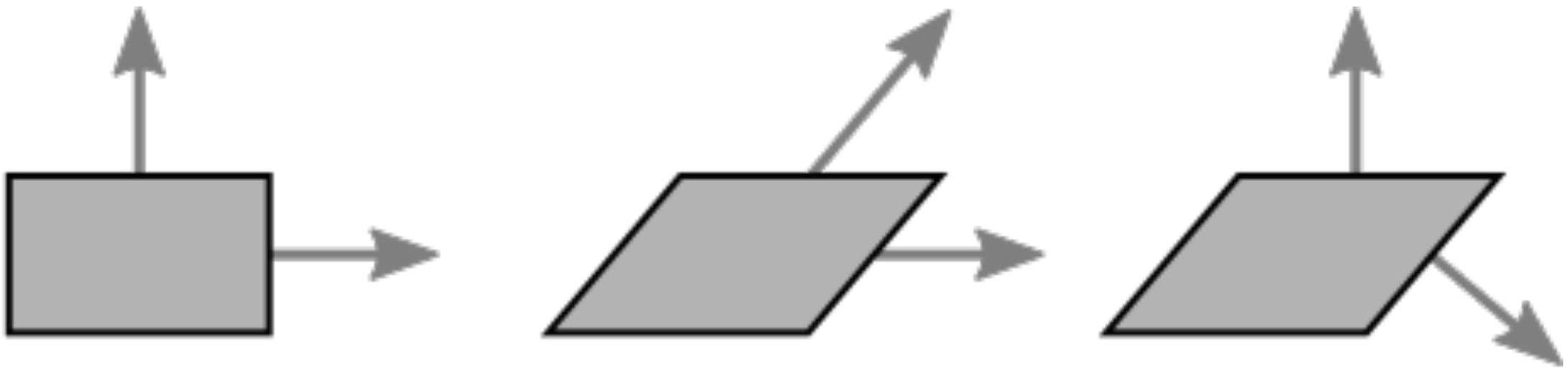
Defining Per-Pixel Normal Vectors

Barycentric interpolation of vertex normals



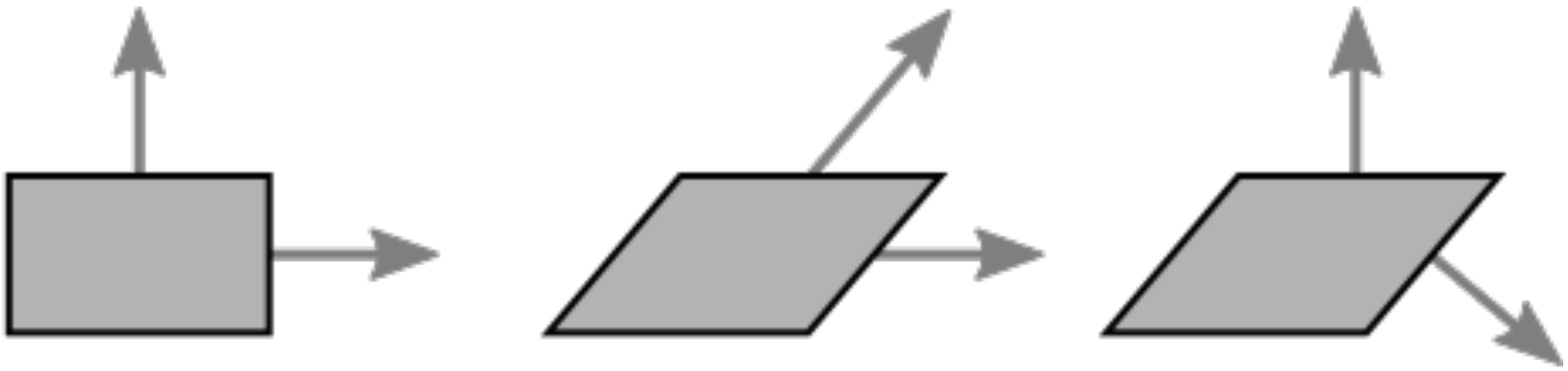
Problem: length of vectors?

Careful: Transforming Normal Vectors



Original object
& normal vectors

Careful: Transforming Normal Vectors

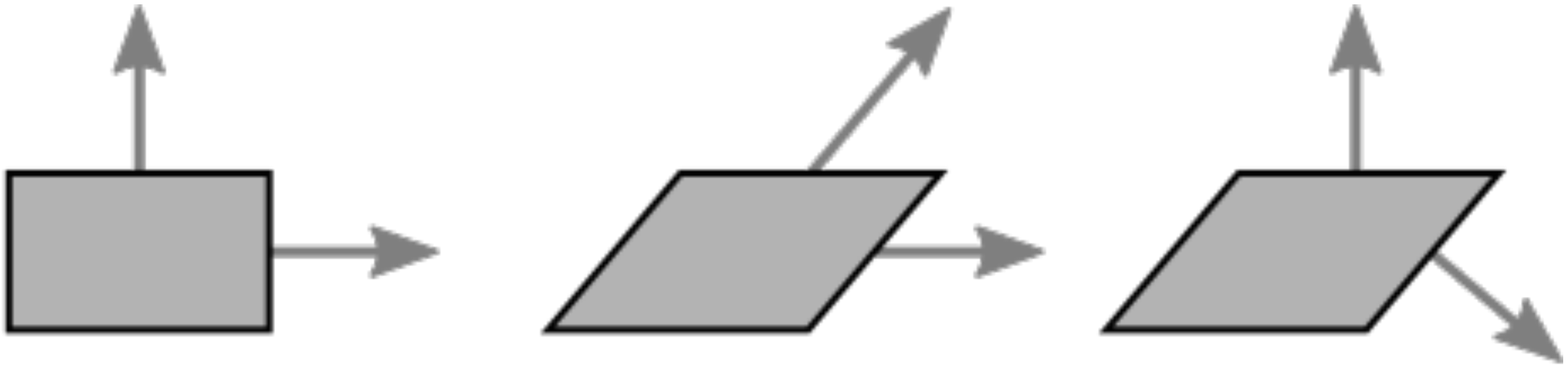


Original object & normal vectors

Transforming vertices & normals with matrix M

Incorrect normals!

Careful: Transforming Normal Vectors



Original object & normal vectors

Transforming vertices & normals with matrix M

Incorrect normals!

Transforming normals with M^{-T}

Correct normals

Careful: Transforming Normal Vectors

Transforming surface normal vectors:

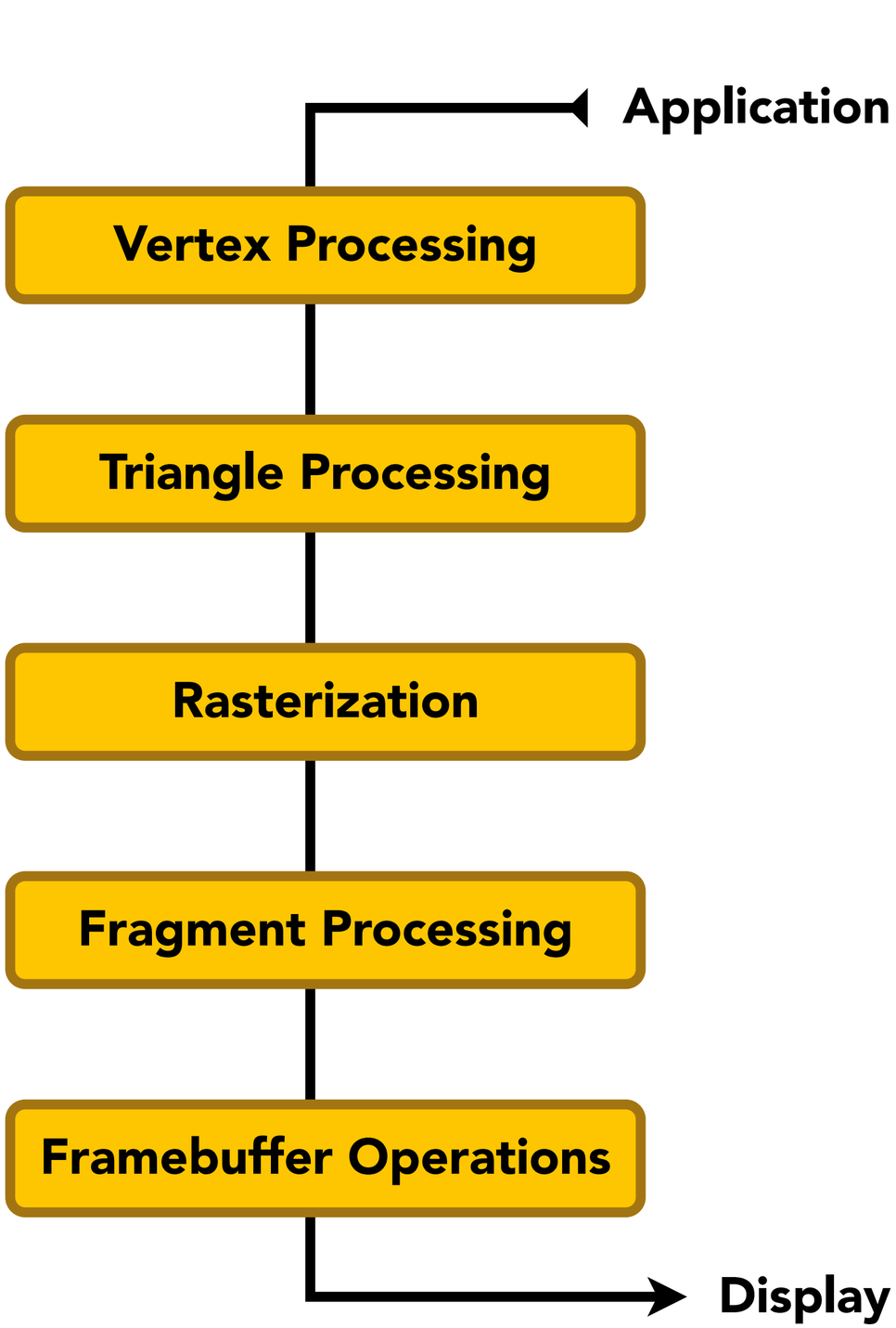
- Must use inverse transpose matrix

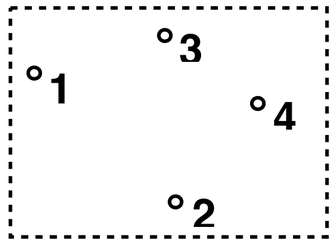
Derivation:

have: $\mathbf{t} \cdot \mathbf{n} = \mathbf{t}^T \mathbf{n} = 0$
want: $M\mathbf{t} \cdot X\mathbf{n} = \mathbf{t}^T M^T X \mathbf{n} = 0$
so set $X = (M^T)^{-1}$
then: $M\mathbf{t} \cdot X\mathbf{n} = \mathbf{t}^T M^T (M^T)^{-1} \mathbf{n} = \mathbf{t}^T \mathbf{n} = 0$

Rasterization Pipeline

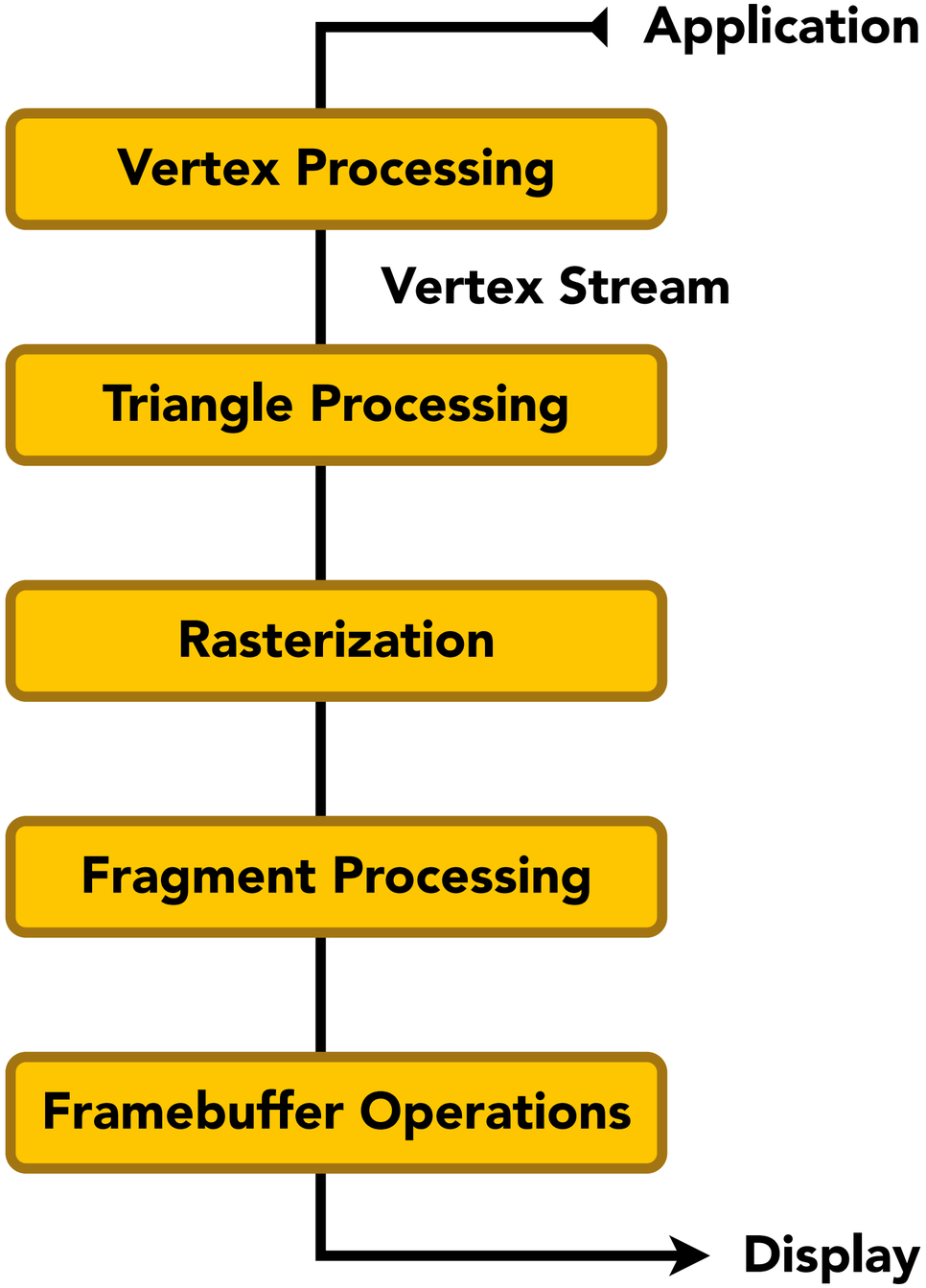
Rasterization Pipeline

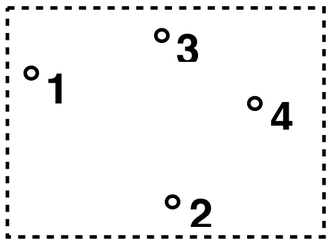




Input: vertices in 3D space

Rasterization Pipeline





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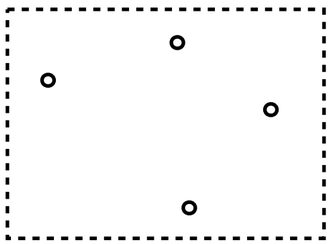
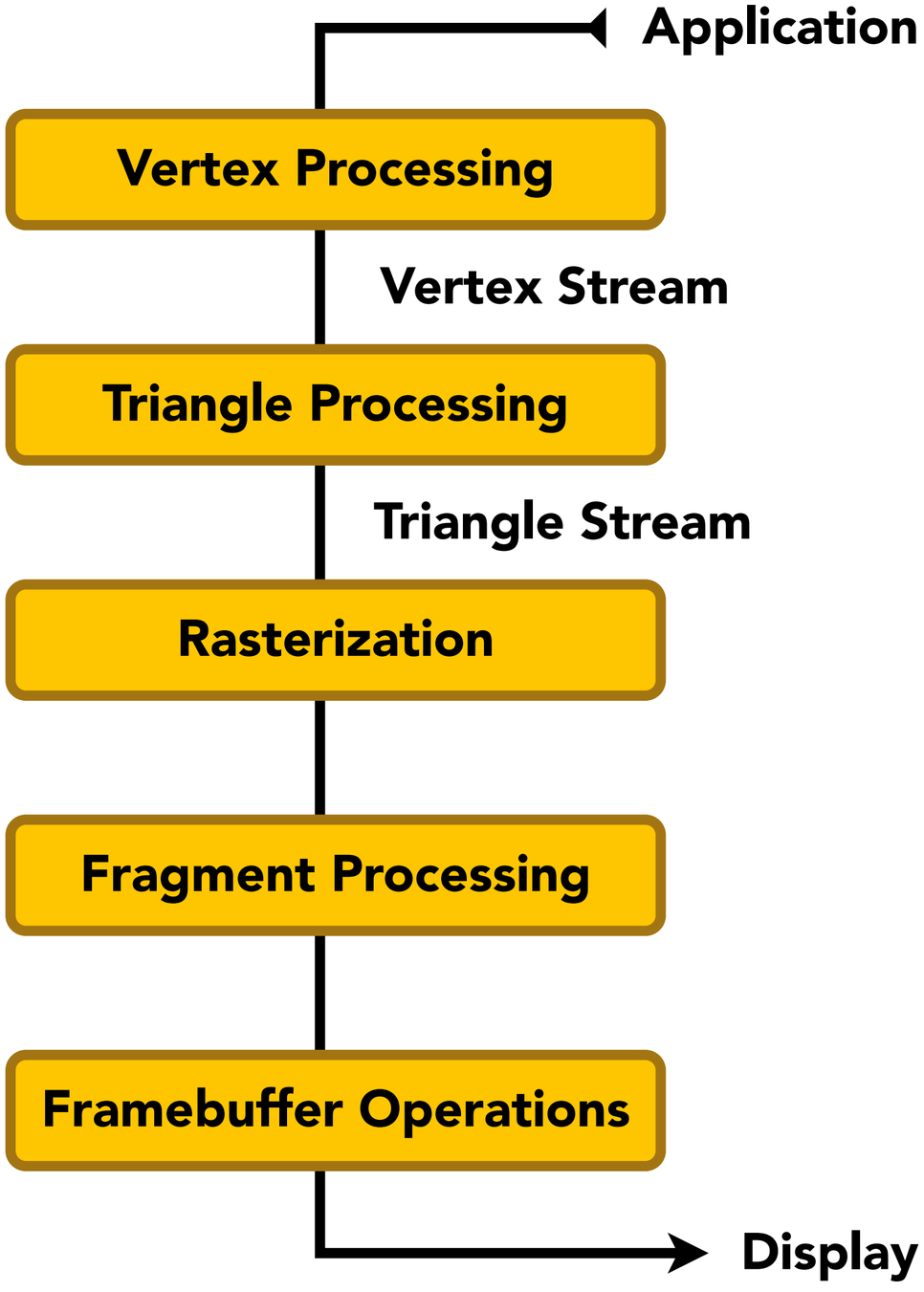
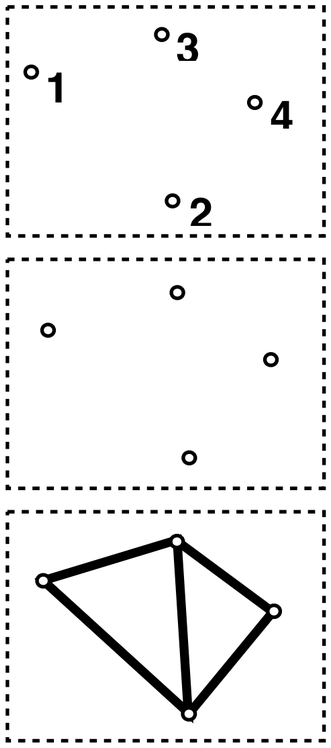


Vertices positioned in screen space

Rasterization Pipeline



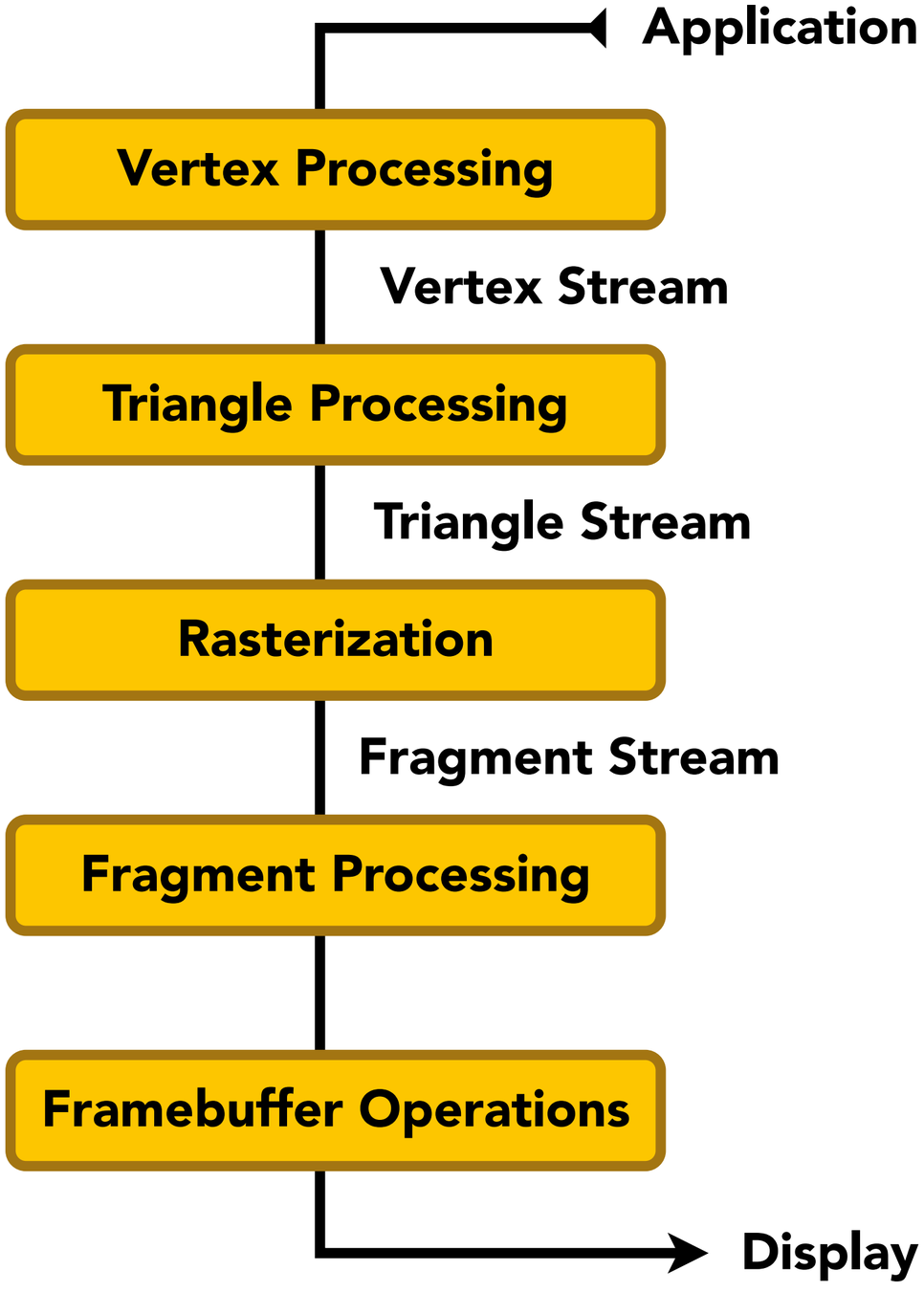


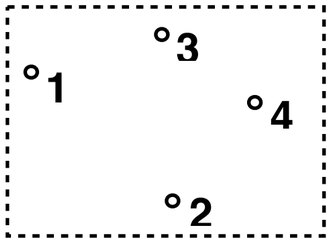
Input: vertices in 3D space

Vertices positioned in screen space

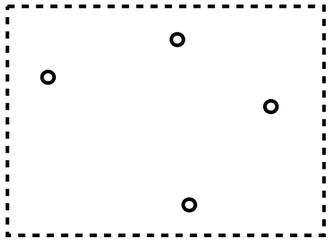
Triangles positioned in screen space

Rasterization Pipeline

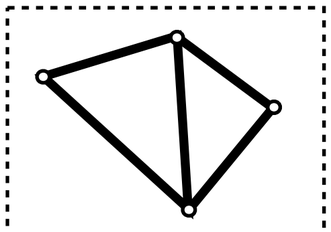




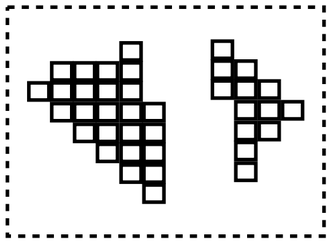
Input: vertices in 3D space



Vertices positioned in screen space

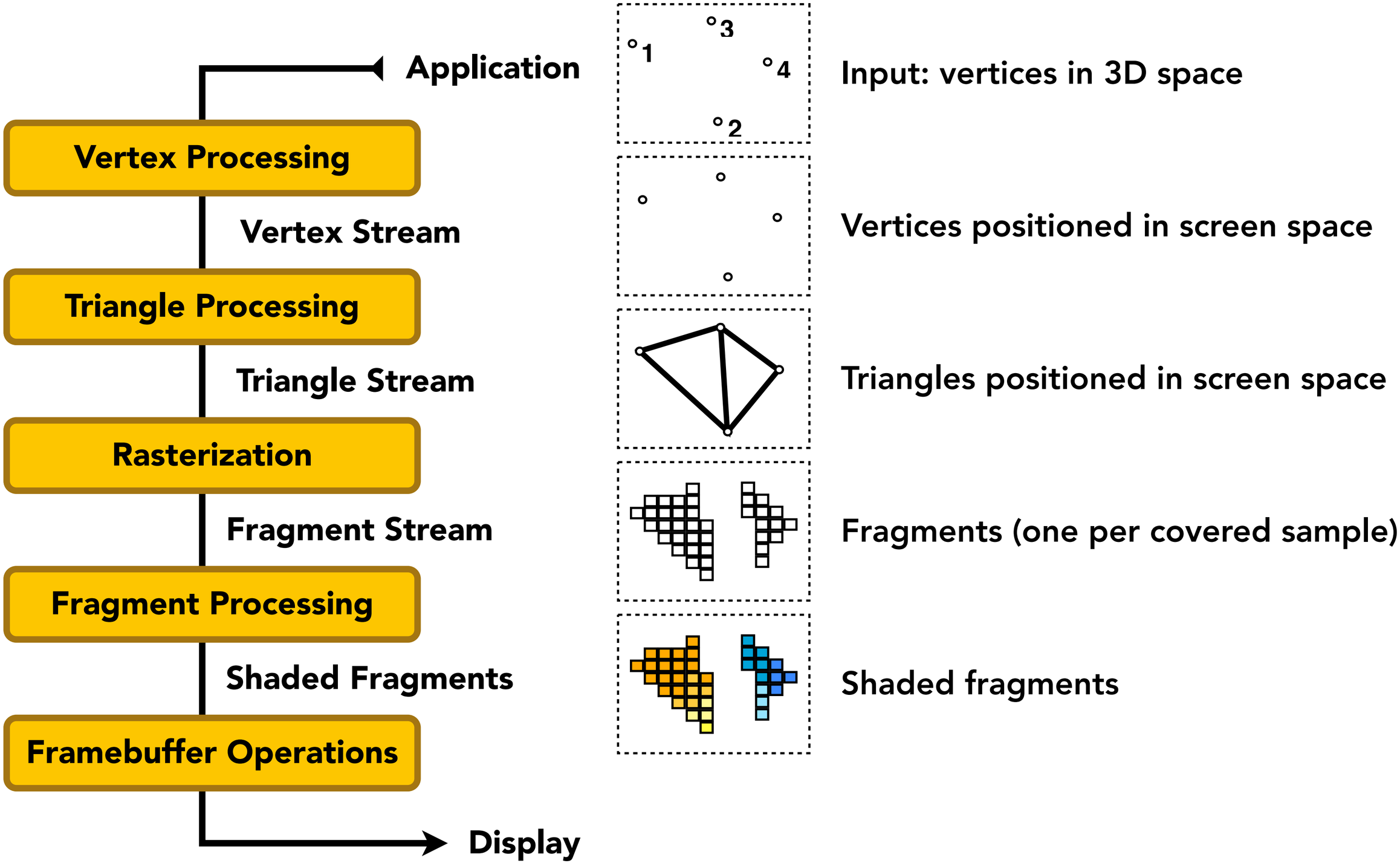


Triangles positioned in screen space

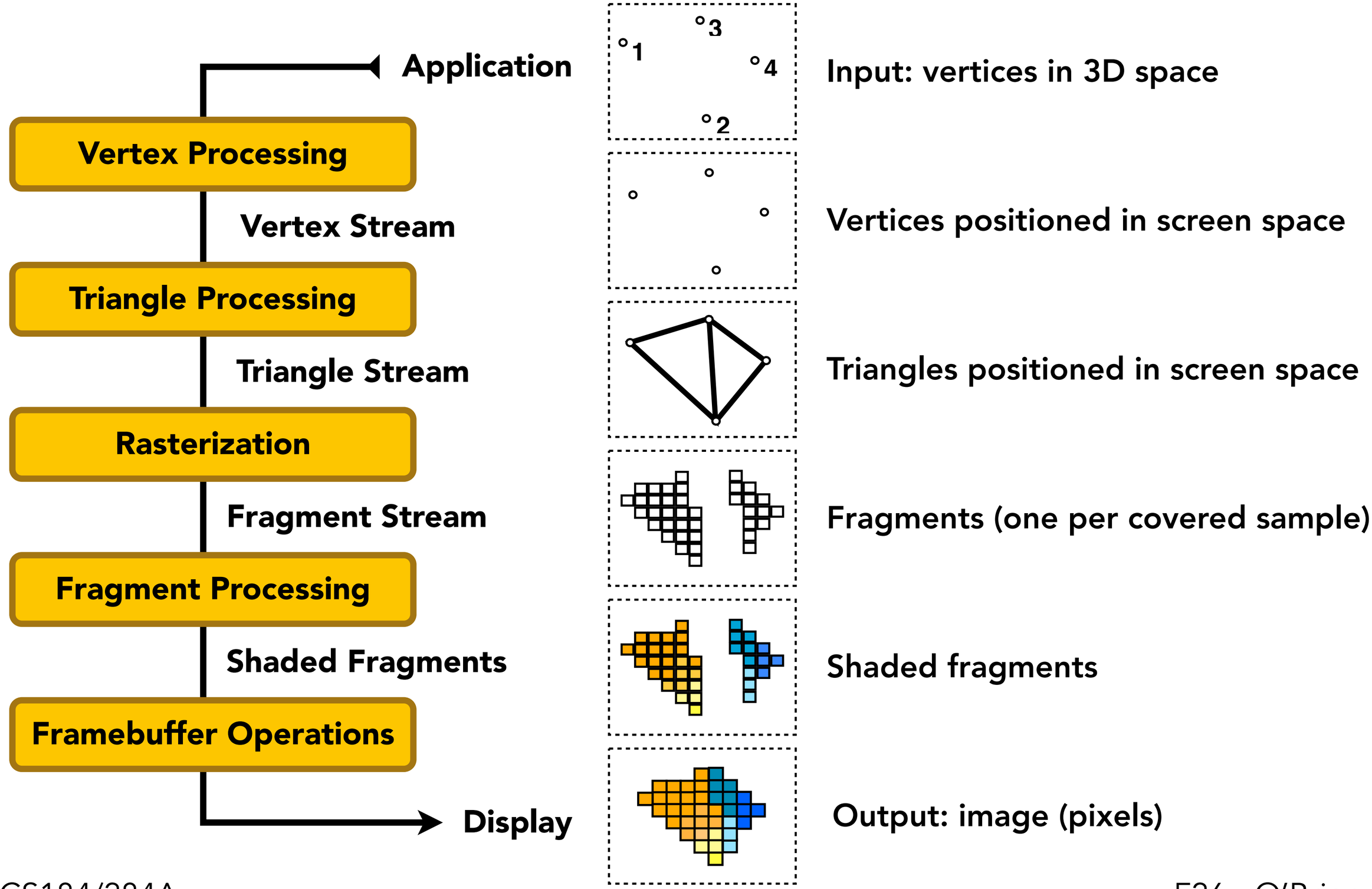


Fragments (one per covered sample)

Rasterization Pipeline



Rasterization Pipeline



Shader Programs

- Program vertex and fragment processing stages
- Describe operation on a single vertex (or fragment)

Example GLSL fragment shader program

```
uniform sampler2D myTexture;
uniform vec3 lightDir;
varying vec2 uv;
varying vec3 norm;

void diffuseShader()
{
  vec3 kd;
  kd = texture2d(myTexture, uv);
  kd *= clamp(dot(–lightDir, norm), 0.0, 1.0);
  gl_FragColor = vec4(kd, 1.0);
}
```

- Shader function executes once per fragment.
- Outputs color of surface at the current fragment's screen sample position.
- This shader performs a texture lookup to obtain the surface's material color at this point, then performs a diffuse lighting calculation.

Shader Programs

- Program vertex and fragment processing stages
- Describe operation on a single vertex (or fragment)

Example GLSL fragment shader program

```
uniform sampler2D myTexture;      // program parameter
uniform vec3 lightDir;            // program parameter
varying vec2 uv;                  // per fragment value (interp. by rasterizer)
varying vec3 norm;                // per fragment value (interp. by rasterizer)

void diffuseShader()
{
  vec3 kd;
  kd = texture2d(myTexture, uv);                  // material color from texture
  kd *= clamp(dot(–lightDir, norm), 0.0, 1.0);    // Lambertian shading model
  gl_FragColor = vec4(kd, 1.0);                   // output fragment color
}
```

Shader Programs

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{
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  kd = texture2d(myTexture, uv);                  // material color from texture
  kd *= clamp(dot(–lightDir, norm), 0.0, 1.0);    // Lambertian shading model
  gl_FragColor = vec4(kd, 1.0);                   // output fragment color
}
```

Shader Programs

Measuring and Modeling the Appearance of Finished Wood

Stephen R. Marschner, Stephen H. Westin, Adam Arbree, and Jonathan T. Moon.
In Proceedings of *SIGGRAPH 2005*. Held in Los Angeles, California, July 2005.

Code on GitHub: https://github.com/mckennapsean/wood-shader

Goal: Highly Complex 3D Scenes in Realtime

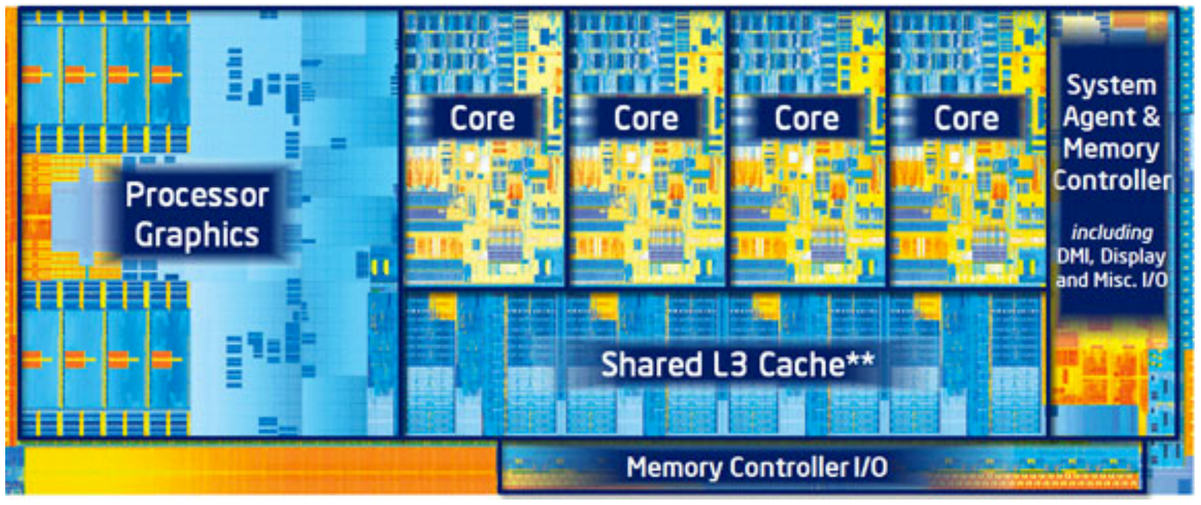
- 100’s of thousands to millions of triangles in a scene
- Complex vertex and fragment shader computations
- High resolution (3-5+ megapixel + supersampling)
- 30-60 frames per second (even higher for VR)

Unreal Engine Kite Demo (Epic Games 2015) [86]

Graphics Pipeline Implementation: GPUs

Specialized processors for executing graphics pipeline computations

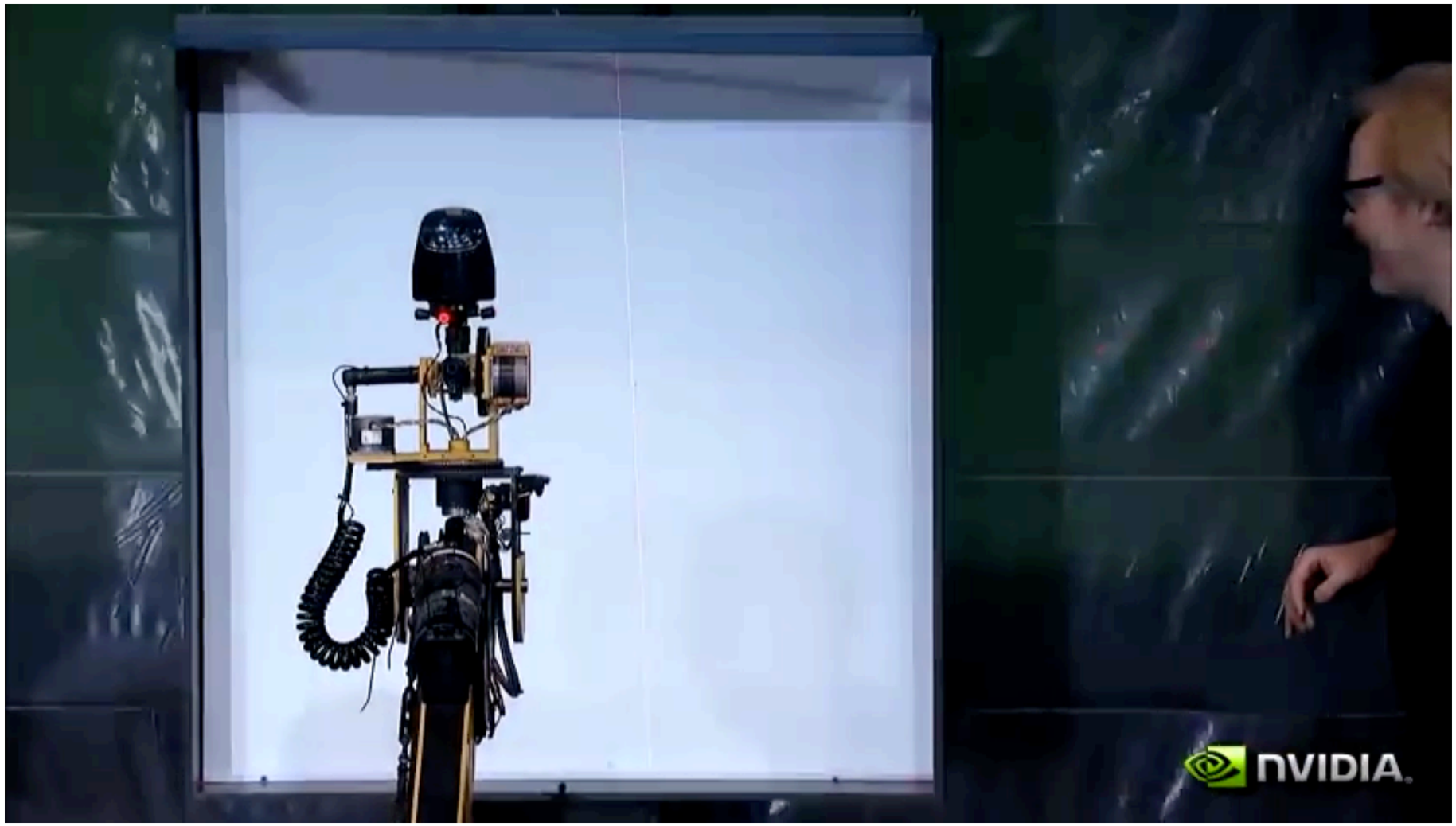
Discrete GPU Card
(NVIDIA GeForce Titan X)



Integrated GPU:
(Part of Intel CPU die)

CPU vs GPU

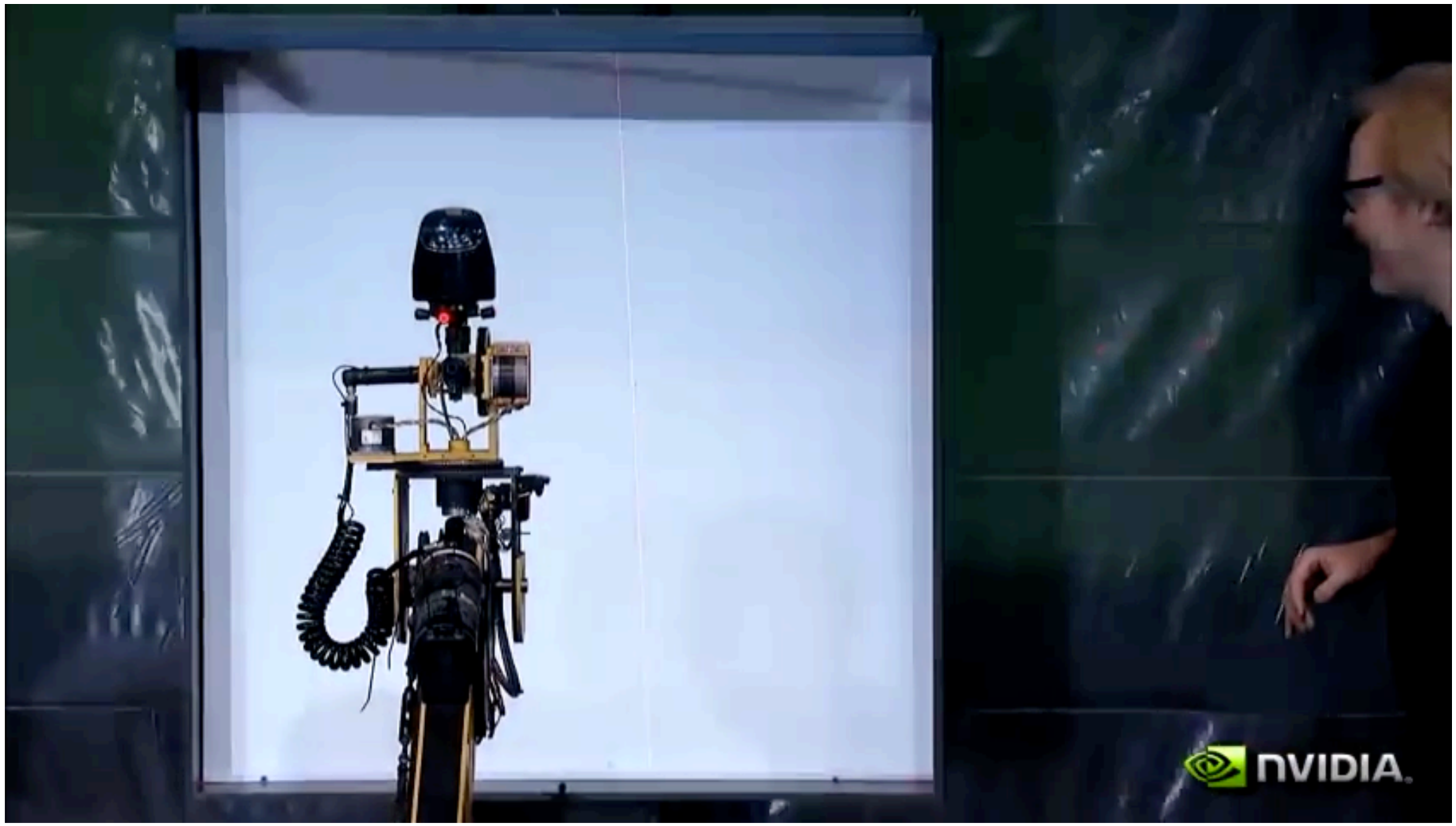
CPU



https://www.youtube.com/watch?v=ZrJeYFxpUyQ

CPU vs GPU

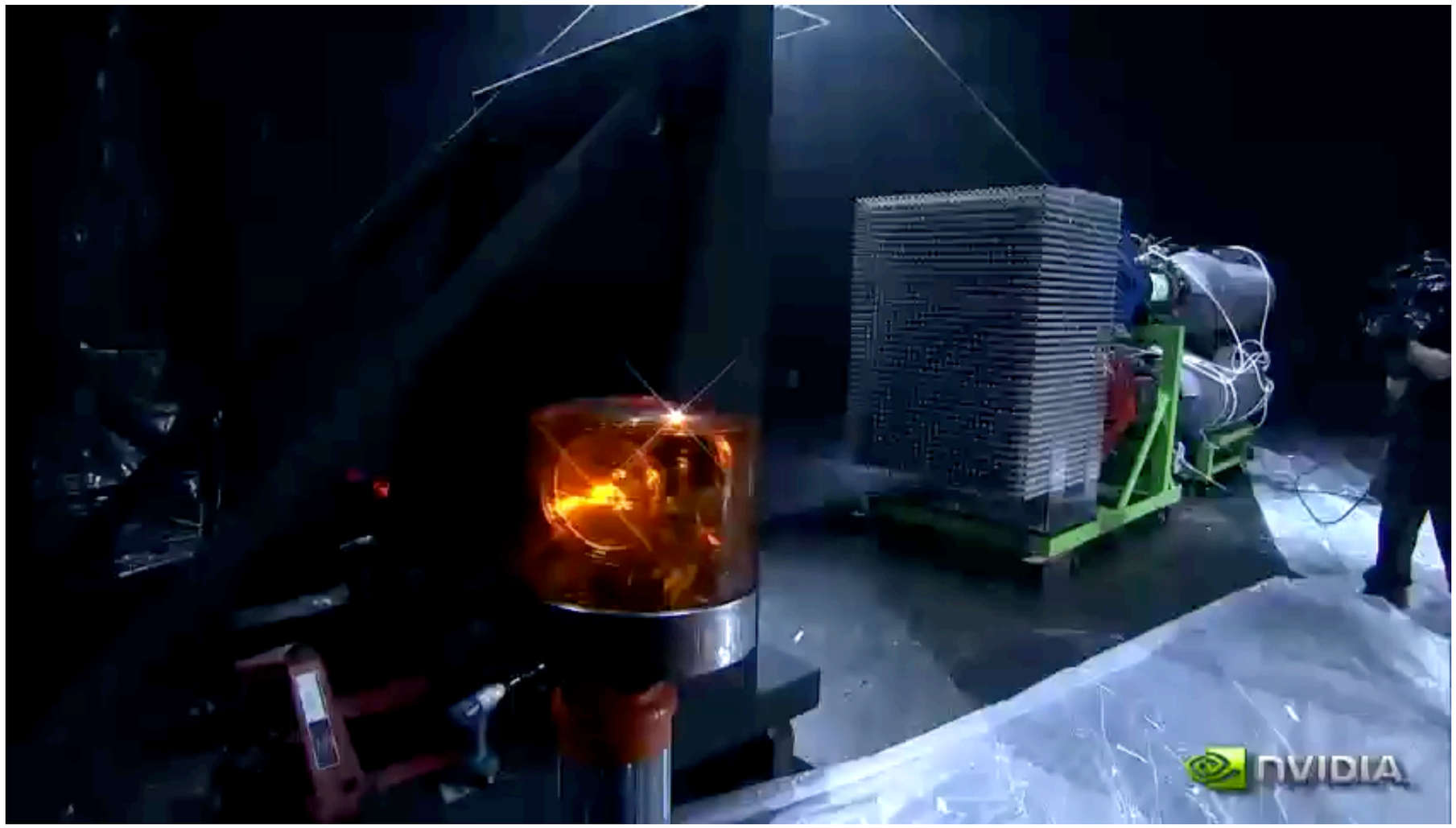
CPU



https://www.youtube.com/watch?v=ZrJeYFxpUyQ

CPU vs GPU

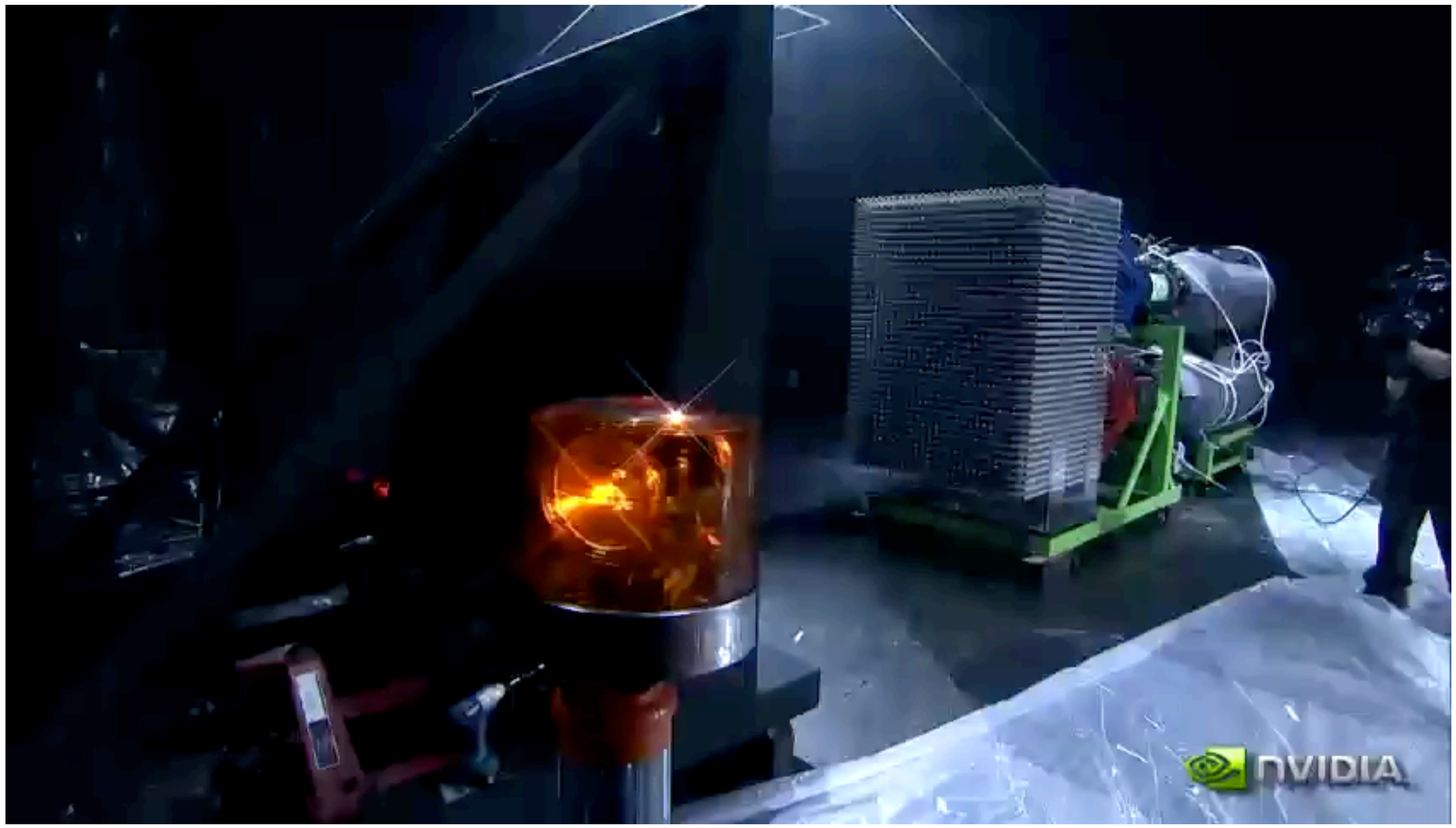
GPU



https://www.youtube.com/watch?v=ZrJeYFxpUyQ

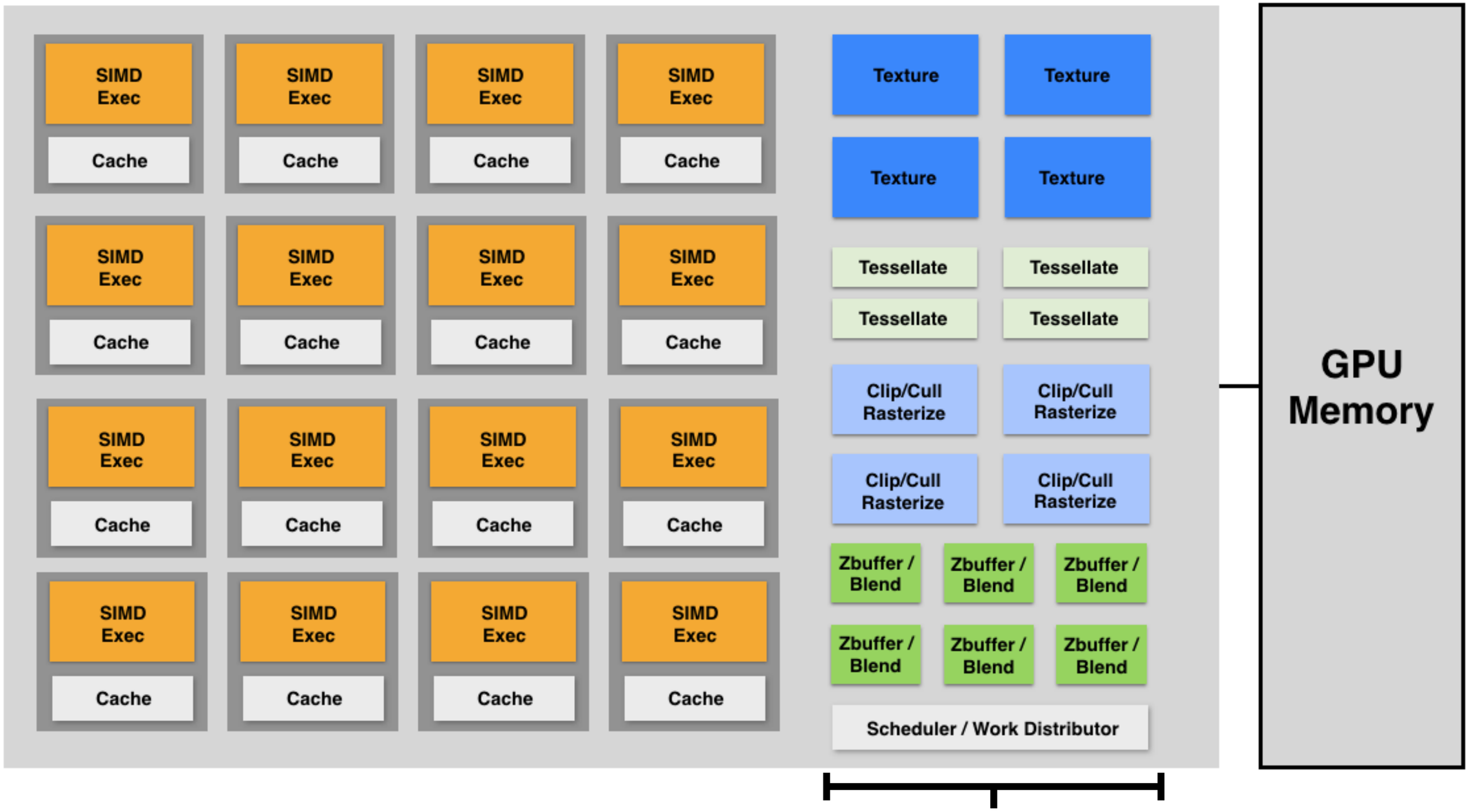
CPU vs GPU

GPU



https://www.youtube.com/watch?v=ZrJeYFxpUyQ

GPU: Heterogeneous, Multi-Core Procesor



Modern GPUs offer ~2-4 Tera-FLOPs of performance for executing vertex and fragment shader programs

Tera-Op's of fixed-function compute capability over here

Things to Remember

Visibility

- **Painter's algorithm and Z-Buffer algorithm**

Simple Shading Model

- **Key geometry: lighting, viewing & normal vectors**
- **Ambient, diffuse & specular reflection functions**
- **Shading frequency: triangle, vertex or fragment**

Graphics Rasterization Pipeline

- **Where do transforms, rasterization, shading, texturing and visibility computations occur?**
- **GPU = parallel processor implementing graphics pipeline**

Acknowledgments

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